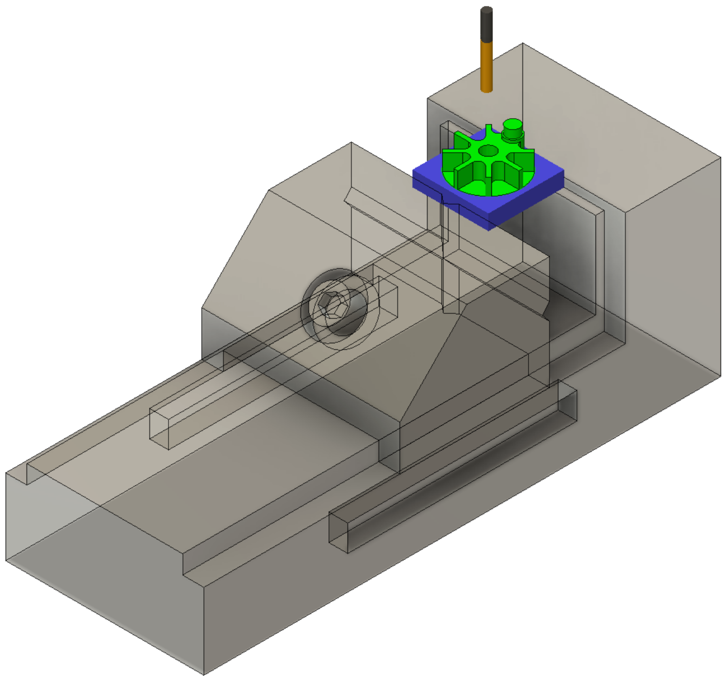
LC-CNC Crank-Slider: Crank
I: General Guidelines



GENERAL SAFETY AND OPERATIONAL INSTRUCTIONS – READ BEFORE USE:

- Users must demonstrate proper personal protective equipment (PPE) use before performing work on this machine.

- Users must wear appropriate PPE (safety glasses, ear protection and mask when necessary) when operating this machine.

- NEVER operate the machine tool without all safety mechanisms (enclosure, door switch, emergency stop, limit switches) connected, complete, and fully operational.

- **NEVER BYPASS ANY SAFETY EQUIPMENT.**

- NEVER place hands, head, or other body parts inside the machine tool enclosure while spindle is powered and door switch / emergency stop are disengaged.

- NEVER handle sharp tools in the spindle without the emergency stop engaged.

- During ANY milling operation, if the cutting tool shows excessive wear or excessive chip build up, or if the spindle's RPM changes noticeably (this is often audible), quickly feed hold or emergency stop the machine (depending on severity) to inspect and determine if it is safe to continue.

- Always allow the spindle to reach its target speed before entering a cut.

- Always double-check setup steps and part programming to minimize machine accidents.

- Always watch and listen to EVERY milling operation closely. Keep a hand resting over the emergency stop button: quick reactions are necessary in mitigating damage to the machine and tools.

II: Operator Checklist

Operator name: ______________________________________

Workpiece:____

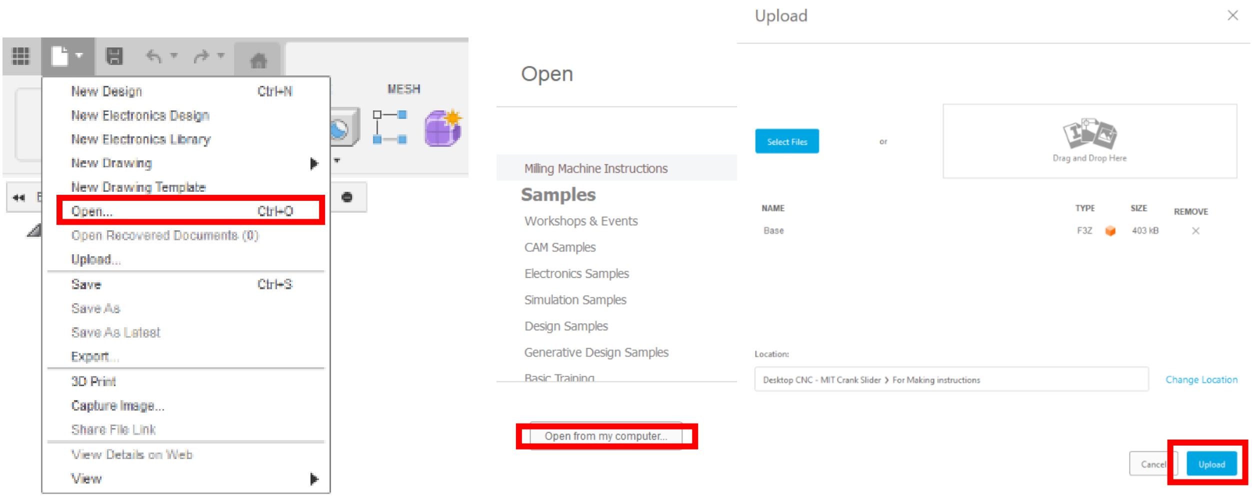
Date:____

Before running the program		
OP10	OP20	
☐	☐	Program has been simulated in CAM
☐	☐	All tool offsets are set per the setup sheet
☐	☐	Workpiece is clamped in the proper position
☐	☐	Correct program is selected on the machine
		Work offsets are set for:
☐	☐	X axis
☐	☐	Y axis
☐	☐	Z axis

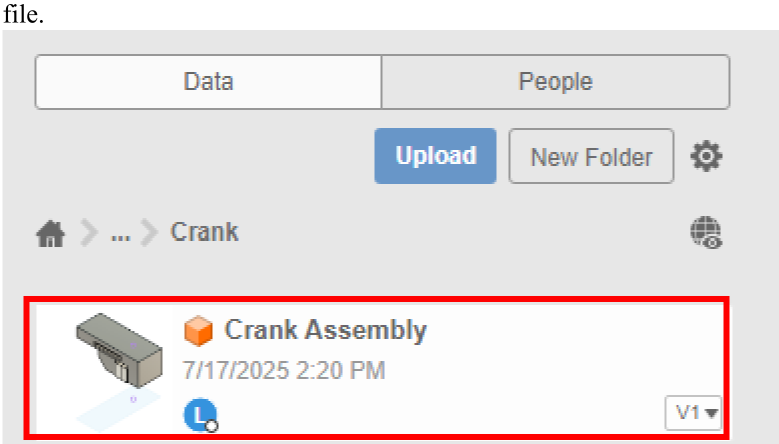
For each tool while running the program		
OP10	OP20	
☐	☐	Check position as the tool approaches the part

III. Part Programming (Autodesk Fusion) OP10

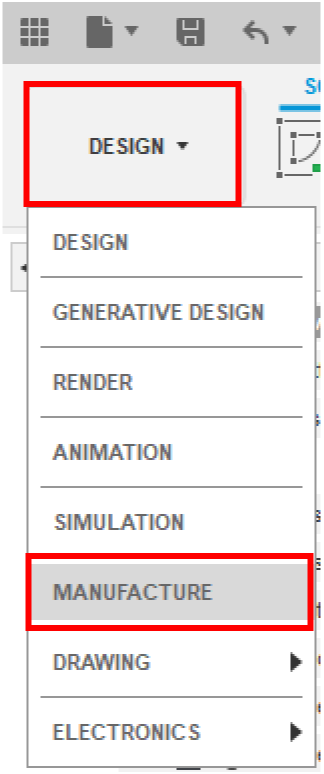
1. Start Fusion and open the provided file. Select the open option from the drop-down menu in the top left corner of Fusion. Select *Open from my computer…* in the following prompt. Find *Crank Assembly.f3z* and select *upload*.



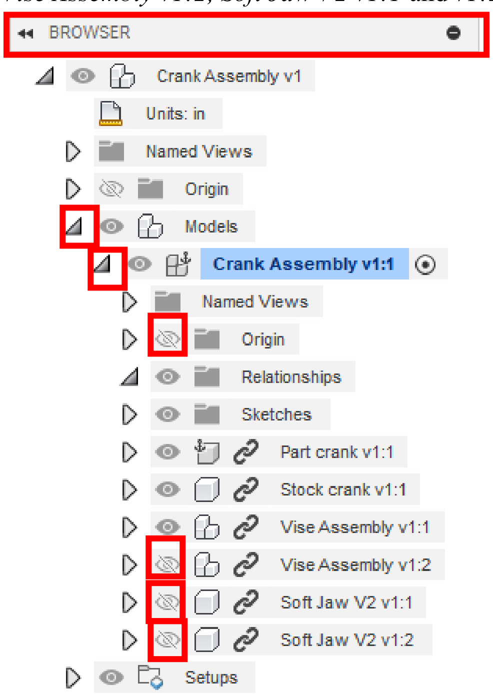
2. After uploading the file, multiple parts and an assembly will appear on the left-hand side. Open the *Cut crank* file.



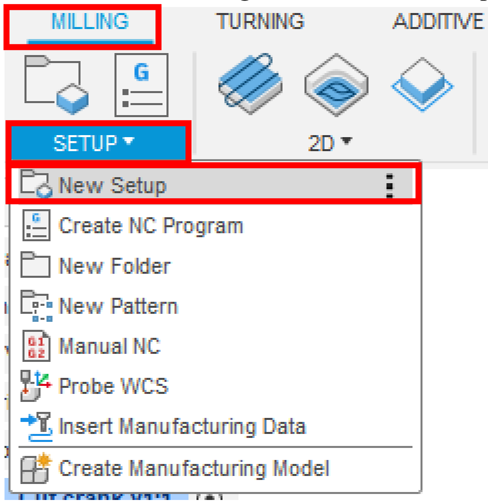
3. From the drop-down menu, select the *Manufacture* tab



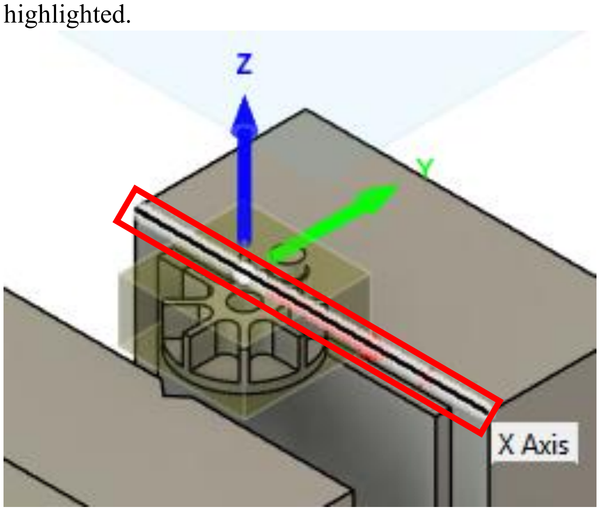
4. In *BROWSER,* drop down *Models then Cut crank v1:1.* With these dropped, *Hide Origin, Stock crank v1:1*, *Vise Assembly v1:2, Soft Jaw V2 v1:1* and *v1:2*. To hide components and sketches select the eye outlined in red.



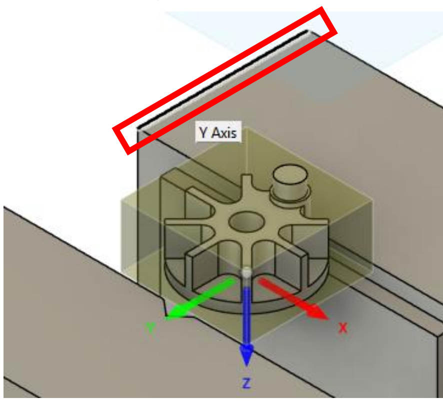
5. Underneath the *Milling* tab, click on the *Setup* dropdown menu. Select *New Setup*.

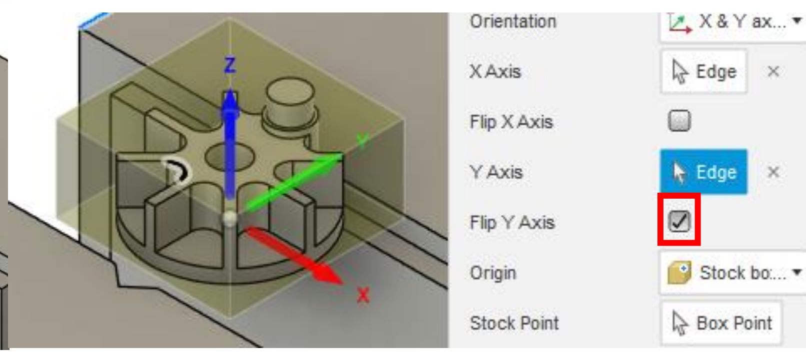


6. From the *Orientation* drop-down menu, select *X and Y axes*. Select the horizontal edge of the vice assembly highlighted.

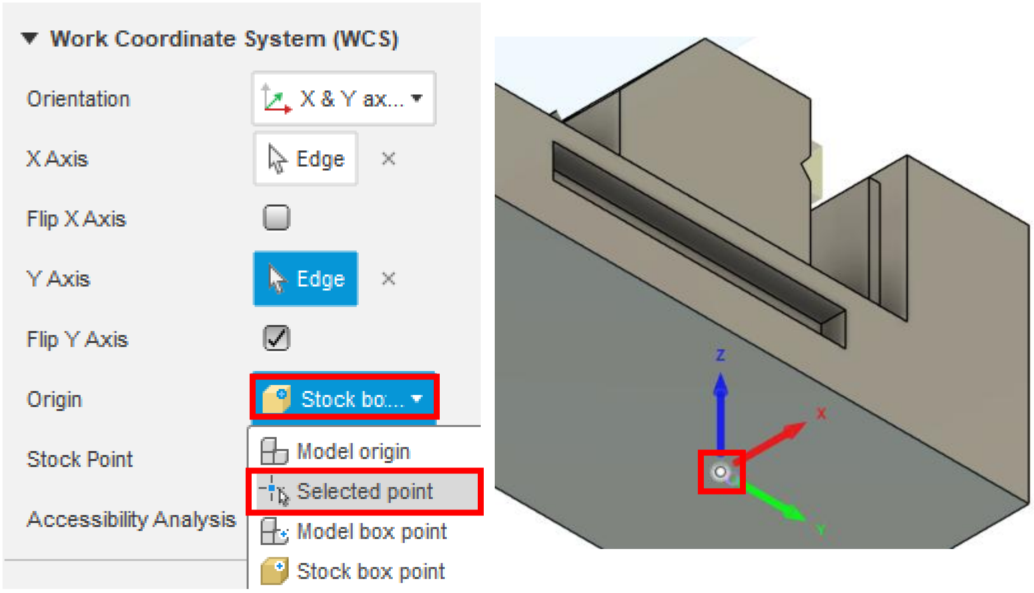


7. After Selecting the X axis, select the perpendicular edge for the Y axis and check off the *Flip Y Axis* box.

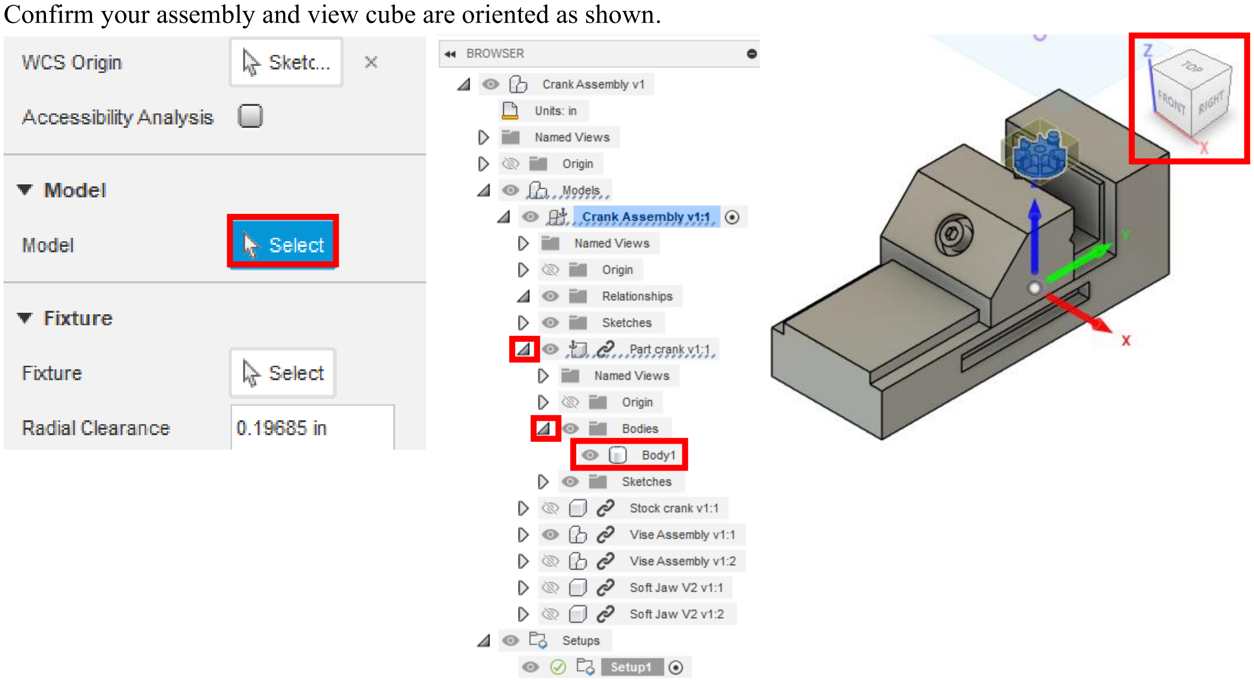




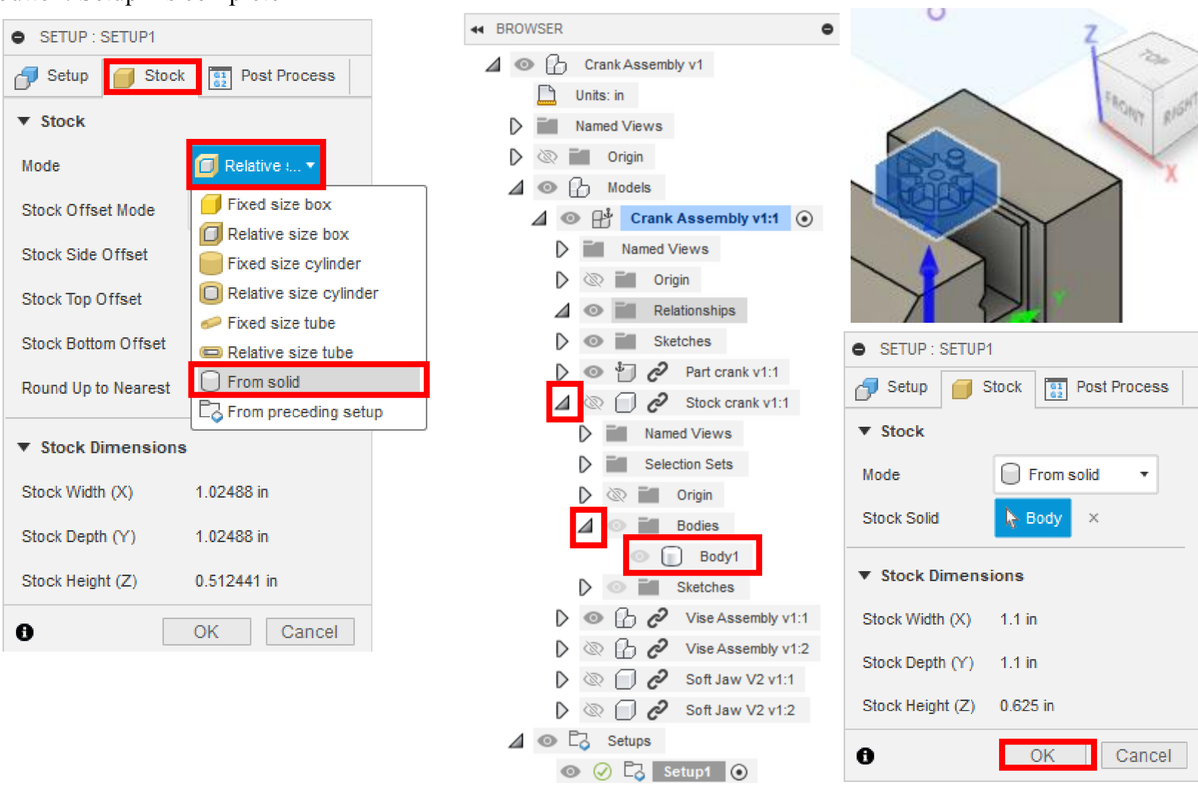
8. Under the *Origin* drop-down menu, select *Selected point*. Select the point on the bottom of the part outlined in blue.



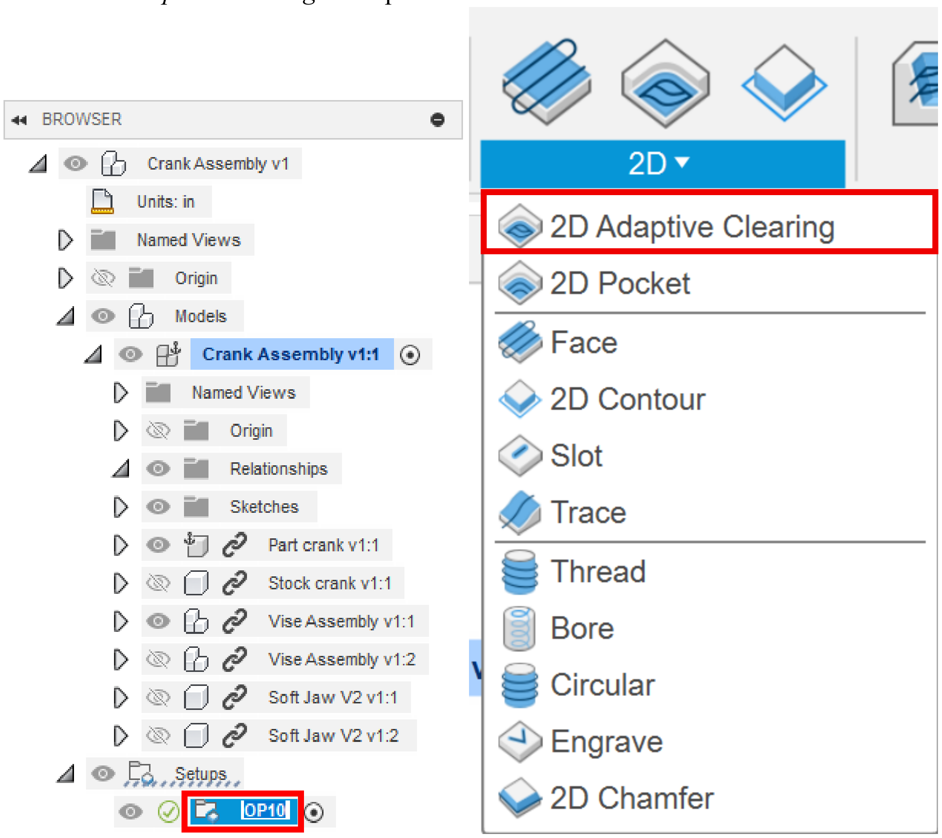
9. Next, in *Model,* click on *Select.* Now in *BROWSER,* drop down *Part crank v1:1>Bodies* and click on *Body1*. Confirm your assembly and view cube are oriented as shown.



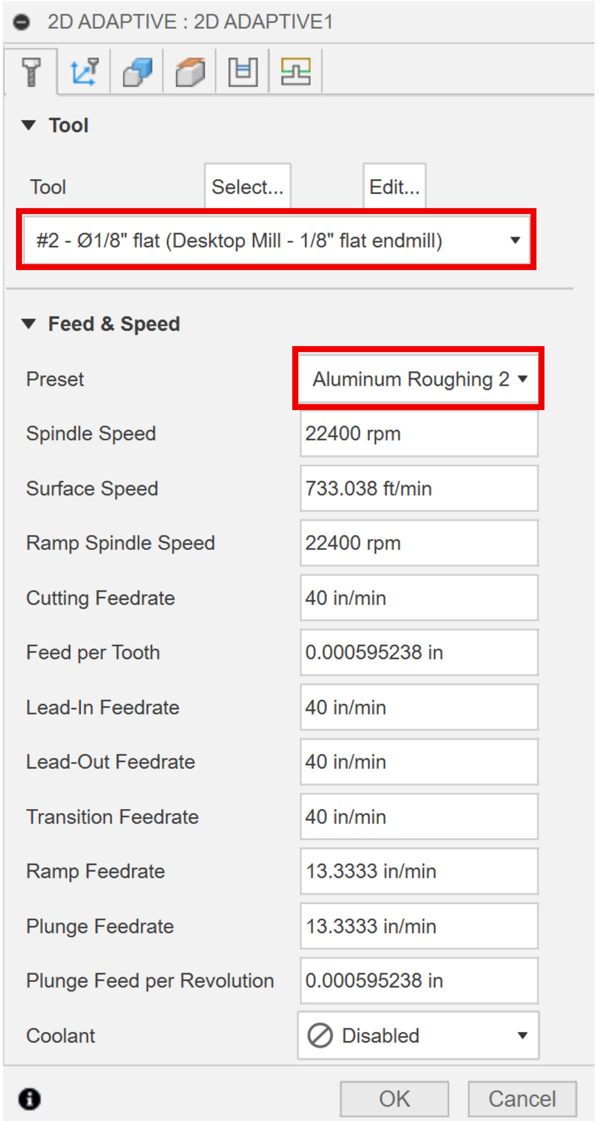
10. Under the *Stock* tab, select *From solid* for the mode. Choose *Body1* under *Stock crank v1:1*. Then, press the *OK* button. Setup1 is complete



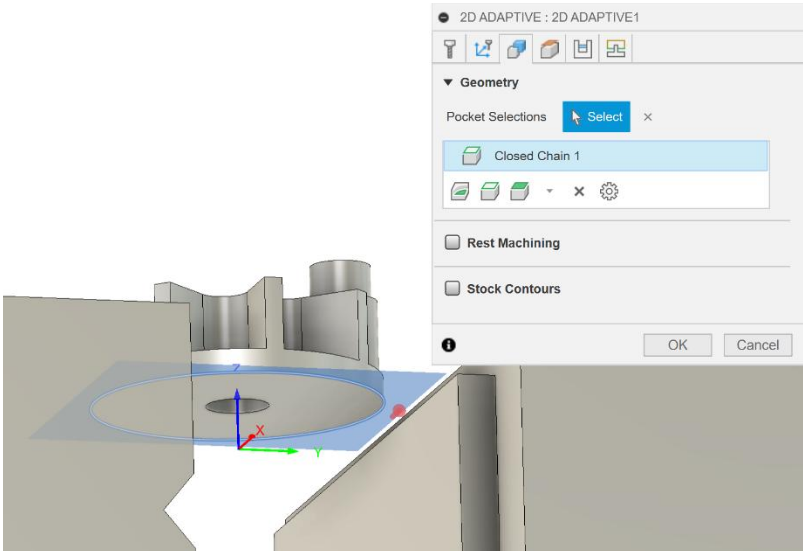
11. Rename *Setup1* as *OP10* by slowly double clicking the name. Next, click on the 2D dropdown menu and select the *2D Adaptive Clearing* tool option.



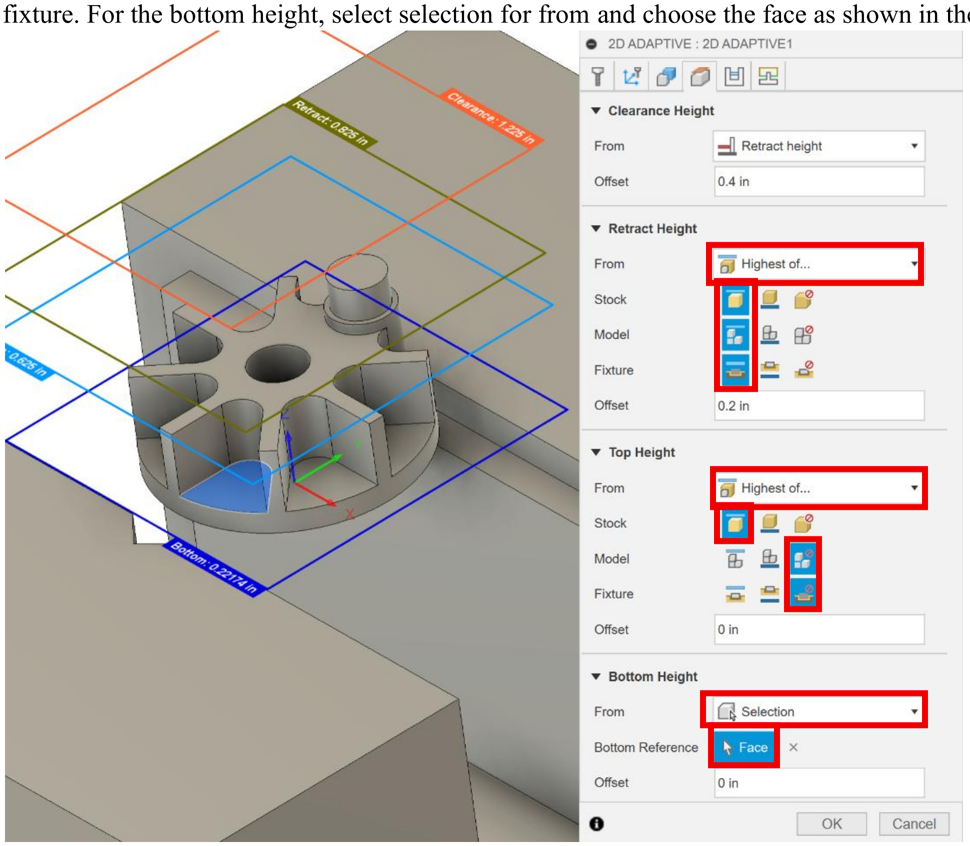
12. In the *tool* tab, select the *1/8" tool* and the *Aluminum roughing 2* preset.



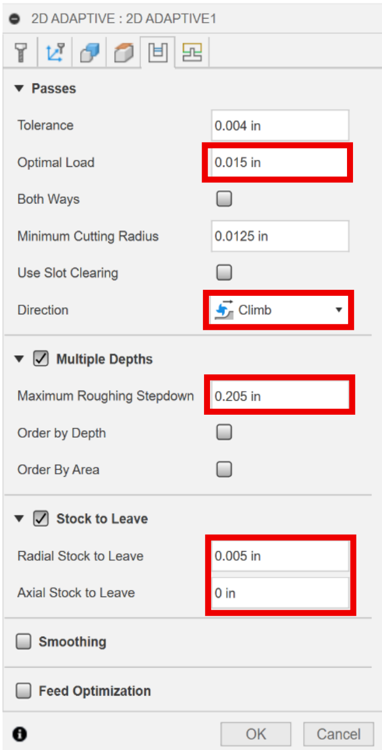
13. Next, click on the *geometry* tab click on the outer edge on the bottom of the part as shown in the figure below.



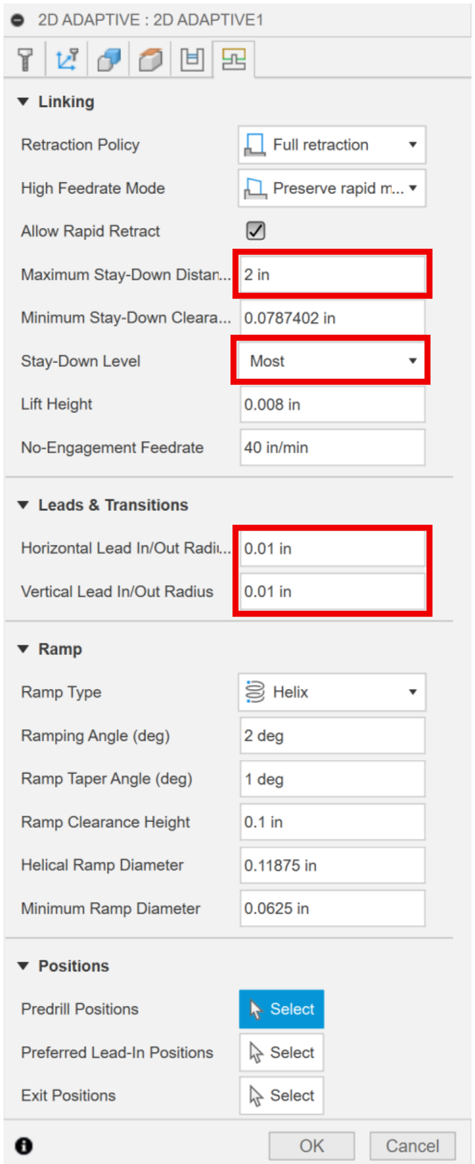
14. Next, click on the *heights* tab. For the retract height, select highest of… for from and select top for stock, model, and fixture. For the top height, select highest of… for from and select top for stock and ignore for model and fixture. For the bottom height, select selection for from and choose the face as shown in the figure below.



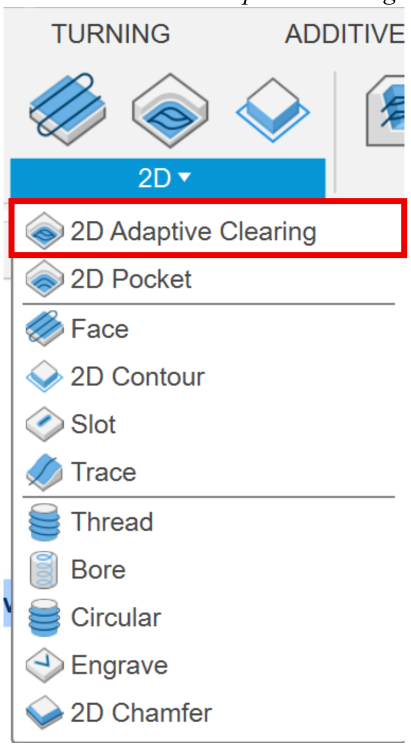
15. In the *passes* tab, check the *multiple depths* tab and change the *maximum roughing stepdown* to 0.205 in. Then, change the *optimal load* to 0.015 in, the *direction* to *climb*, the *radial stock to leave* to 0.005 in, and the *axial stock to leave* to 0 in.


2D ADAPTIVE : 2D ADAPTIVE1
Passes
Tolerance
0.004 in
Optimal Load
0.015 in
Both Ways
Minimum Cutting Radius
0.0125 in
Use Slot Clearing
Direction
Climb
Multiple Depths
Maximum Roughing Stepdown
0.205 in
Order by Depth
Order By Area
Stock to Leave
Radial Stock to Leave
0.005 in
Axial Stock to Leave
0 in
Smoothing
Feed Optimization
OK
Cancel

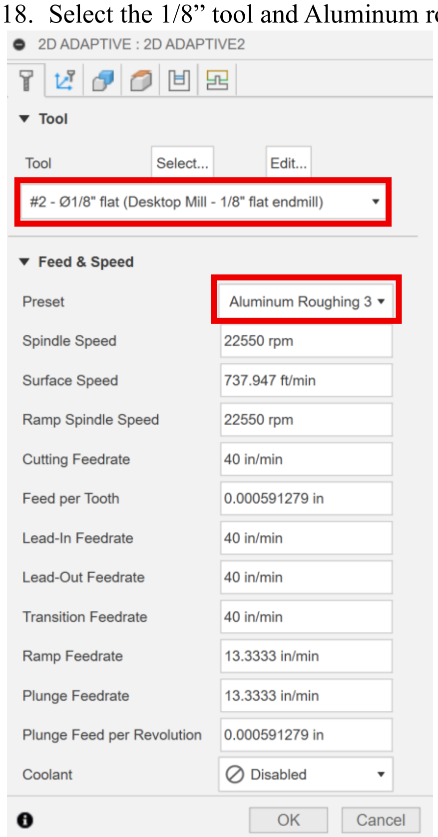
16. In the *linking* tab, change the *maximum stay-down distance* to 2 in, the *stay-down level* to *most*, and *leads & transitions* to 0.01 in. Then click *ok*.



17. Select the *2D Adaptive Clearing* tool from the 2D dropdown menu.

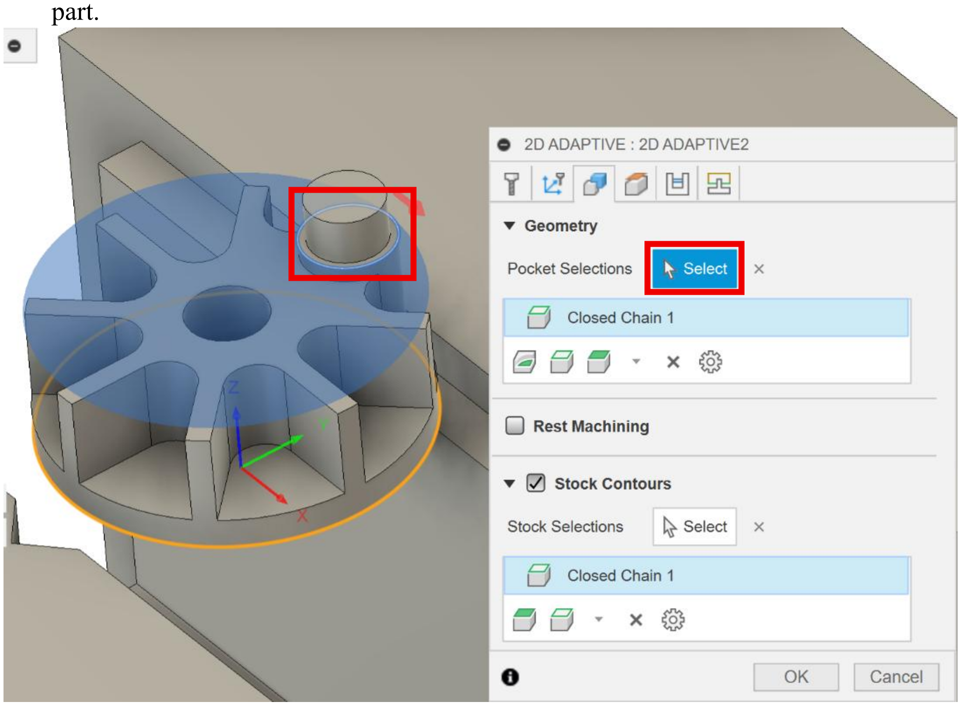


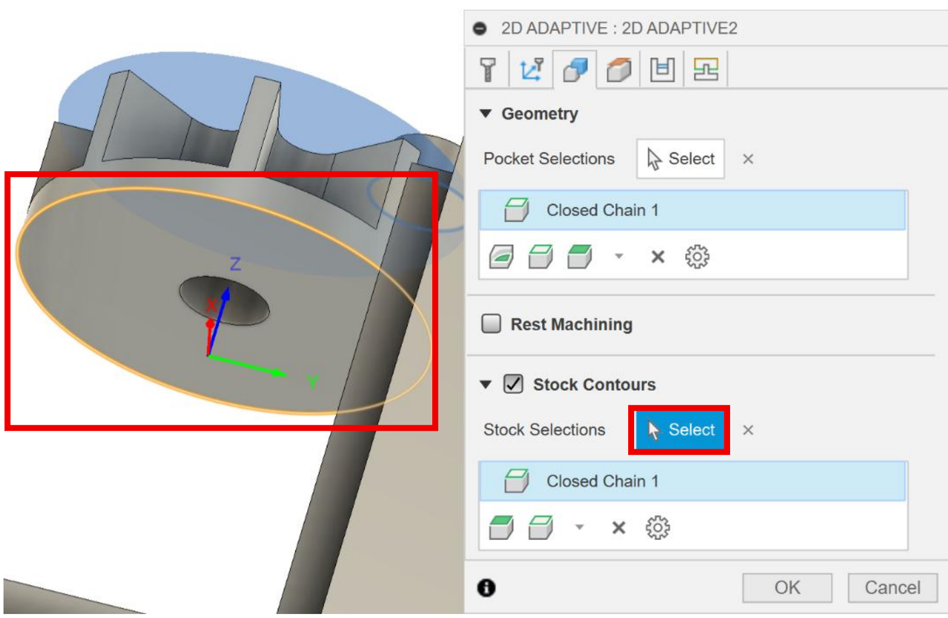
18. Select the 1/8" tool and Aluminum roughing 3 preset.



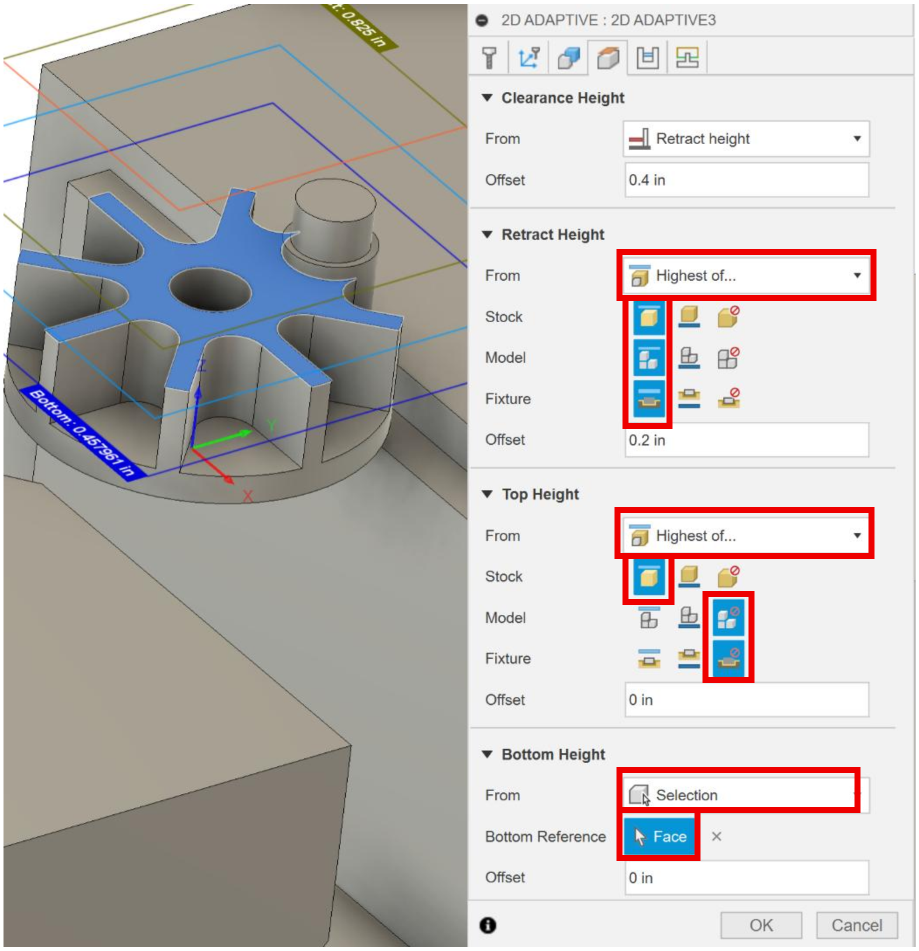
.

19. Next, move on to the *geometry* tab and check the *stock contours* box. For the *pocket selections*, select the larger radius below the cylinder on the part. For the *stock selections*, select the outer edge of the bottom face of the part.



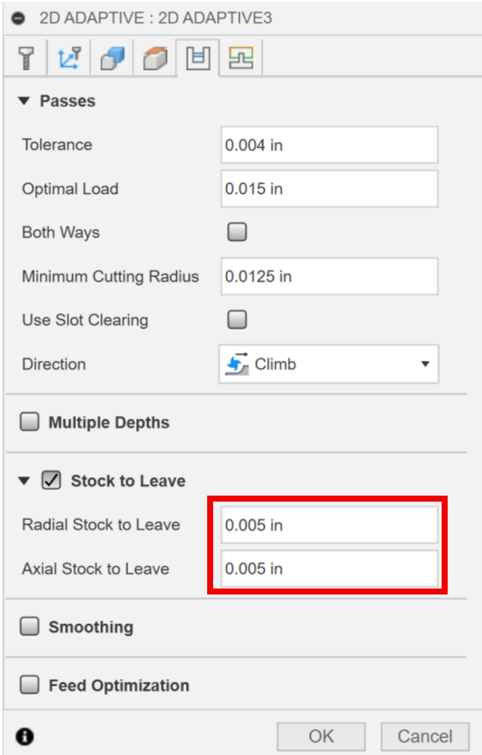


20. In the *heights* tab, for the *retract height* select *highest of…* for *from*, then *top* for *stock*, *model* and *fixture*. For the *top height*, select *highest of…* for *from*, then *top* for *stock* and *ignore* for *model* and *fixture*. For the *bottom height*, select *selection* for *from* then select the top face of the part as shown in the figure below.

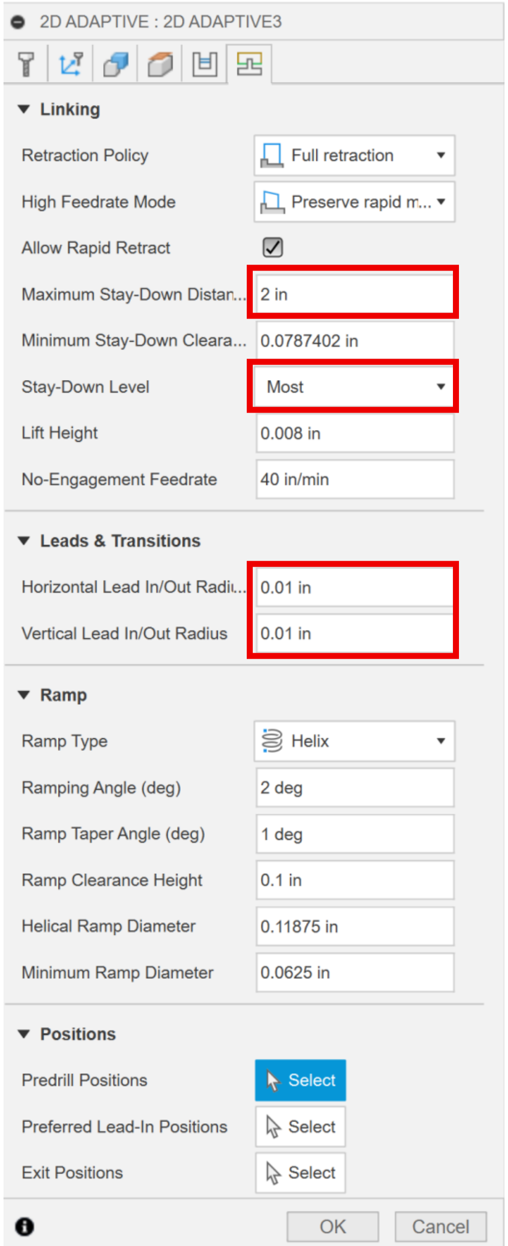


Bottom: 0.457961 in
2D ADAPTIVE : 2D ADAPTIVE3
Clearance Height
From
Retract height
Offset
0.4 in
Retract Height
From
Highest of...
Stock
Model
Fixture
Offset
0.2 in
Top Height
From
Highest of...
Stock
Model
Fixture
Offset
0 in
Bottom Height
From
Selection
Bottom Reference
Face
Offset
0 in
OK
Cancel

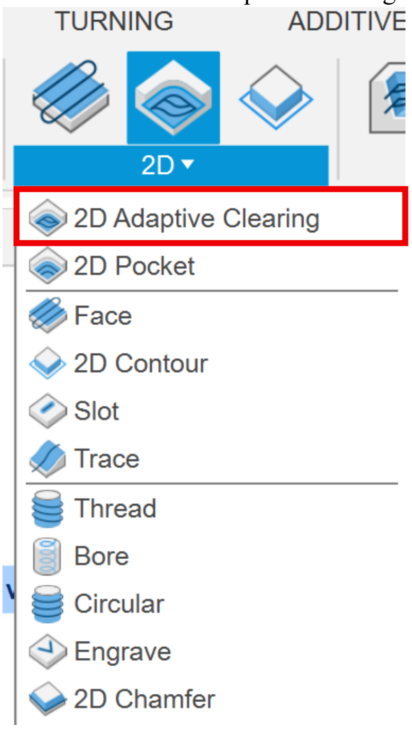
21. In the *passes* tab, change the *radial and axial stock to leave* to 0.005 in.



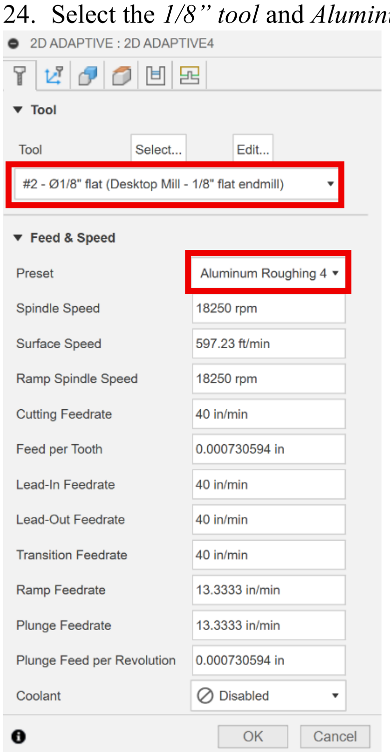
22. In the *linking* tab, change the *maximum stay-down distance* to 2 in, the *stay-down level* to *most*, and the *leads & transitions* to 0.01 in. Then click *ok*.



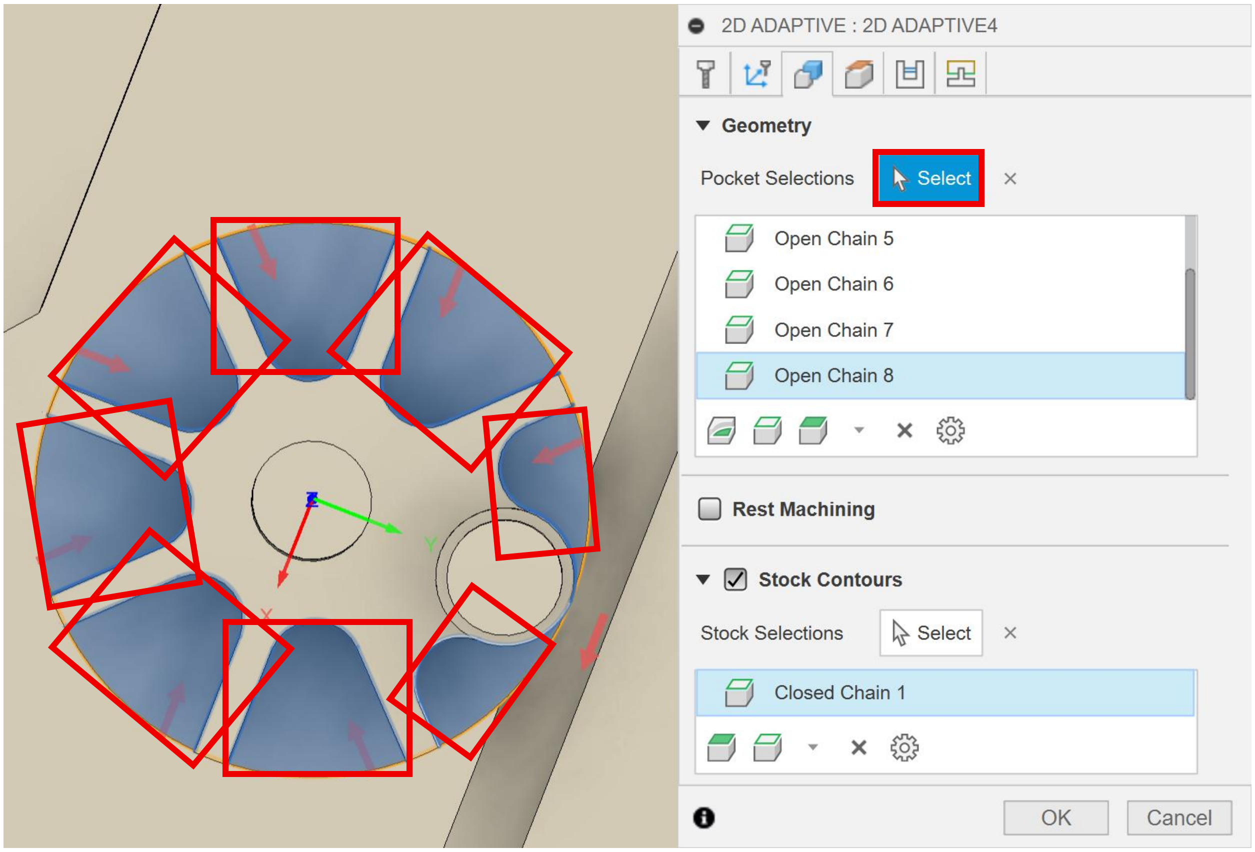
23. Select the 2D Adaptive Clearing tool from the 2D dropdown menu.

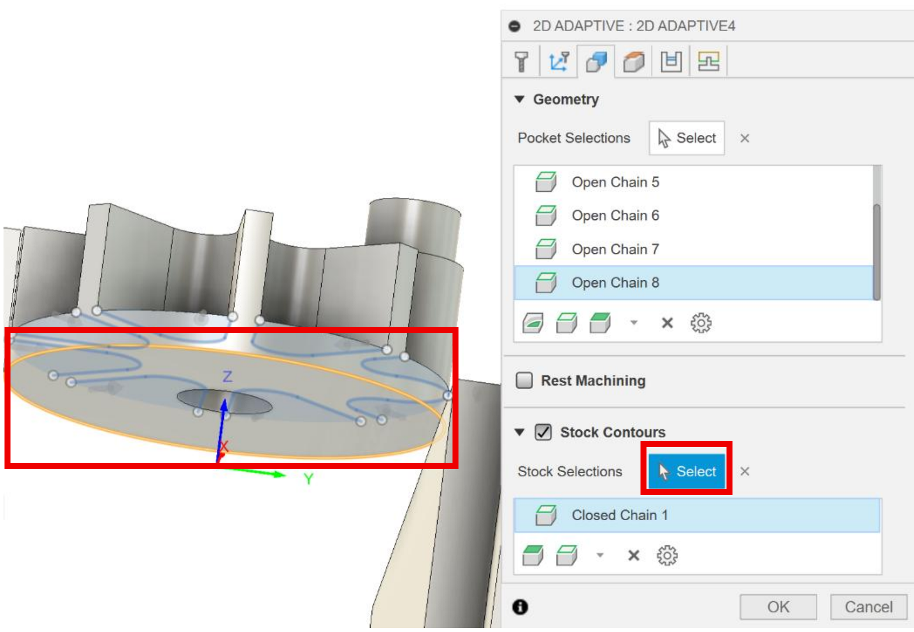


24. Select the *1/8" tool* and *Aluminum roughing 4* preset.

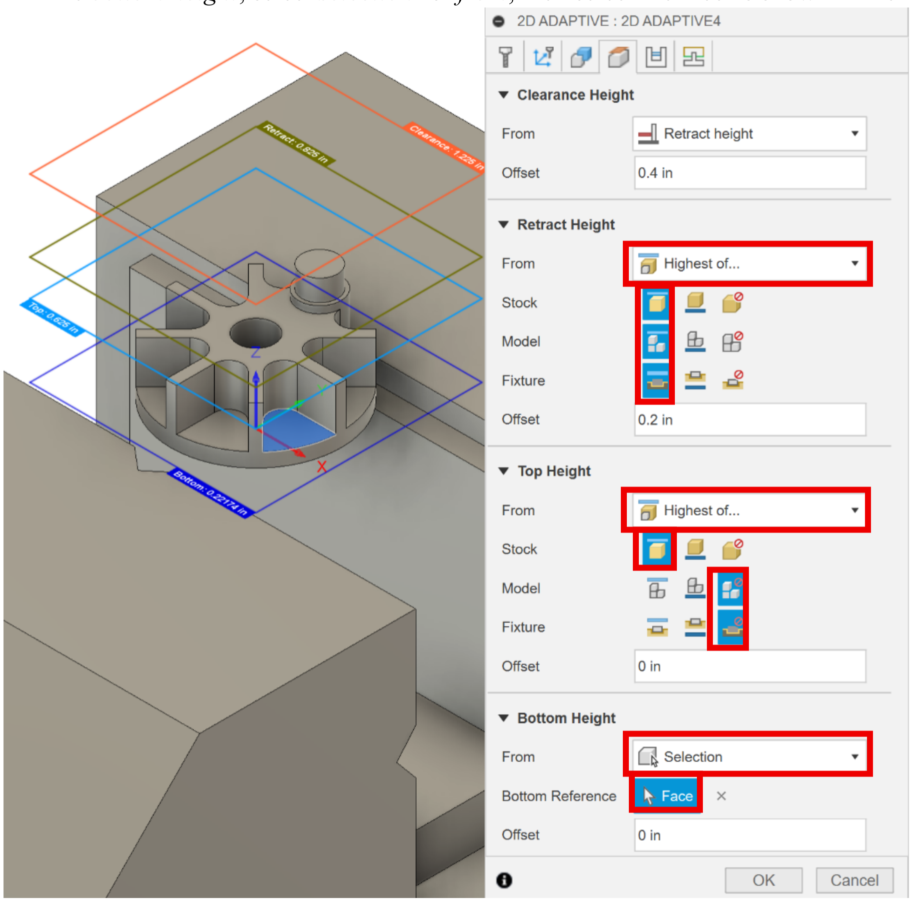


25. In the *geometry* tab, check the *stock contours* box. For *pocket selections*, select the 8 open pockets at the top of the part. For *stock selections* select the outer edge at the bottom of the part.

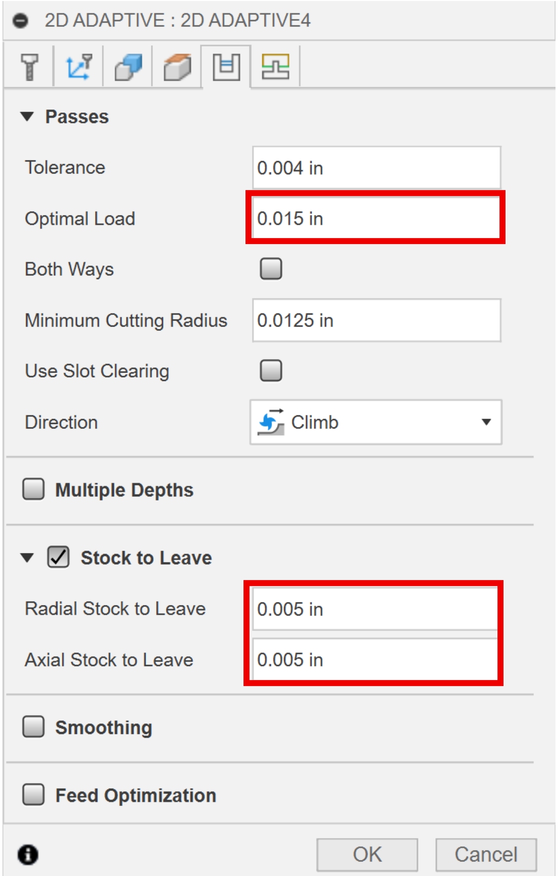

2D ADAPTIVE : 2D ADAPTIVE4
Geometry
Pocket Selections
Select
Open Chain 5
Open Chain 6
Open Chain 7
Open Chain 8
Rest Machining
Stock Contours
Stock Selections
Select
Closed Chain 1
OK
Cancel


2D ADAPTIVE : 2D ADAPTIVE4
Geometry
Pocket Selections
Select
Open Chain 5
Open Chain 6
Open Chain 7
Open Chain 8
Rest Machining
Stock Contours
Stock Selections
Select
Closed Chain 1
OK
Cancel

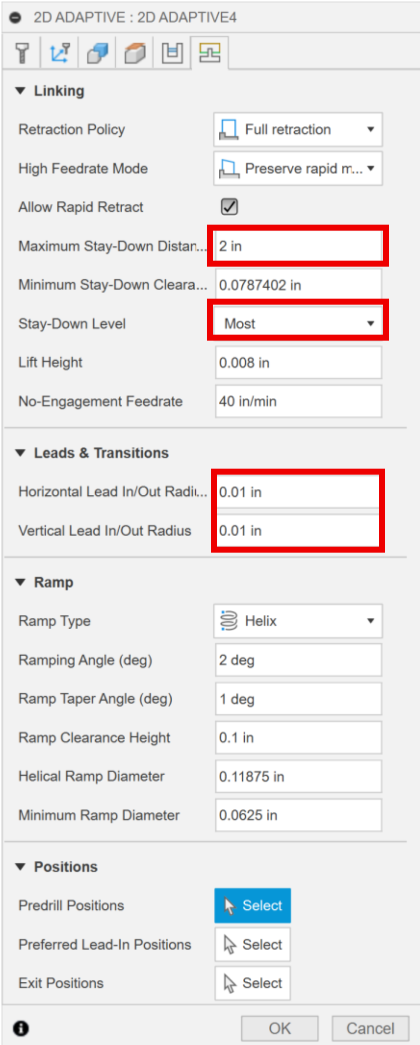
26. In the *heights* tab, for the *retract height* select *highest of…* for *from*, then select *top* for *stock*, *model* and *fixture*. For the *top height*, select *highest of…* for *from*, then select *top* for the *stock* and *ignore* for *model* and *fixture*. For the *bottom height*, select *selection* for *from*, then select the face as shown in the figure below.



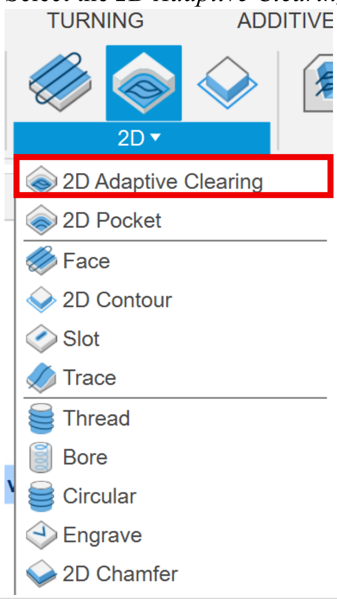
27. In the *passes* tab, change the *optimal load* to 0.015 in and the *radial* and *axial stock to leave* to 0.005 in.

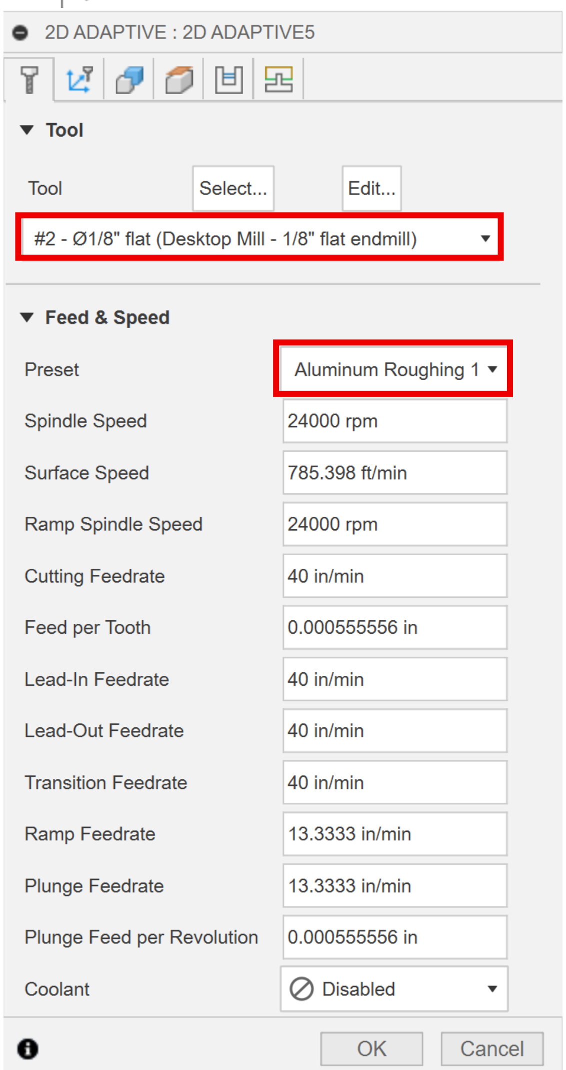


28. In the *linking* tab, change the *maximum stay-down distance* to 2 in, the *stay-down level* to *most*, and the *leads & transitions* to 0.01 in. Then click *ok*.

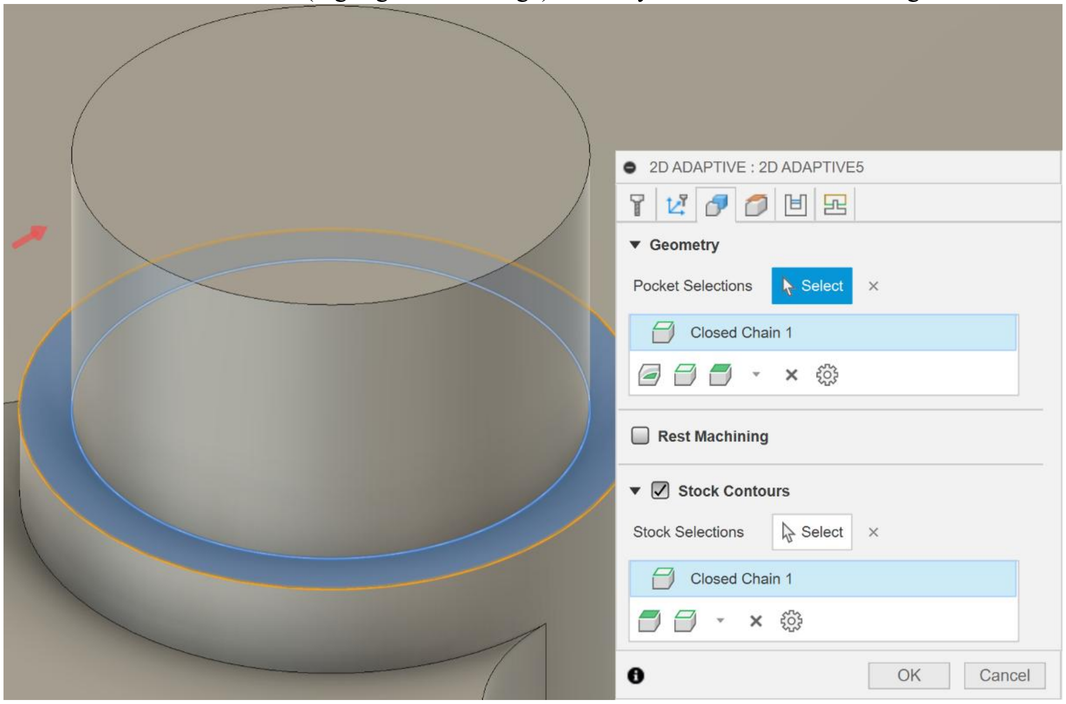


29. Select the *2D Adaptive Clearing* tool, then select the *1/8" tool* and *Aluminum roughing 1* preset.

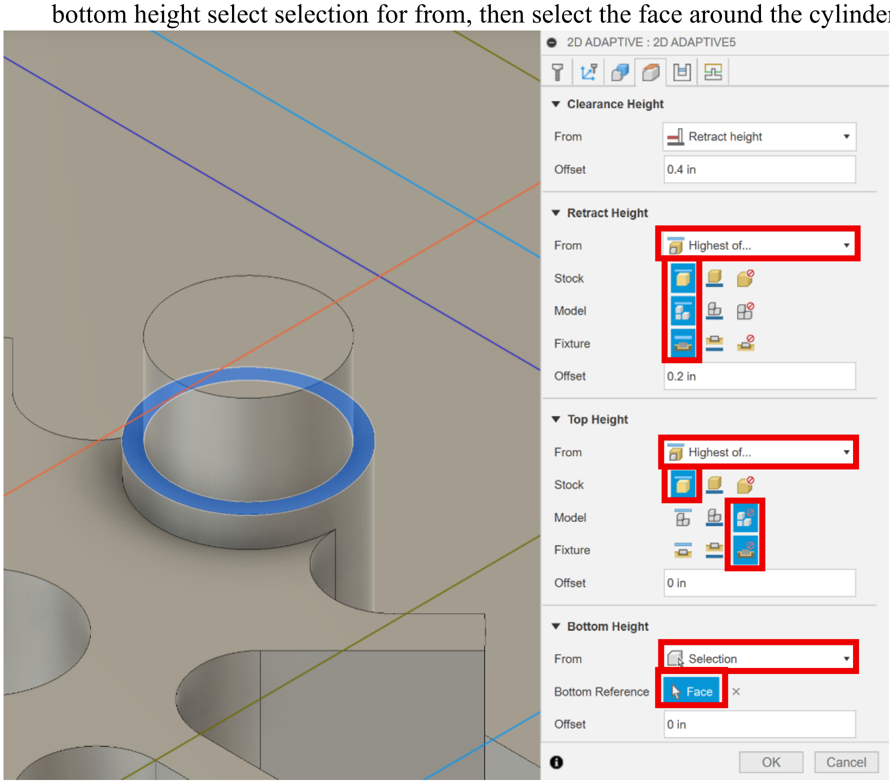




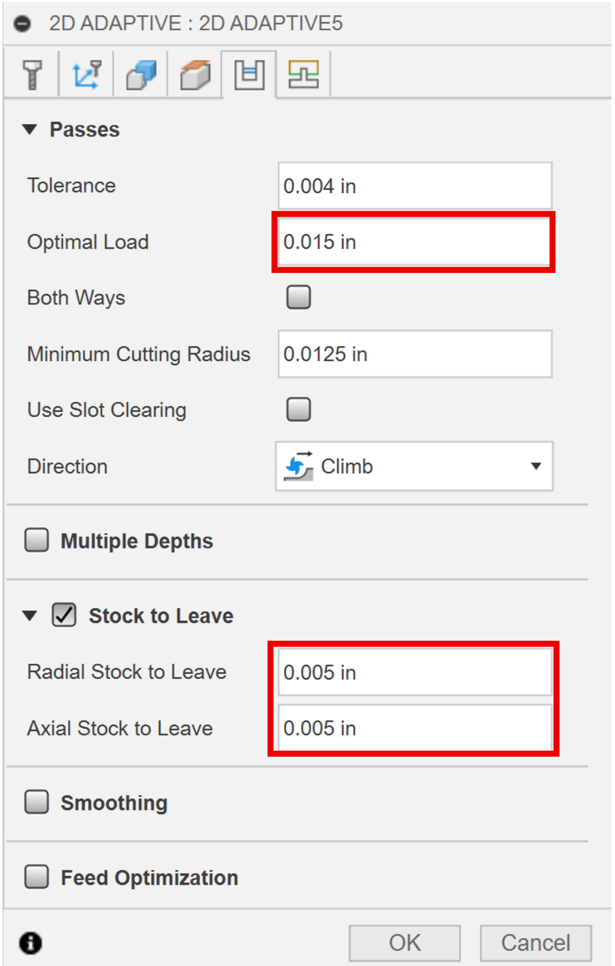
30. In the geometry tab, check the stock contours box. For pocket selections, choose the inner radius (highlighted in the lighter shade of blue) at the bottom of the cylinder as shown in the figure below. For the stock selections, choose the outer radius (highlighted in orange) of the cylinder as shown in the figure below.



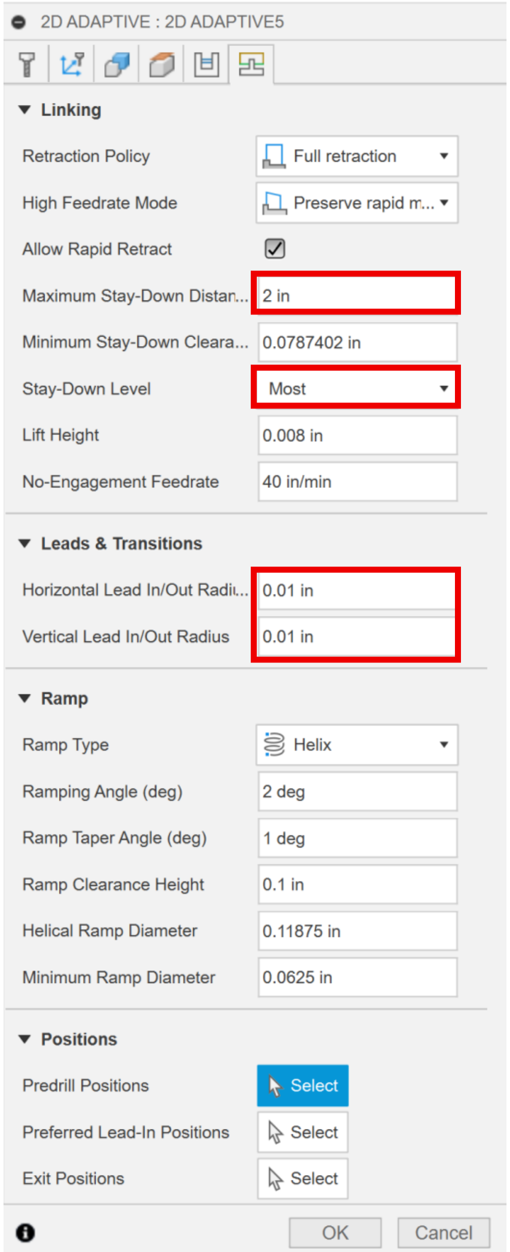
31. In the heights tab, for the retract height select the highest of… for from, then top for stock, model and fixture. For the top height, select highest of… for from, then top for stock and ignore for model and fixture. For the bottom height select selection for from, then select the face around the cylinder as shown in the figure below.



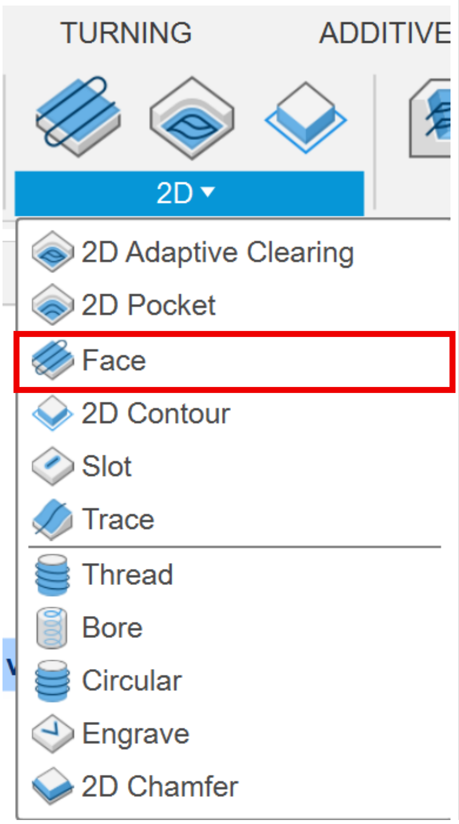
32. In the passes tab, change the *optimal load* to 0.015 in then change the *radial* and *axial stock to leave* to 0.005 in.

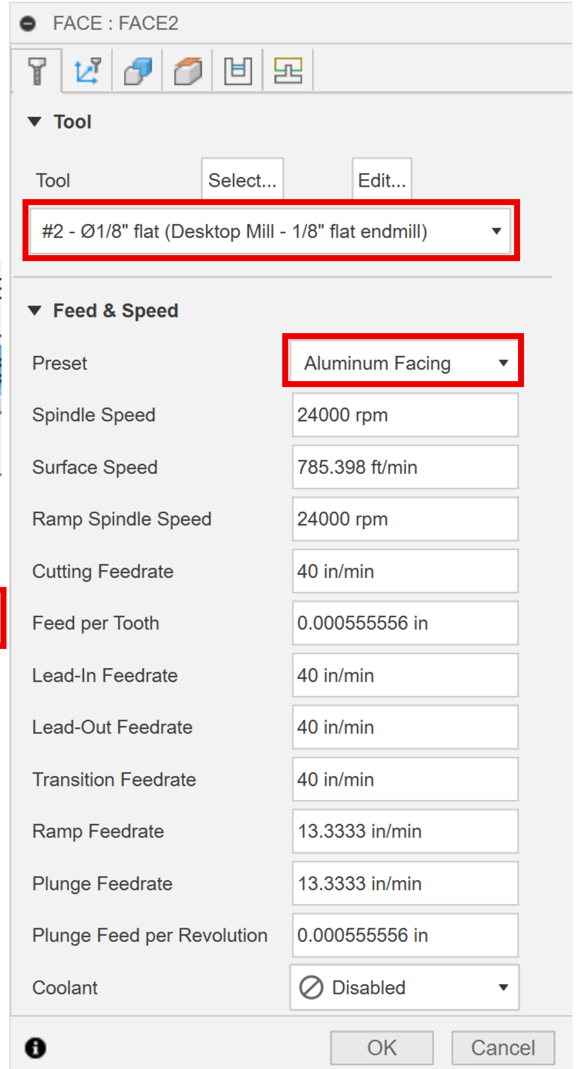


33. In the *linking* tab, change the *maximum stay-down distance* to 2 in, the *stay-down level* to *most* and the *leads & transitions* to 0.01 in. Then click *ok*.

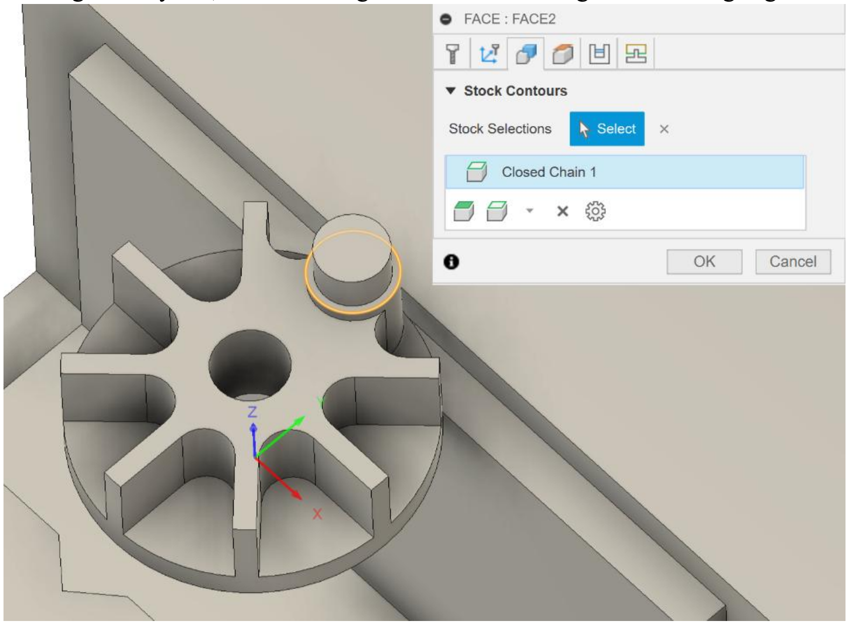


34. Select the *2D Face* tool, then select the *1/8" tool* and *Aluminum facing* preset.

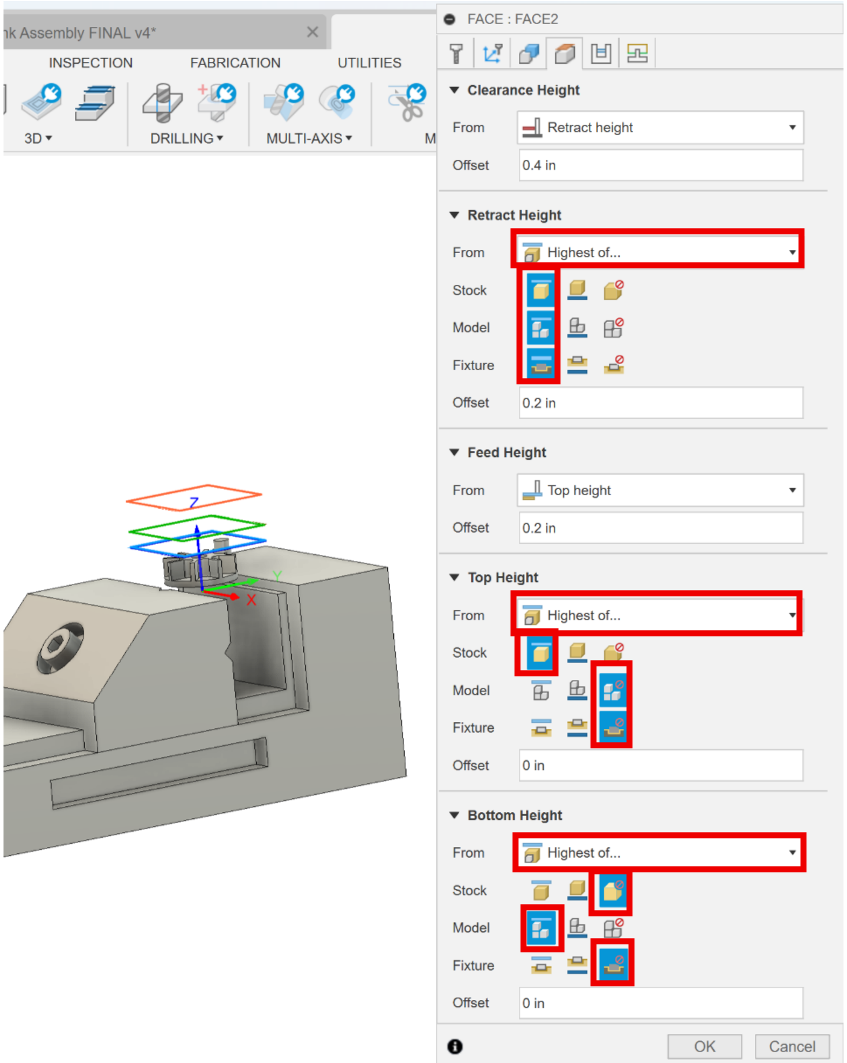




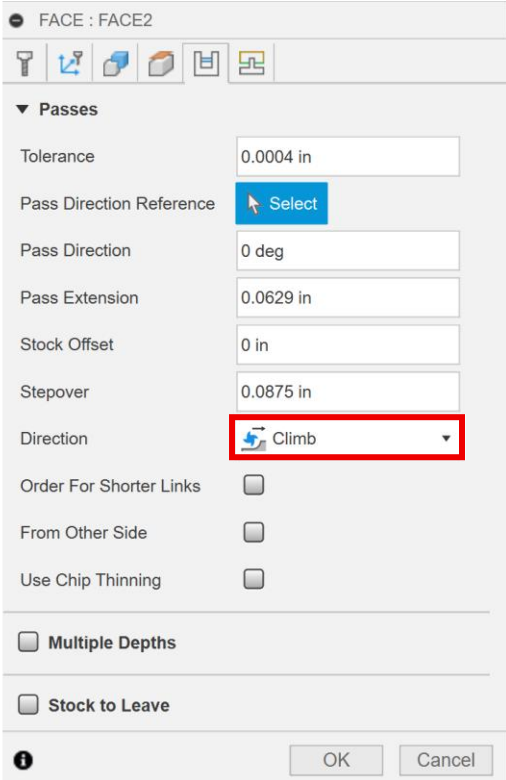
35. In the geometry tab, select the edge as shown in the figure below highlighted in orange.



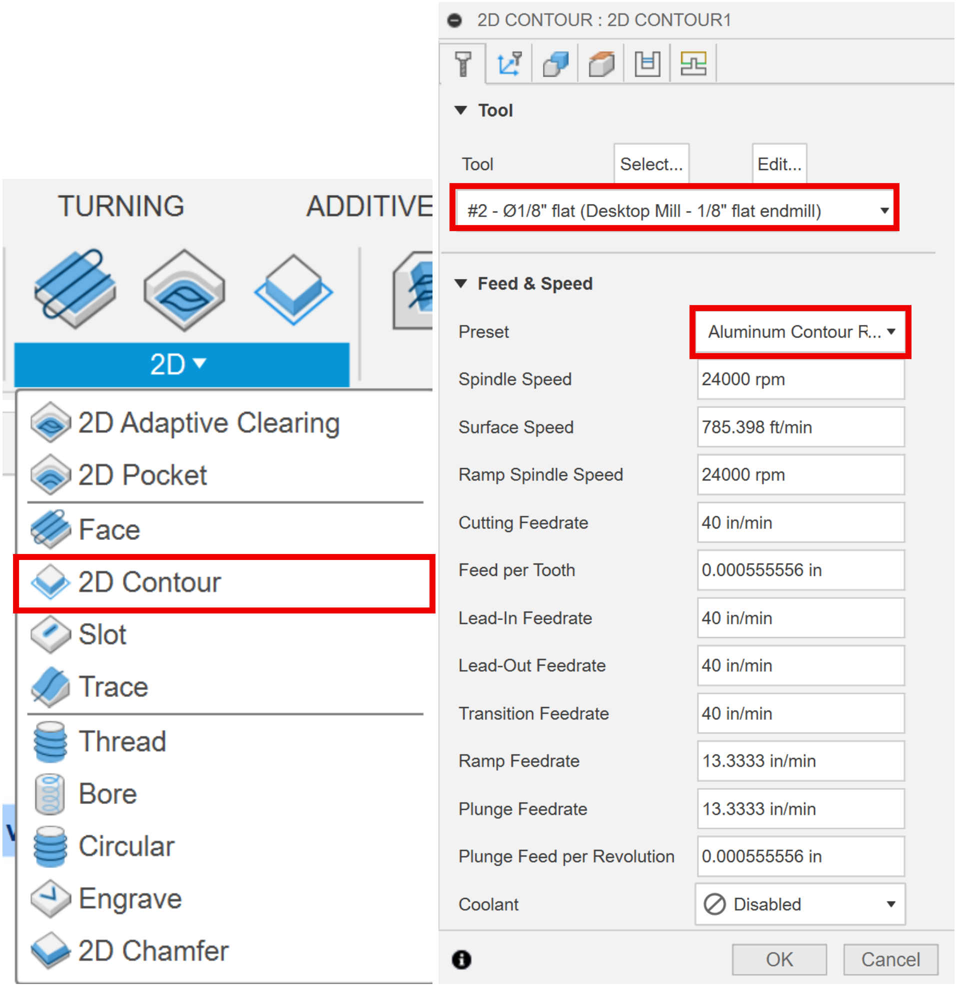
36. Next, in the *heights* tab, for the *retract height* select *highest of…* for *from*, then *top* for *stock*, *model* and *fixture*. For the *top height*, select *highest of…* for *from*, then *top* for *stock* and *ignore* for *model* and *fixture*. For the *bottom height* select *highest of…* for *from*, then *ignore* for *stock* and *fixture* and *top* for *model*.



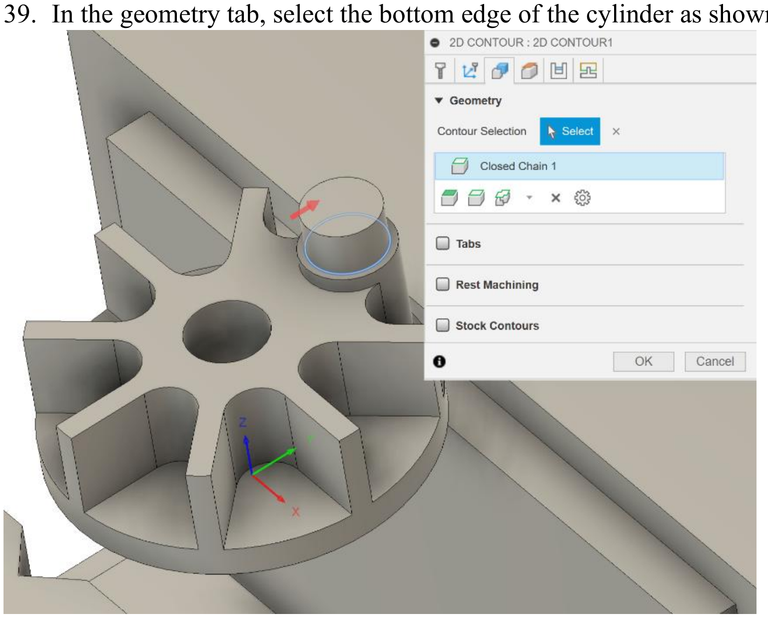
37. In the *passes* tab, change the *direction* to *climb*. Then click *ok*.



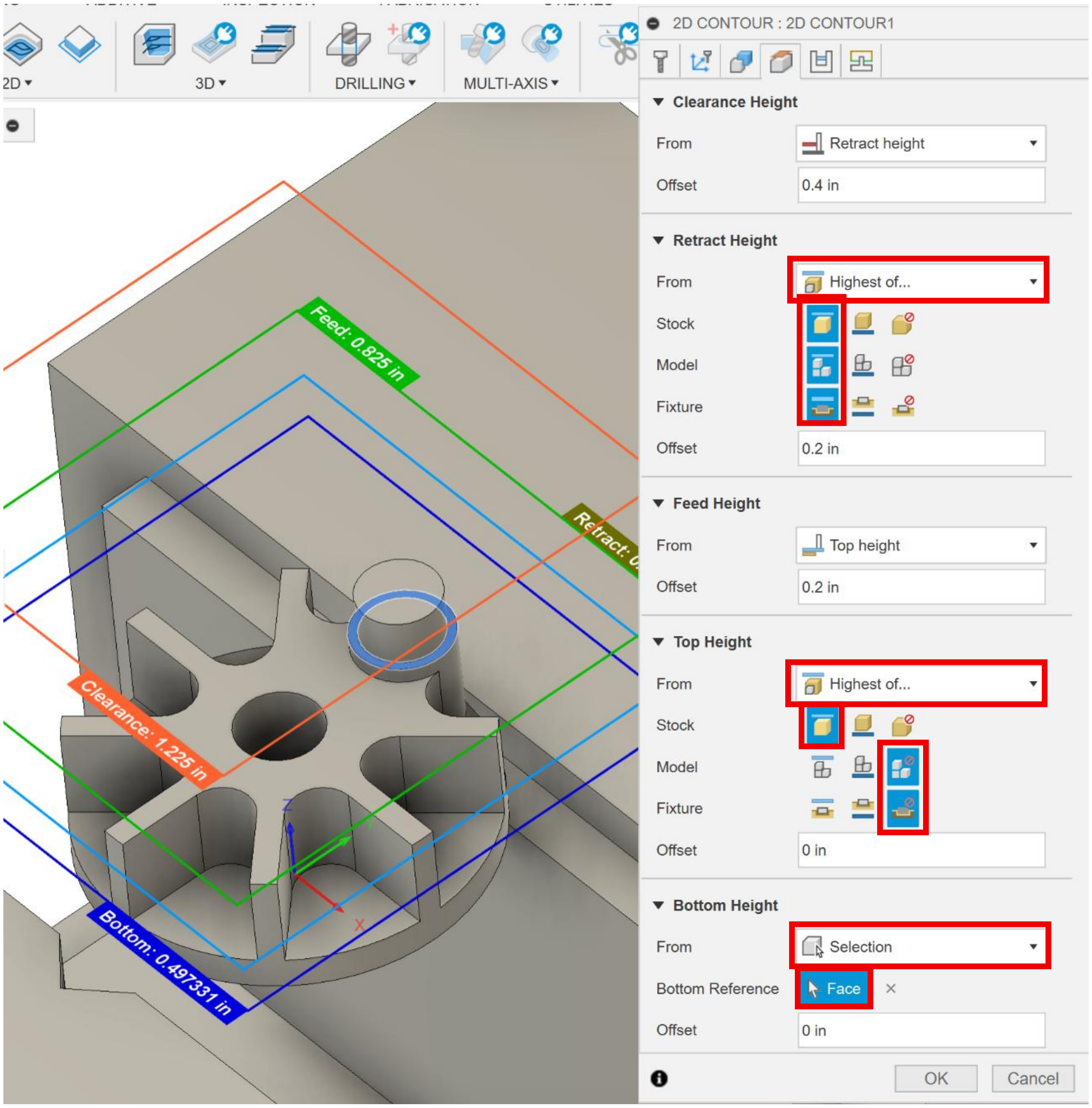
38. Select the *2D contour tool*, then select the *1/8" tool* and the *Aluminum contour roughing* preset.



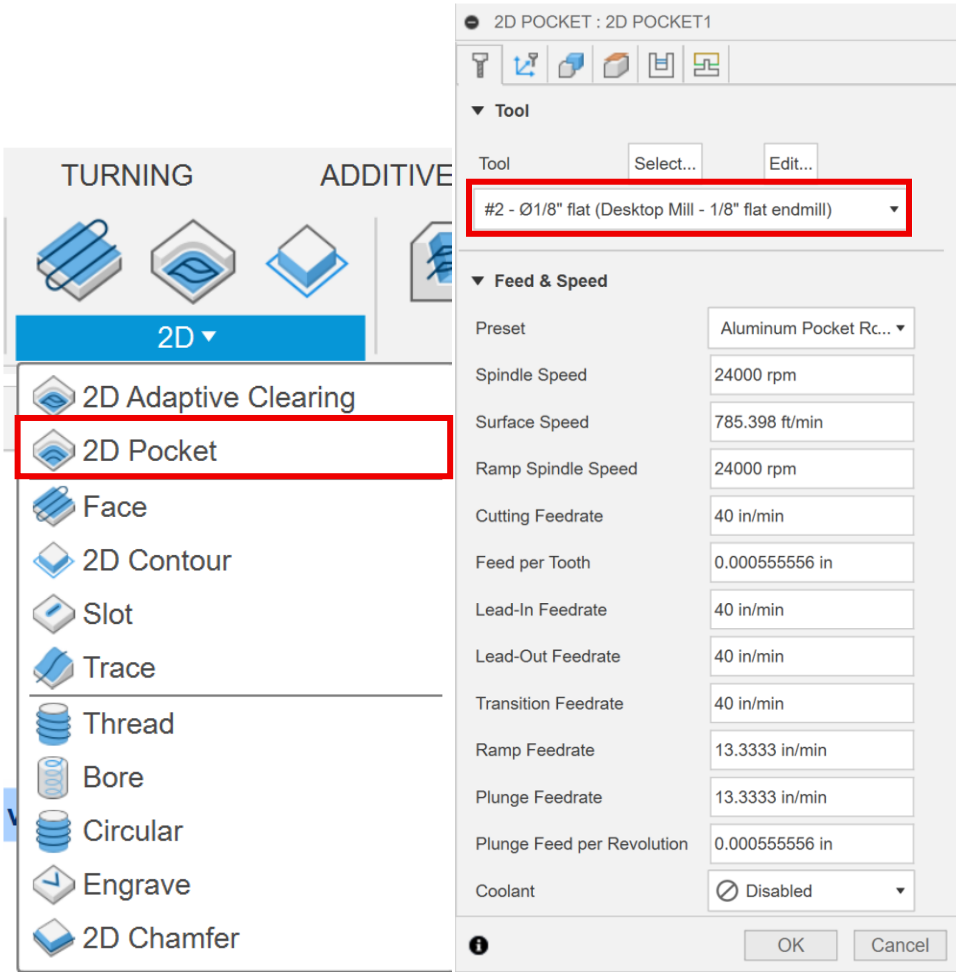
39. In the geometry tab, select the bottom edge of the cylinder as shown in the figure below.



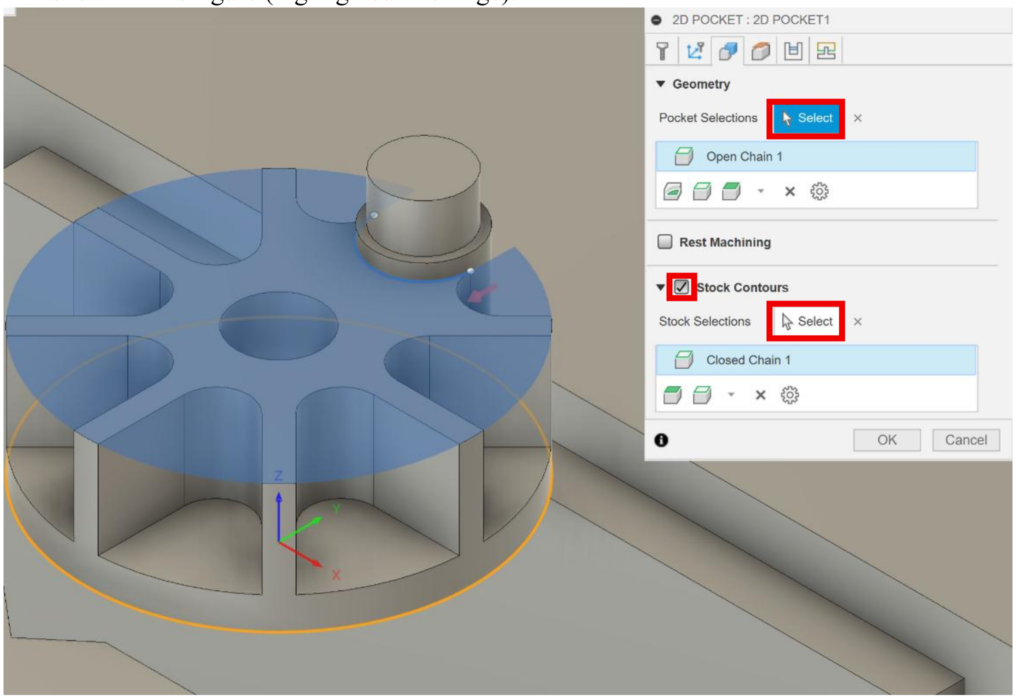
40. In the *heights* tab, for the *retract height* select *highest of…* for *from*, then *top* for *stock*, *model* and *fixture*. For the *top height* select *highest of…* for *from*, then *top* for *stock* and *ignore* for *model* and *fixture*. For the *bottom height* select *selection* for *from* and select the face surrounding the cylinder as shown in the figure below. Then click *ok*.



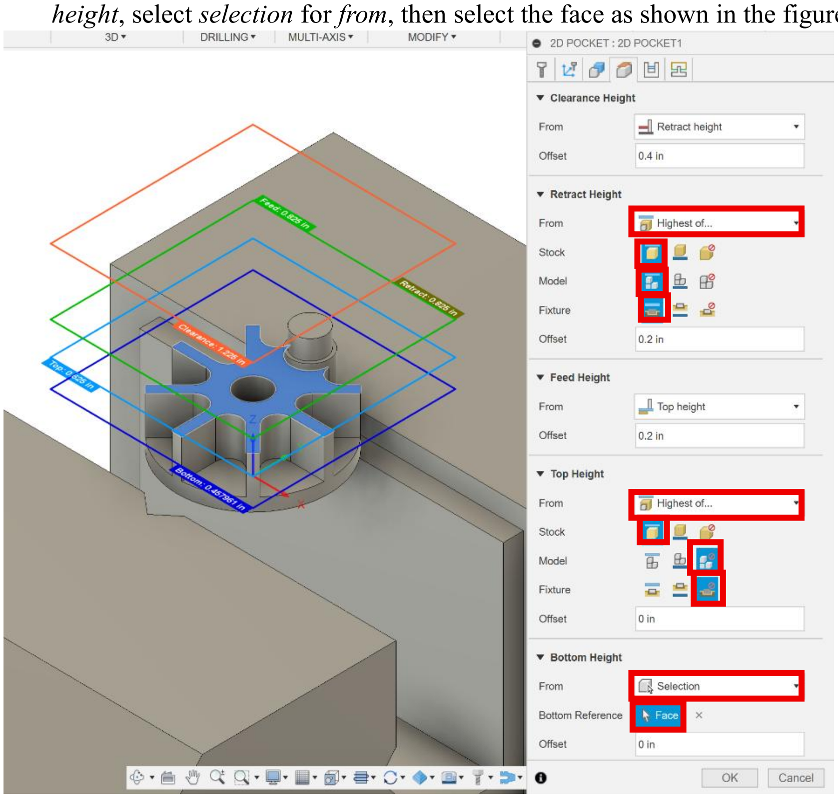
41. Select *2D Pocket* tool from the 2D dropdown menu. Make sure the *1/8" tool* and *Aluminum pocket roughing* preset are selected.


TURNING
ADDITIVE
2D
2D Adaptive Clearing
2D Pocket
Face
2D Contour
Slot
Trace
Thread
Bore
Circular
Engrave
2D Chamfer
2D POCKET : 2D POCKET1
Tool
Tool
Select...
Edit...
#2 - Ø1/8" flat (Desktop Mill - 1/8" flat endmill)
Feed & Speed
Preset
Aluminum Pocket Rc...
Spindle Speed
24000 rpm
Surface Speed
785.398 ft/min
Ramp Spindle Speed
24000 rpm
Cutting Feedrate
40 in/min
Feed per Tooth
0.000555556 in
Lead-In Feedrate
40 in/min
Lead-Out Feedrate
40 in/min
Transition Feedrate
40 in/min
Ramp Feedrate
13.3333 in/min
Plunge Feedrate
13.3333 in/min
Plunge Feed per Revolution
0.000555556 in
Coolant
Disabled
OK
Cancel

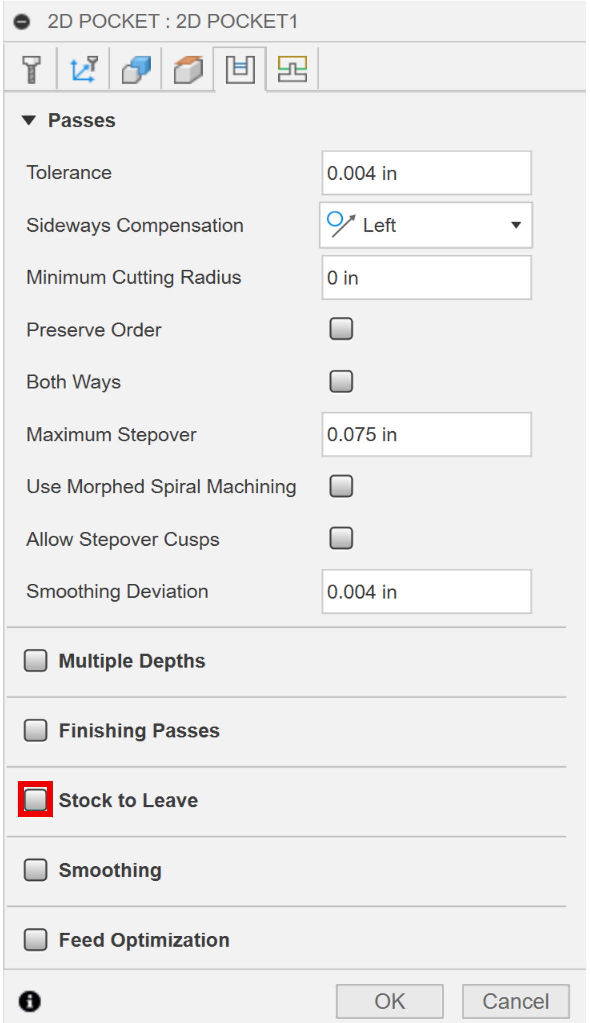
42. In the *geometry* tab, check the *stock contours* box. For *pocket selections*, select the edge (highlighted in blue with dots at the end) as shown in the figure below. For *stock selections*, select the bottom edge of the part as shown in the figure (highlighted in orange).



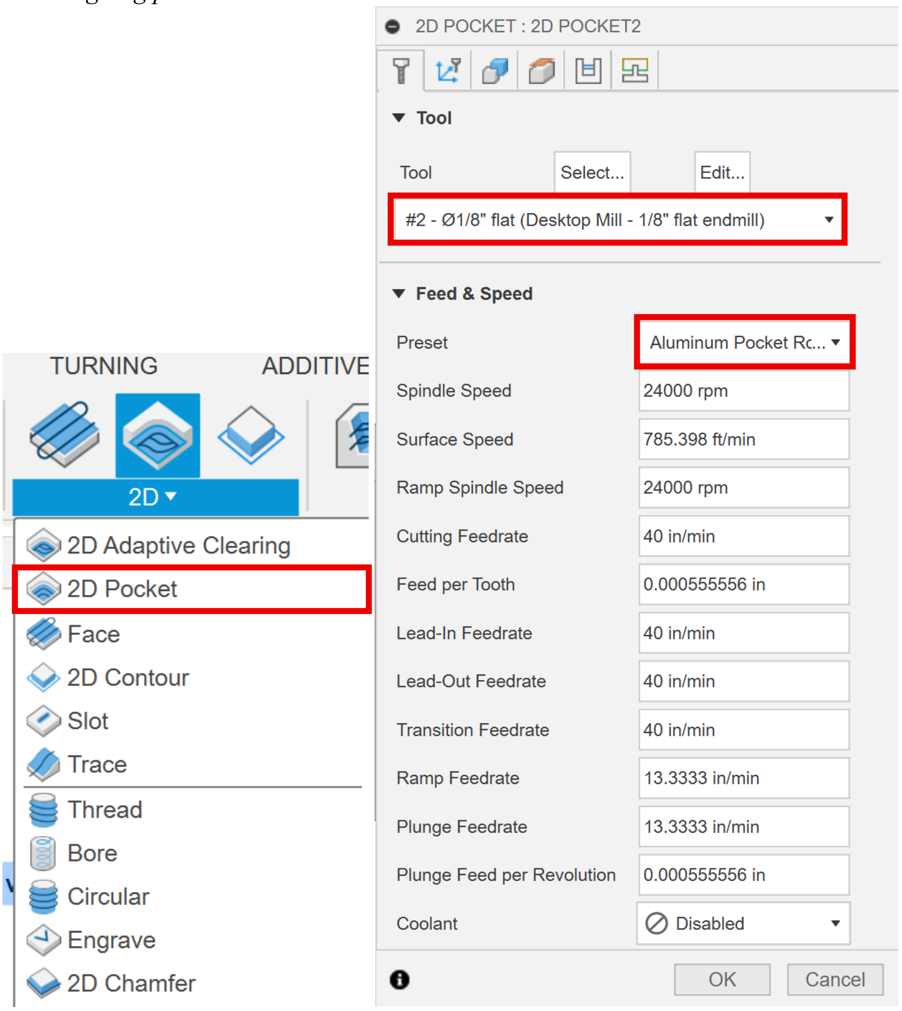
43. In the *heights* tab, for the *retract height* select *highest of…* for *from*, then *top* for *stock*, *model* and *fixture*. For the *top height*, select *highest of…* for *from* then *top* for *stock* and ignore for *model* and *fixture*. For the *bottom height*, select *selection* for *from*, then select the face as shown in the figure below.



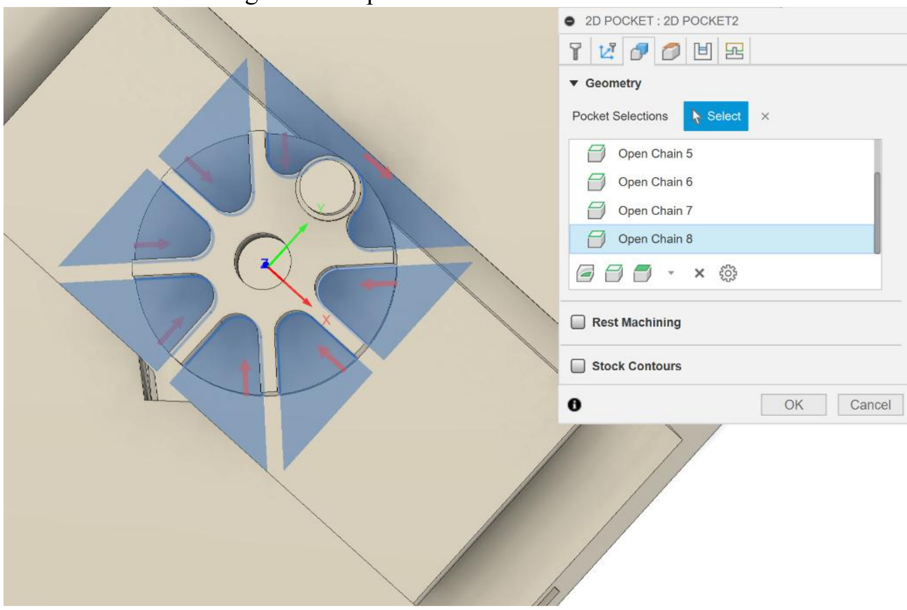
44. In the *passes* tab, uncheck the *stock to leave* box. Then click *ok*.



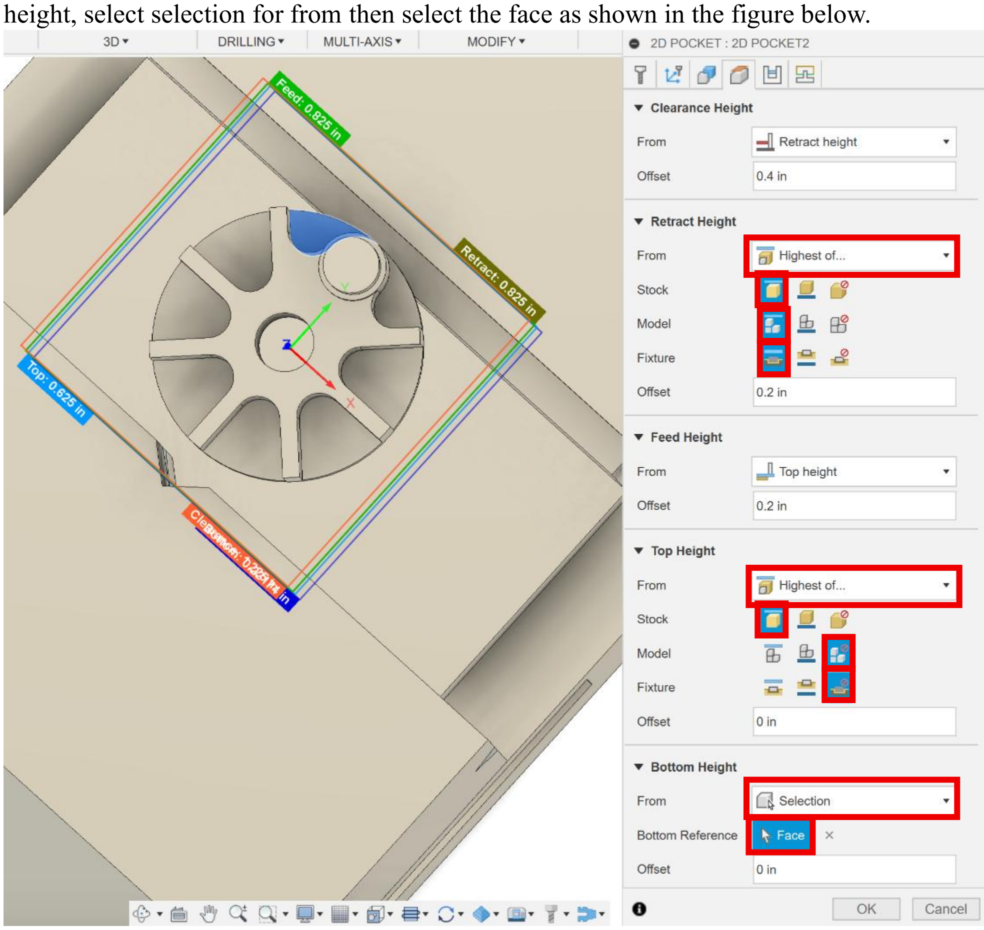
45. Select the *2D Pocket* tool again in the 2D dropdown menu. Then make sure the *1/8" tool* and *Aluminum pocket roughing preset* are selected.



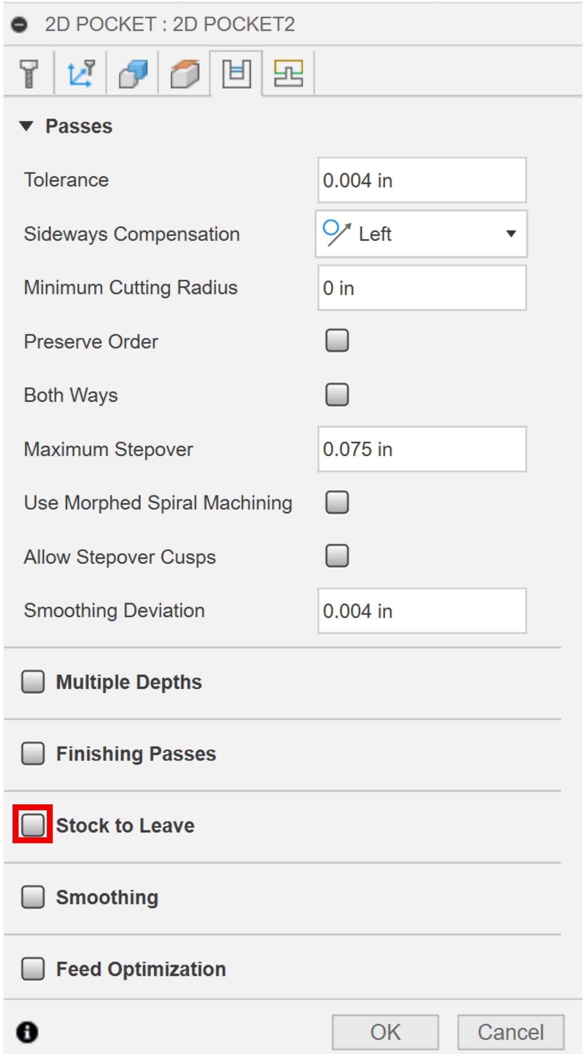
46. In the geometry tab, for pocket selections, select the 8 pockets as shown in the figure below. Make sure that you select the inner edge of these pockets.



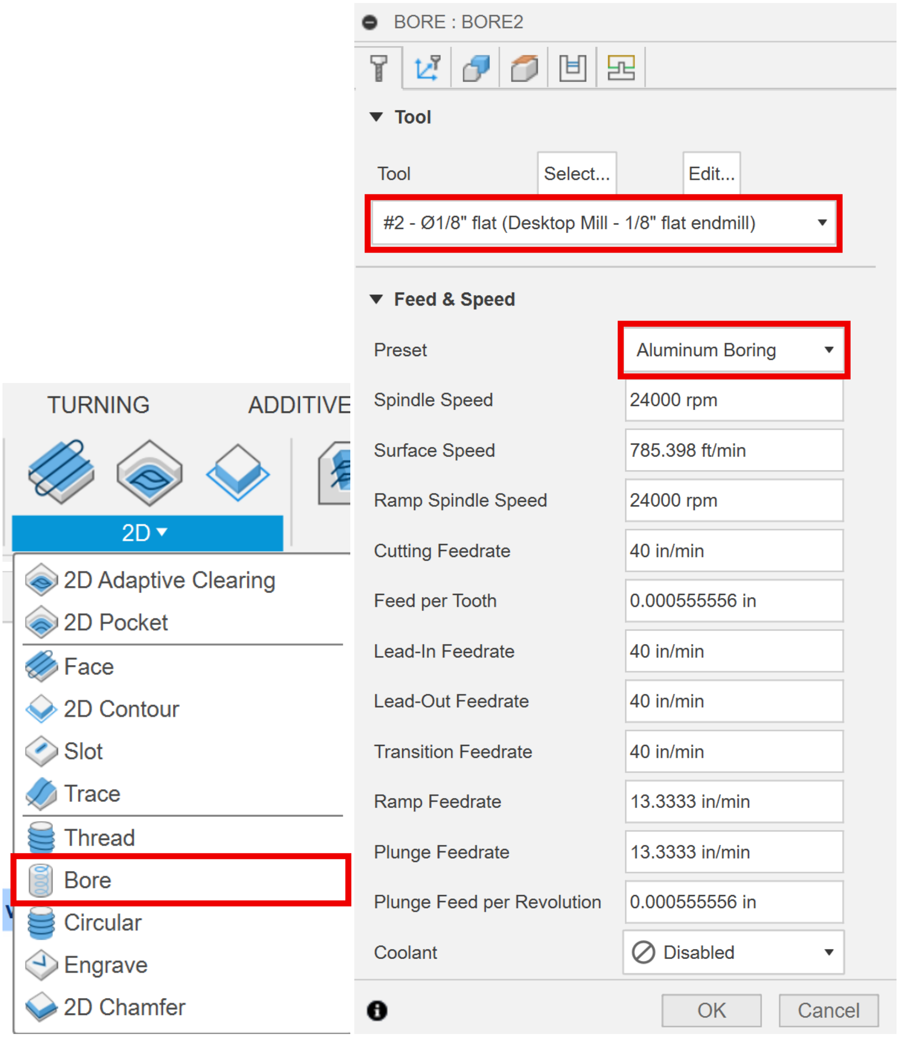
47. In the heights tab, for the retract height select highest of… for from, the top for stock, model and fixture. For the top height, select highest of… for from, then top for stock and ignore for model and fixture. For the bottom height, select selection for from then select the face as shown in the figure below.



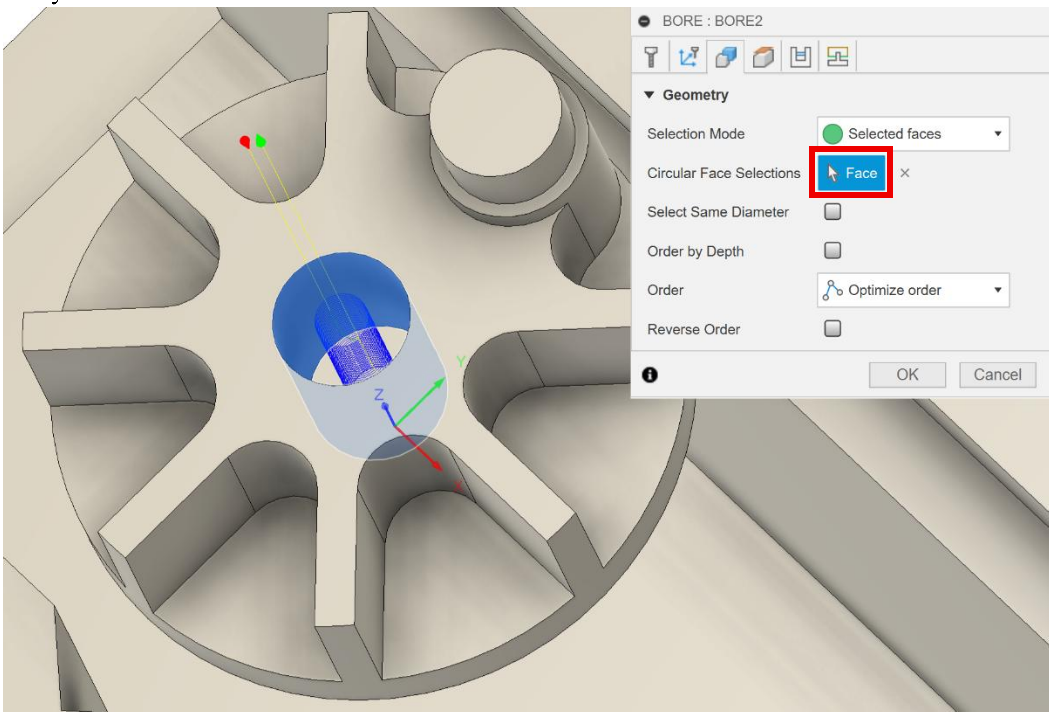
48. In the *passes* tab, uncheck the *stock to leave* box. Then click *ok*.



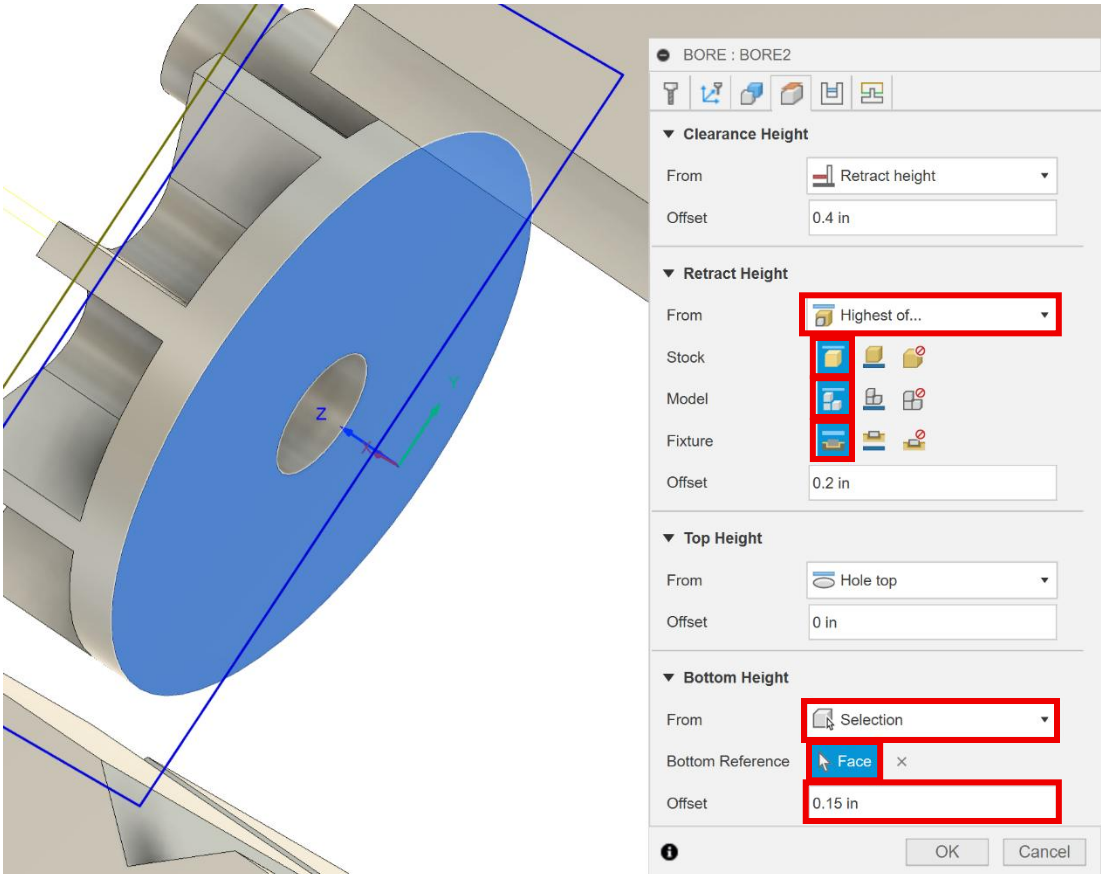
49. Select the *2D Bore* tool in the 2D dropdown menu. Make sure the *1/8" tool* and the *Aluminum boring* preset are selected.



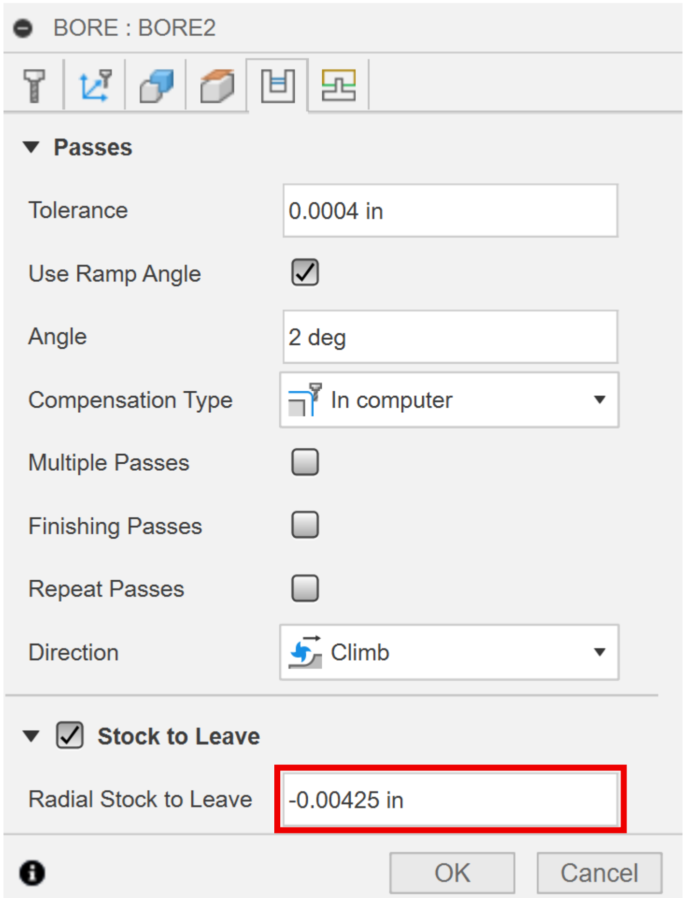
50. In the *geometry* tab, select the inner face of the hole as shown in the figure below. *Zoom* and *pan* as much as you need.



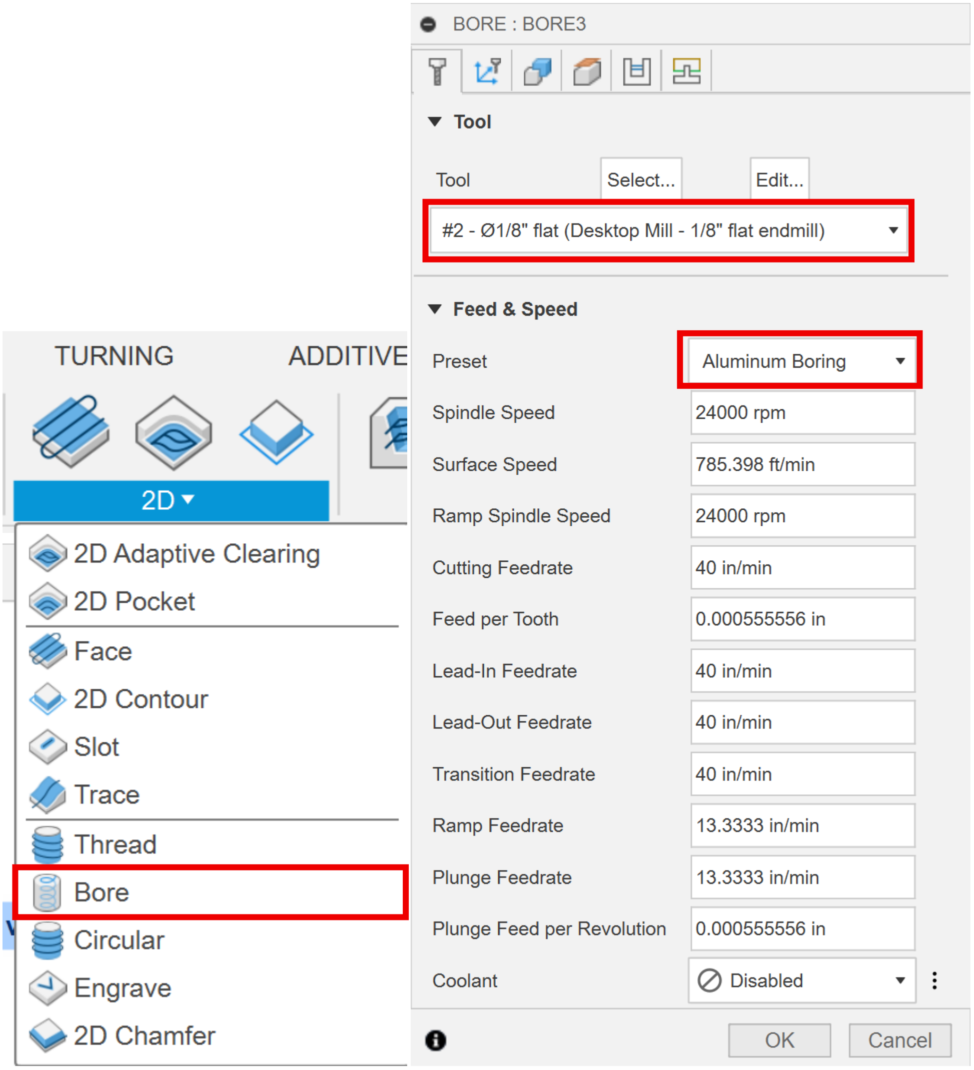
51. In the *heights* tab, for the retract height select highest of… for from, then top for stock, model and fixture. For the bottom height, select selection for from the select the bottom face of the part as shown in the figure below. Change the offset to 0.15 in.



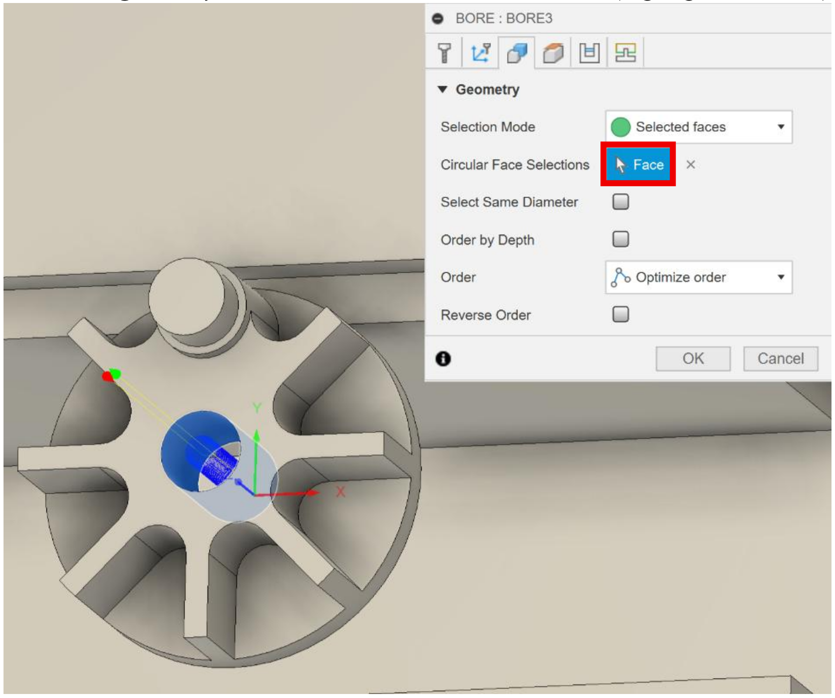
52. In the *passes* tab, check the *stock to leave* box and change the *radial stock to leave* to -0.00425 in. Then click ok.


BORE : BORE2
Passes
Tolerance
0.0004 in
Use Ramp Angle
Angle
2 deg
Compensation Type
In computer
Multiple Passes
Finishing Passes
Repeat Passes
Direction
Climb
Stock to Leave
Radial Stock to Leave
-0.00425 in
OK
Cancel

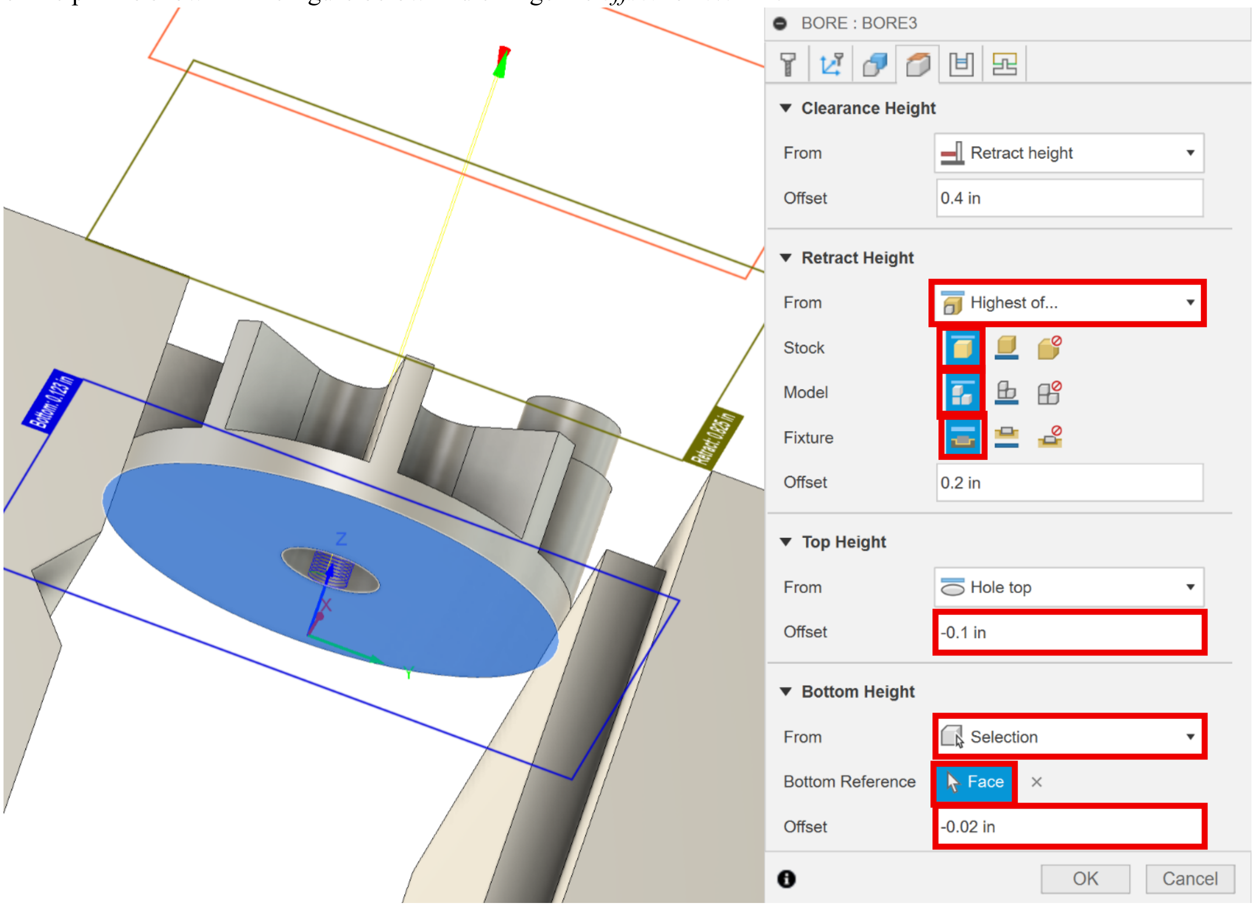
53. Select *2D Bore* tool again in the 2D dropdown menu. Make sure the *1/8" tool* and *Aluminum boring* preset are selected.



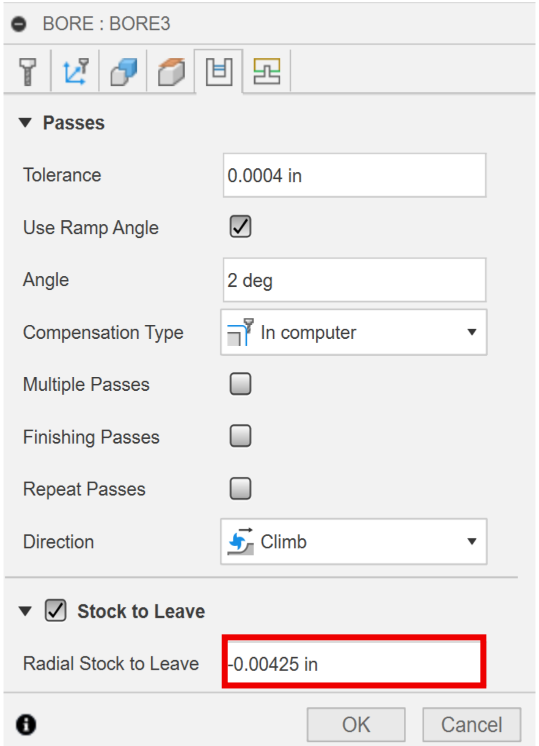
54. In the *geometry* tab, select the inner face of the hole (highlighted in blue) as shown in the figure below.



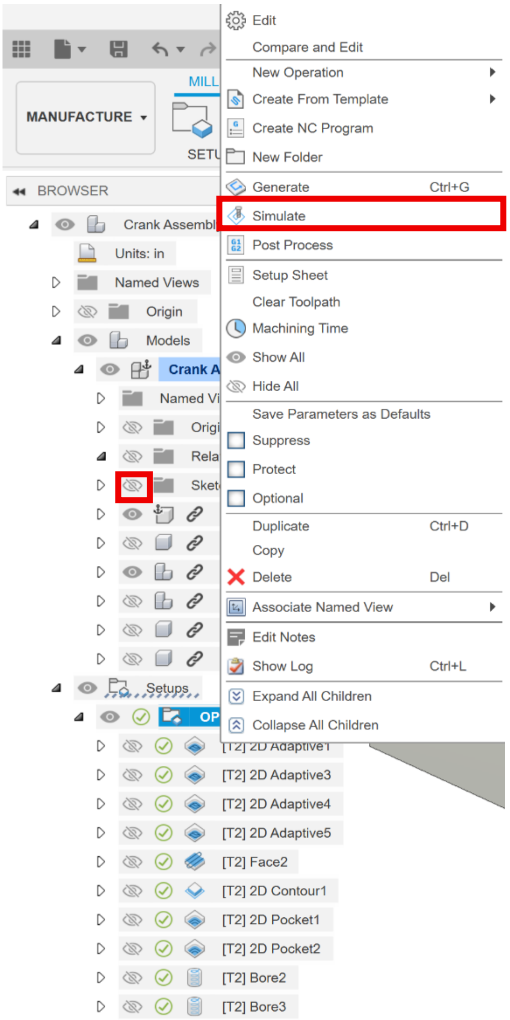
55. In the *heights* tab, for the *retract height* select *highest of…* for *from*, then *top* for *stock*, *model* and *fixture*. For the top height change the offset to -0.1 in. For the *bottom height*, select *selection* for *from*, then the bottom face of the part as shown in the figure below and change the *offset* to -0.02 in.

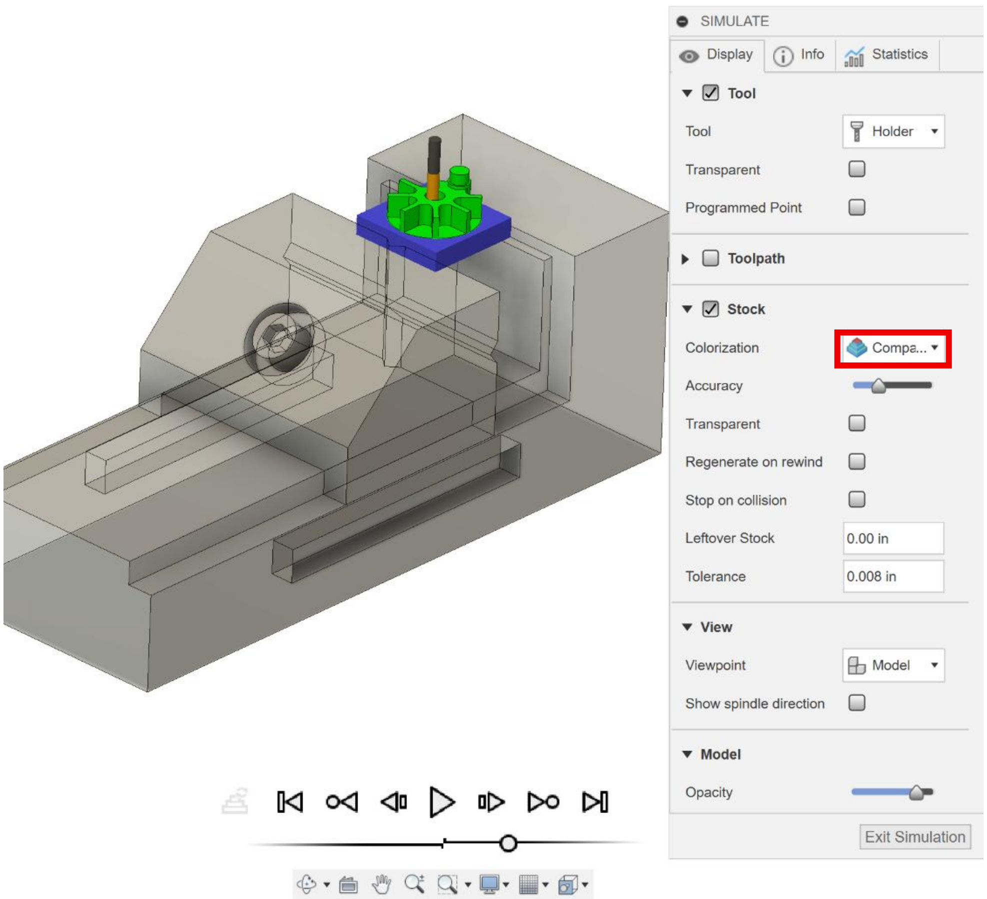


56. In the *passes* tab, check the *stock to leave* box then change the *radial stock to leave* to -0.00425 in. Then click *ok*.



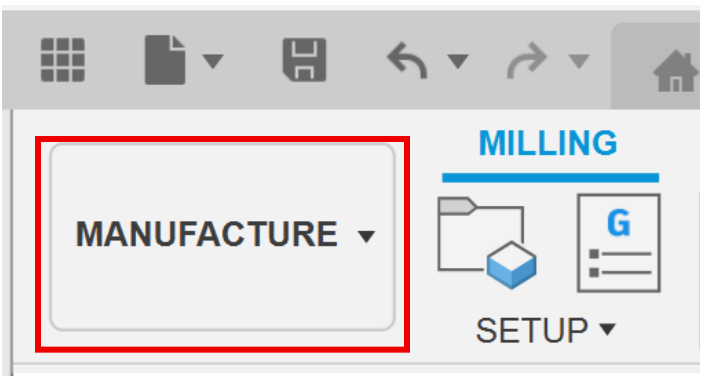
57. *OP10* should now be finished. Hide *Sketches* in *Base Assembly v1:1.* Right click on *OP10* and select *Simulate*. Under *Colorization* choose *Comparison.* Press the play button at the bottom center of the window. Adjust the *Accuracy* slider until simulation runs smoothly. Once the simulation is complete, the part should be green if programmed correctly. <u>Note: If *Accuracy* is set to the lowest value, blue may appear around the part.</u>


MANUFACTURE
MILL
SETU
BROWSER
Crank Assembl
Units: in
Named Views
Origin
Models
Crank A
Named Vi
Origi
Rela
Sket
Setups
OP
Edit
Compare and Edit
New Operation
Create From Template
Create NC Program
New Folder
Generate Ctrl+G
Simulate
Post Process
Setup Sheet
Clear Toolpath
Machining Time
Show All
Hide All
Save Parameters as Defaults
Suppress
Protect
Optional
Duplicate Ctrl+D
Copy
Delete Del
Associate Named View
Edit Notes
Show Log Ctrl+L
Expand All Children
Collapse All Children
[T2] 2D Adaptive3
[T2] 2D Adaptive4
[T2] 2D Adaptive5
[T2] Face2
[T2] 2D Contour1
[T2] 2D Pocket1
[T2] 2D Pocket2
[T2] Bore2
[T2] Bore3

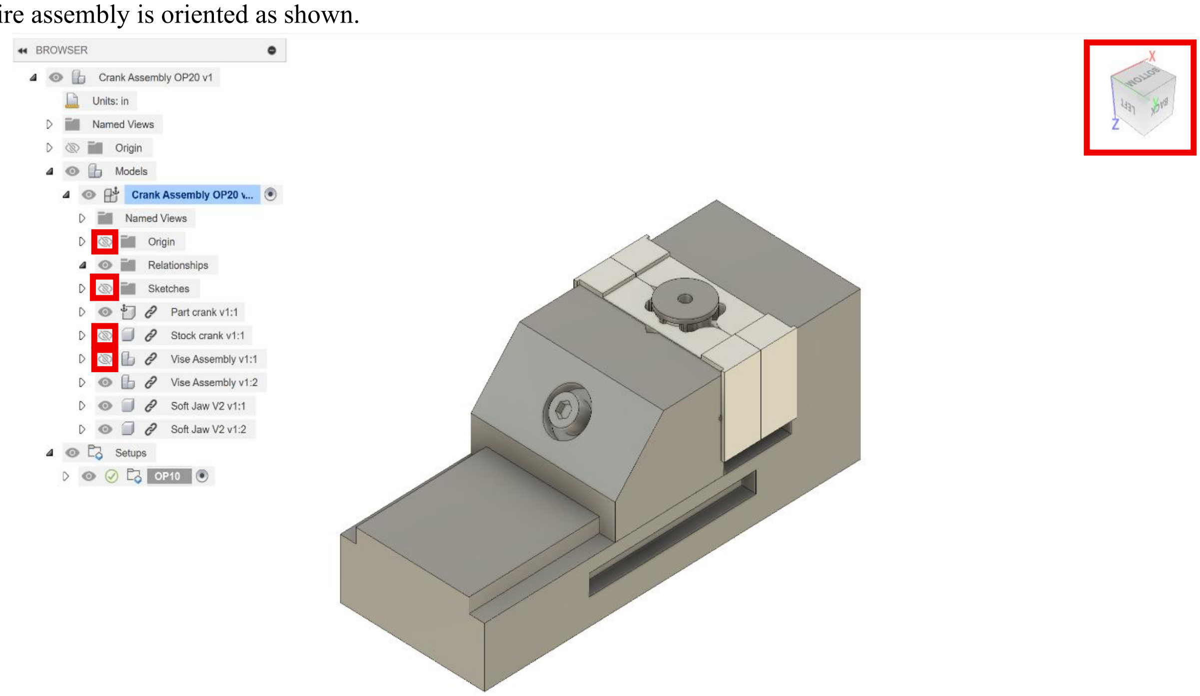

SIMULATE
Display
Info
Statistics
Tool
Tool
Holder
Transparent
Programmed Point
Toolpath
Stock
Colorization
Compa...
Accuracy
Transparent
Regenerate on rewind
Stop on collision
Leftover Stock
0.00 in
Tolerance
0.008 in
View
Viewpoint
Model
Show spindle direction
Model
Opacity
Exit Simulation

Part Programming (Autodesk Fusion) OP20

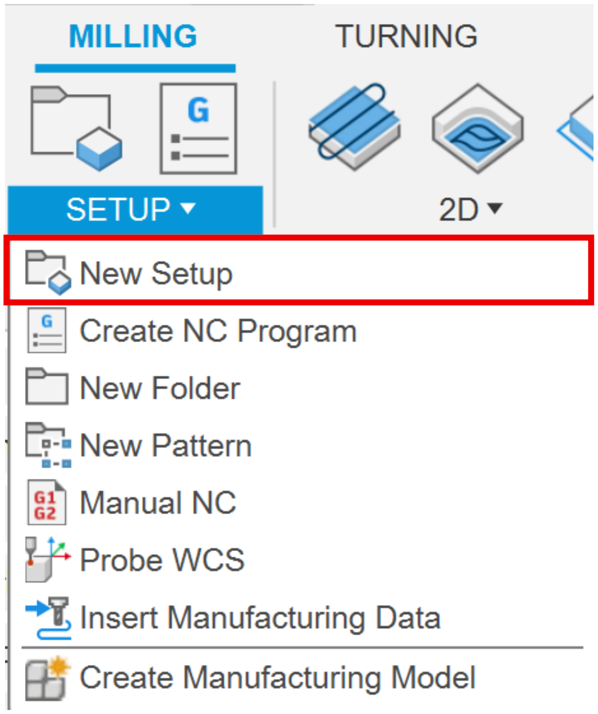
1. OP10 is now complete. Make sure you are in the Manufacture workspace.



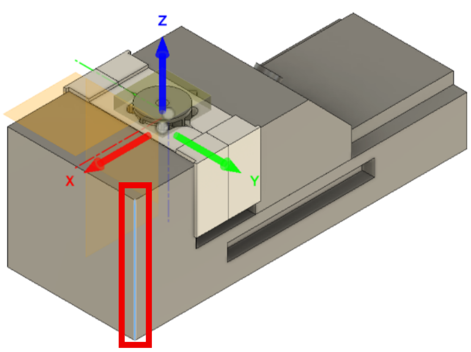
2. Make sure *Vise Assembly v1:2* is visible. Hide *sketches*, *stock crank v1:1* and *Vise Assembly v1:1*. Make sure the entire assembly is oriented as shown.

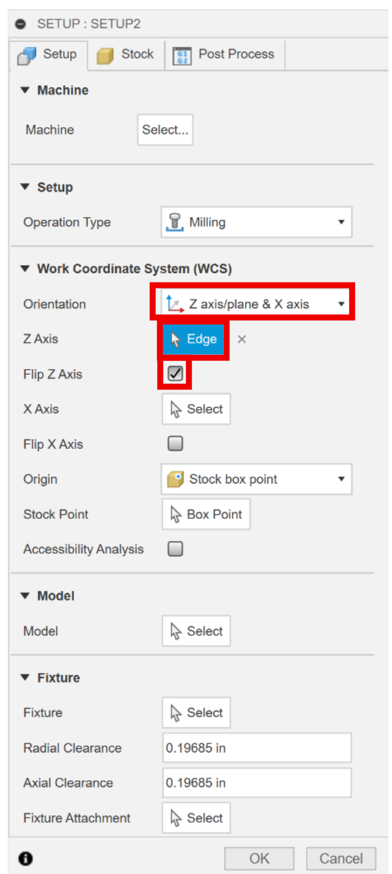


3. Create OP20 by selecting New Setup.

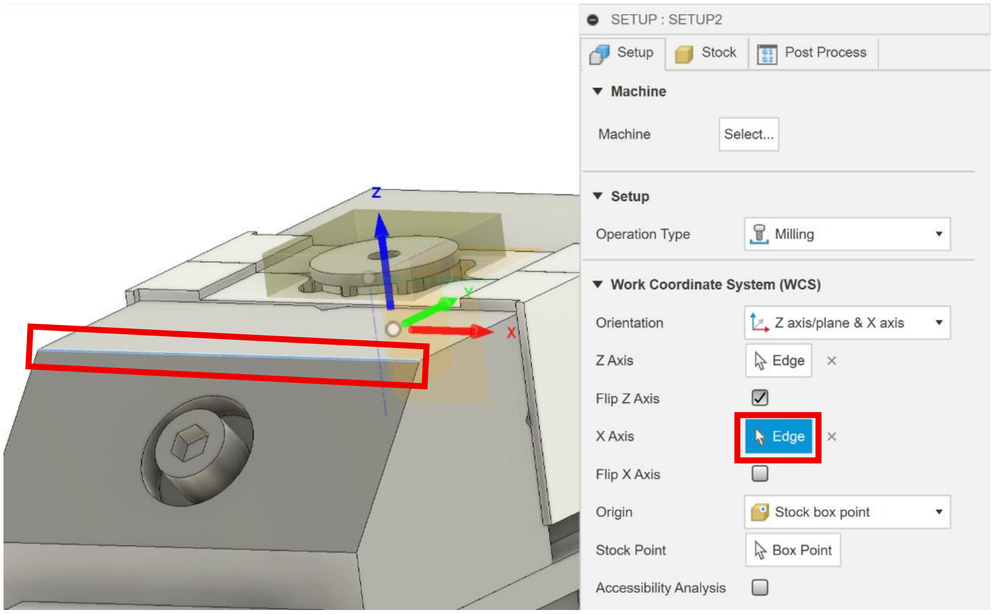


4. Set the orientation to *Z axis/plane & X axis*. Then select the edge shown for the Z axis and check *the Flip Z Axis* box.

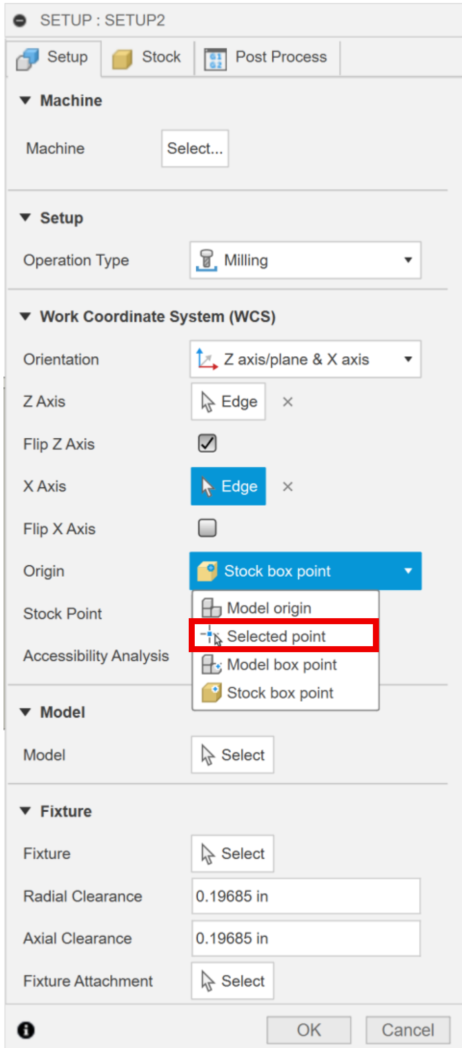


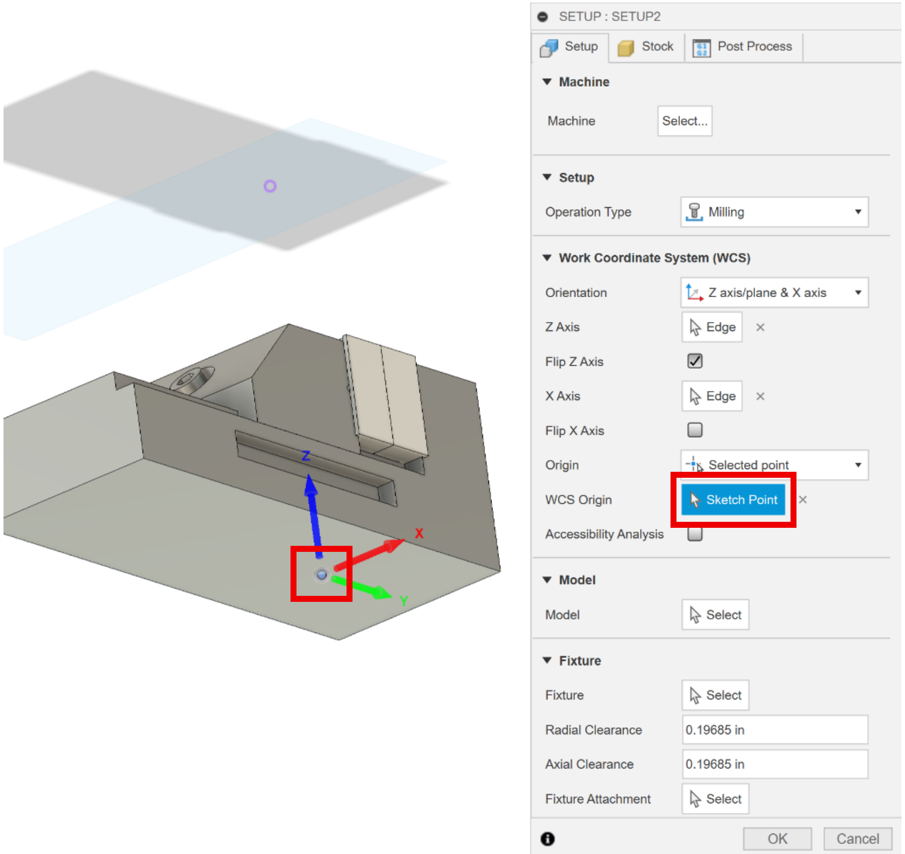


5. For X axis, select the edge shown in the figure.

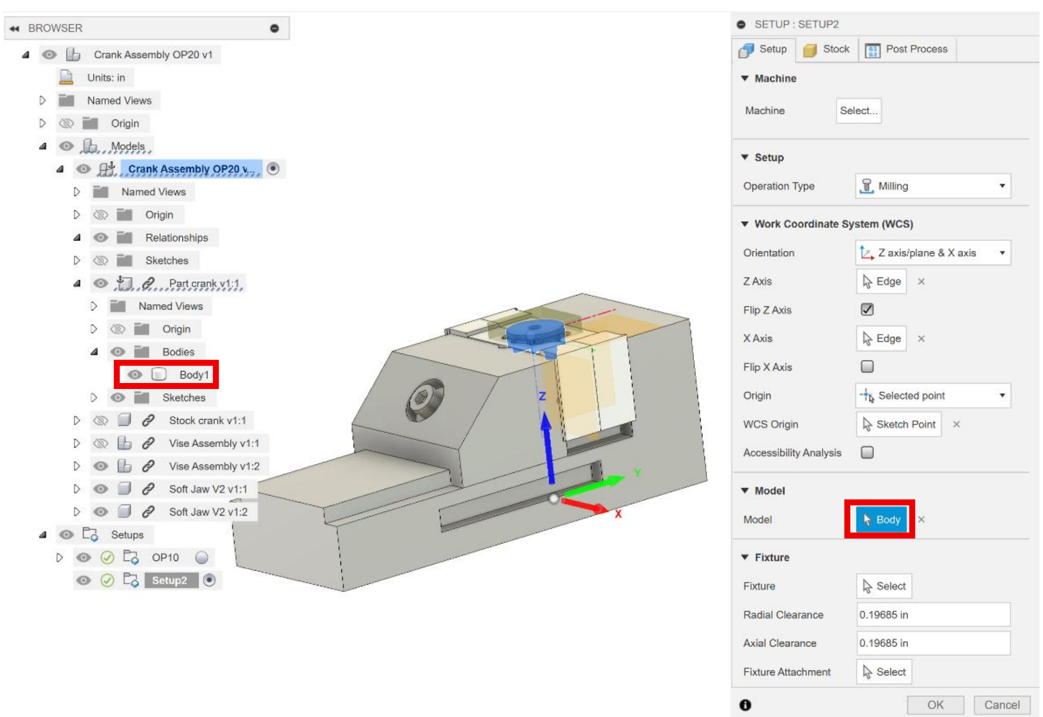


6. For the origin, click *selected point*. Turn on *sketches* and pick the point below the part.

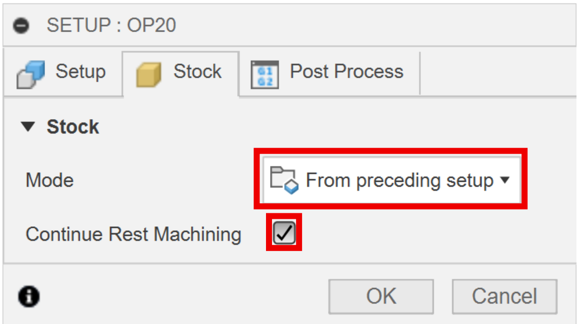

SETUP : SETUP2
Setup
Stock
Post Process
Machine
Machine
Select...
Setup
Operation Type
Milling
Work Coordinate System (WCS)
Orientation
Z axis/plane & X axis
Z Axis
Edge
Flip Z Axis
X Axis
Edge
Flip X Axis
Origin
Stock box point
Stock Point
Model origin
Selected point
Model box point
Stock box point
Accessibility Analysis
Model
Model
Select
Fixture
Fixture
Select
Radial Clearance
0.19685 in
Axial Clearance
0.19685 in
Fixture Attachment
Select
OK
Cancel



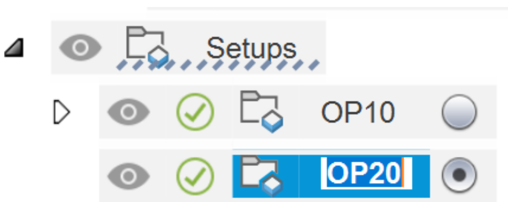
7. For the *model*, drop down *part crank v1:1*, drop down *bodies*, then select *Body 1*.



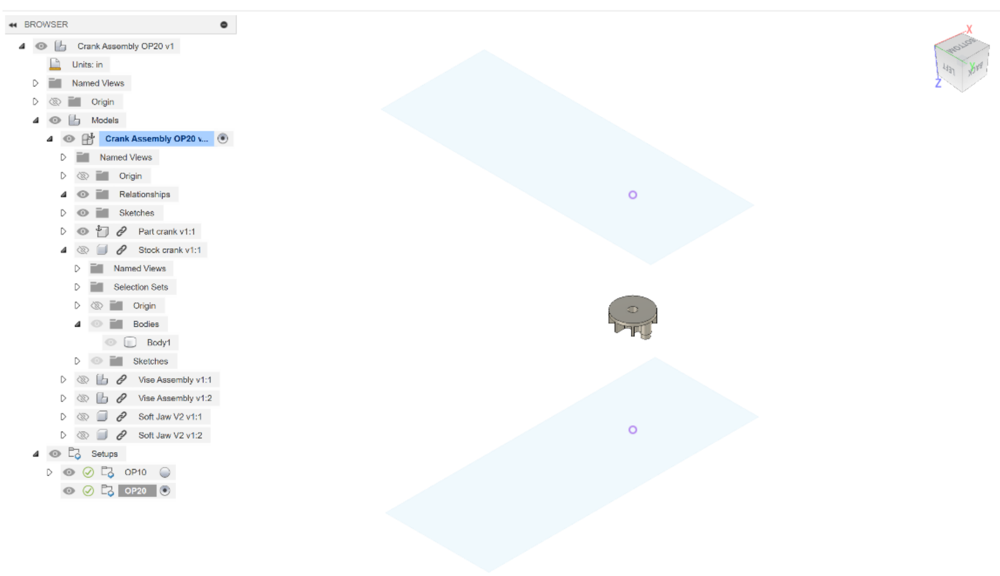
8. In the *stock* tab select *from preceding setup* and then check the *continue* rest *machining* box.

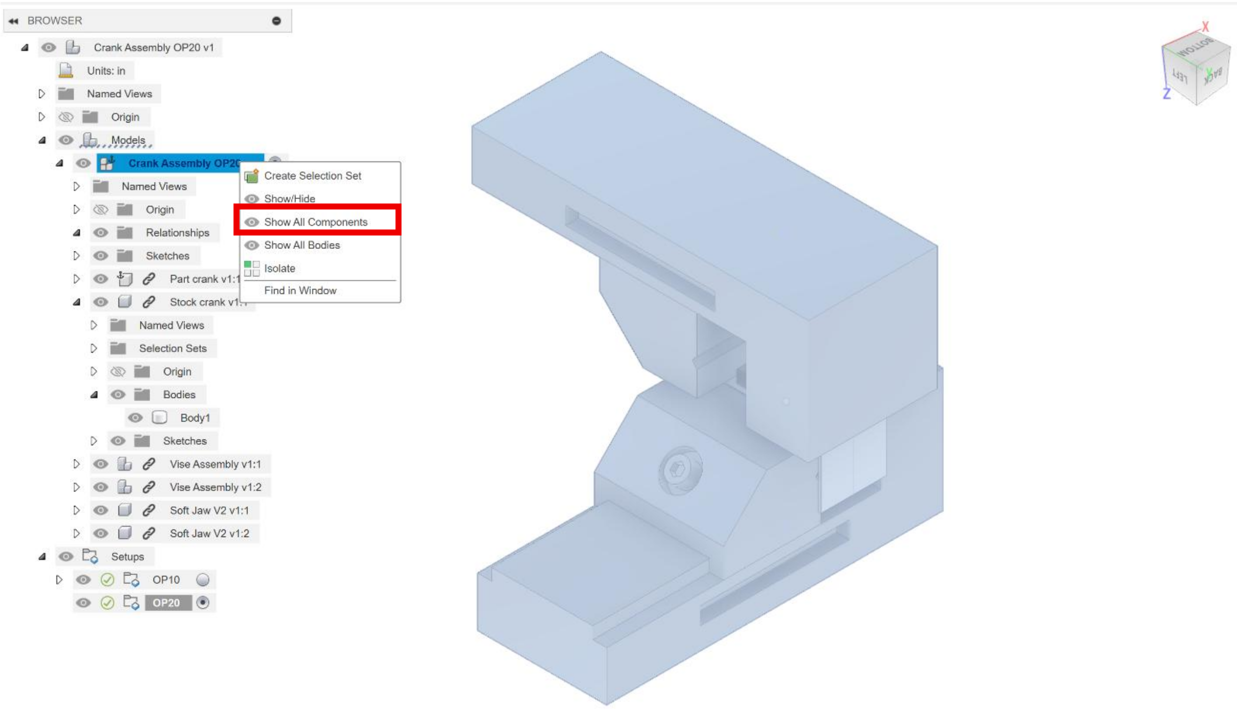


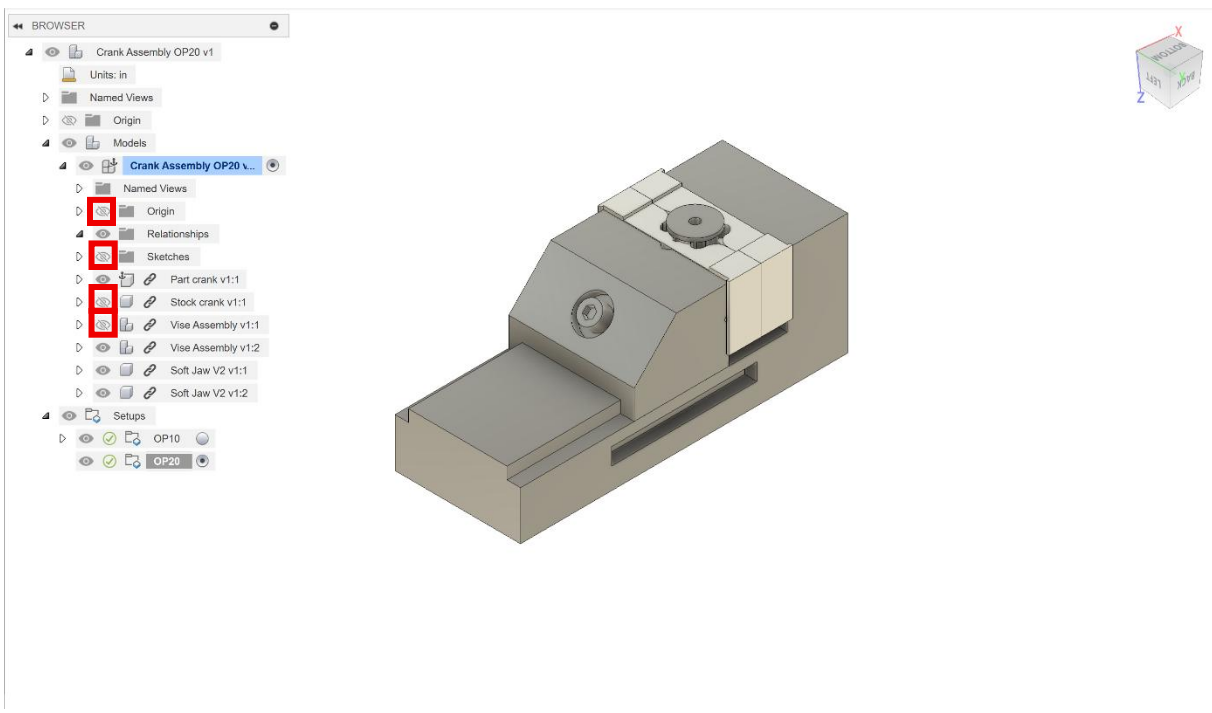
9. Once you have completed your set up rename it to OP20 by selecting the setup then selecting it again, then you can rename it.



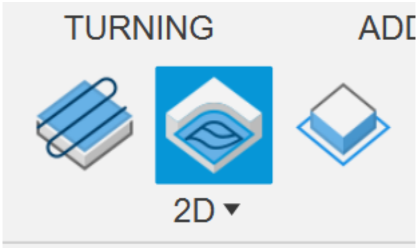
10. You may have realized that your components have disappeared. If this happens, click on the name of your file under models, then select *show all components*. Then hide all the components mentioned in step 2.

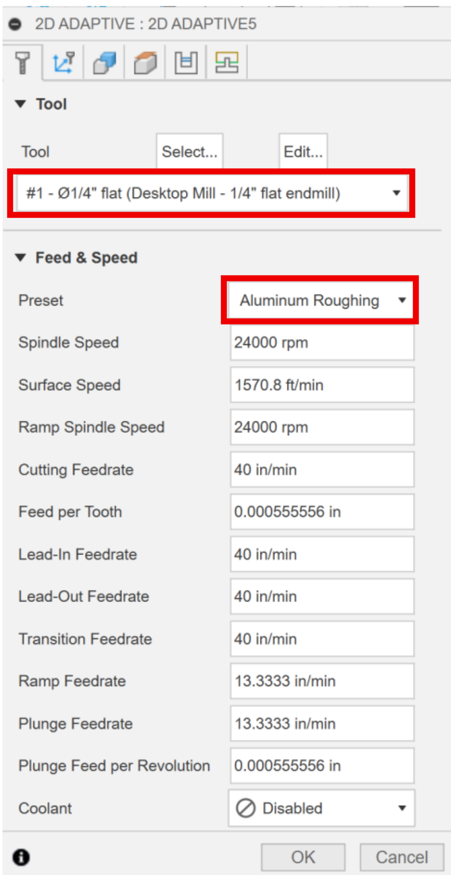




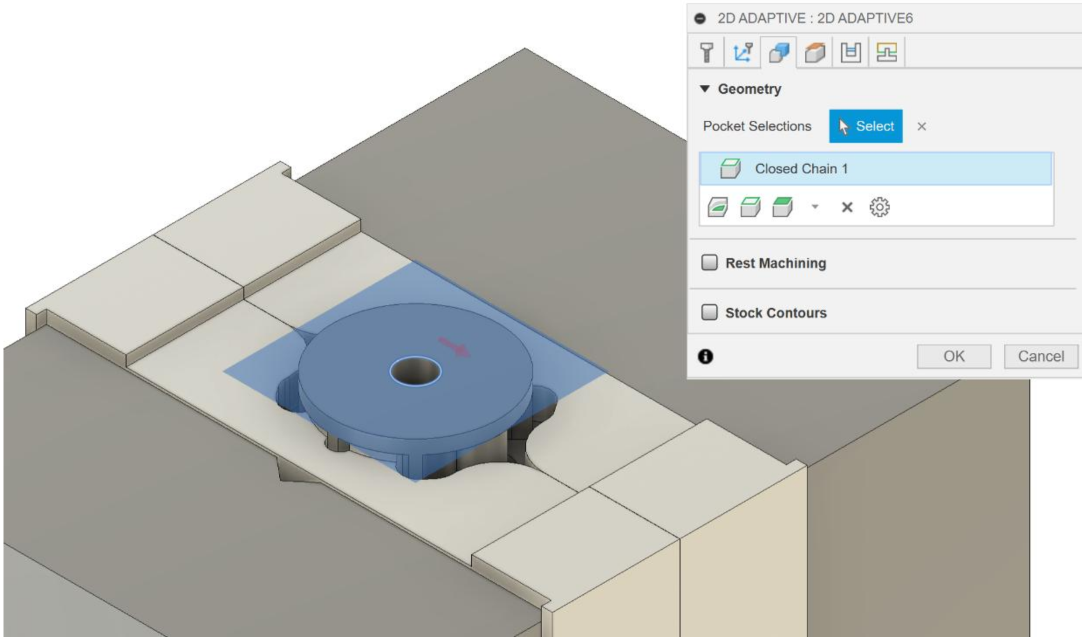


11. Select the *2D Adaptive Clearing* tool option. Make sure the *1/4" tool* and *Aluminum Roughing* preset are selected.

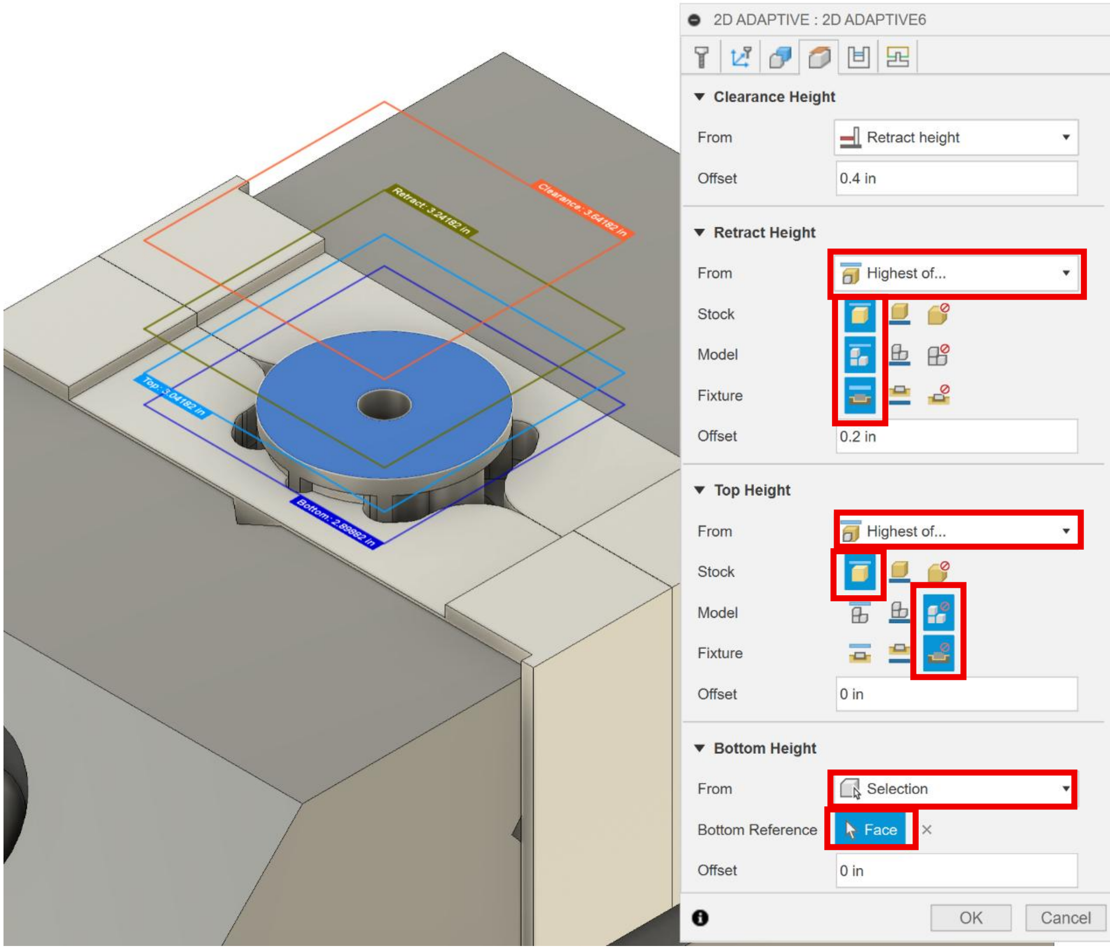




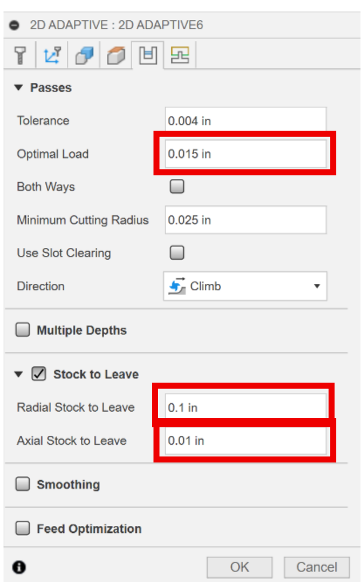
12. In the *geometry* tab, select the edge selected below. You will have to click the red arrow to make the selection look as is in the figure.



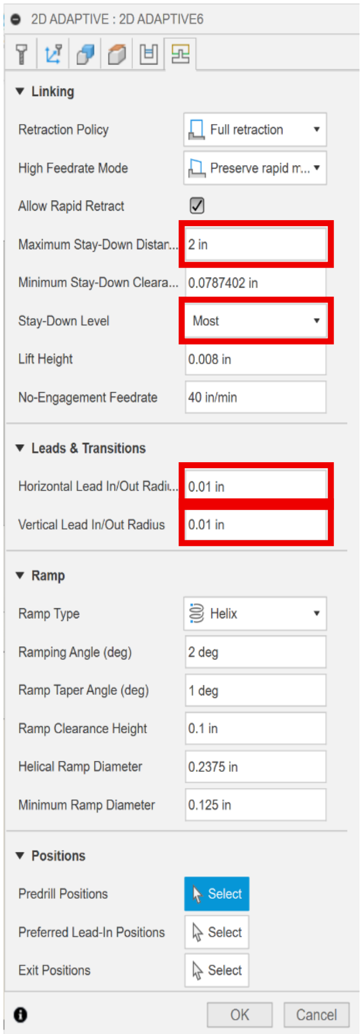
13. In the heights tab, for *retract height* select *highest of…* for *from* then *top* for *stock*, *model* and *fixture*. For *top height* select highest of for *from* then *top* for *stock*, *ignore* for *model* and *fixture*. For *bottom height* select *selection* for *from*, then click the top face of the part as shown in the figure below.



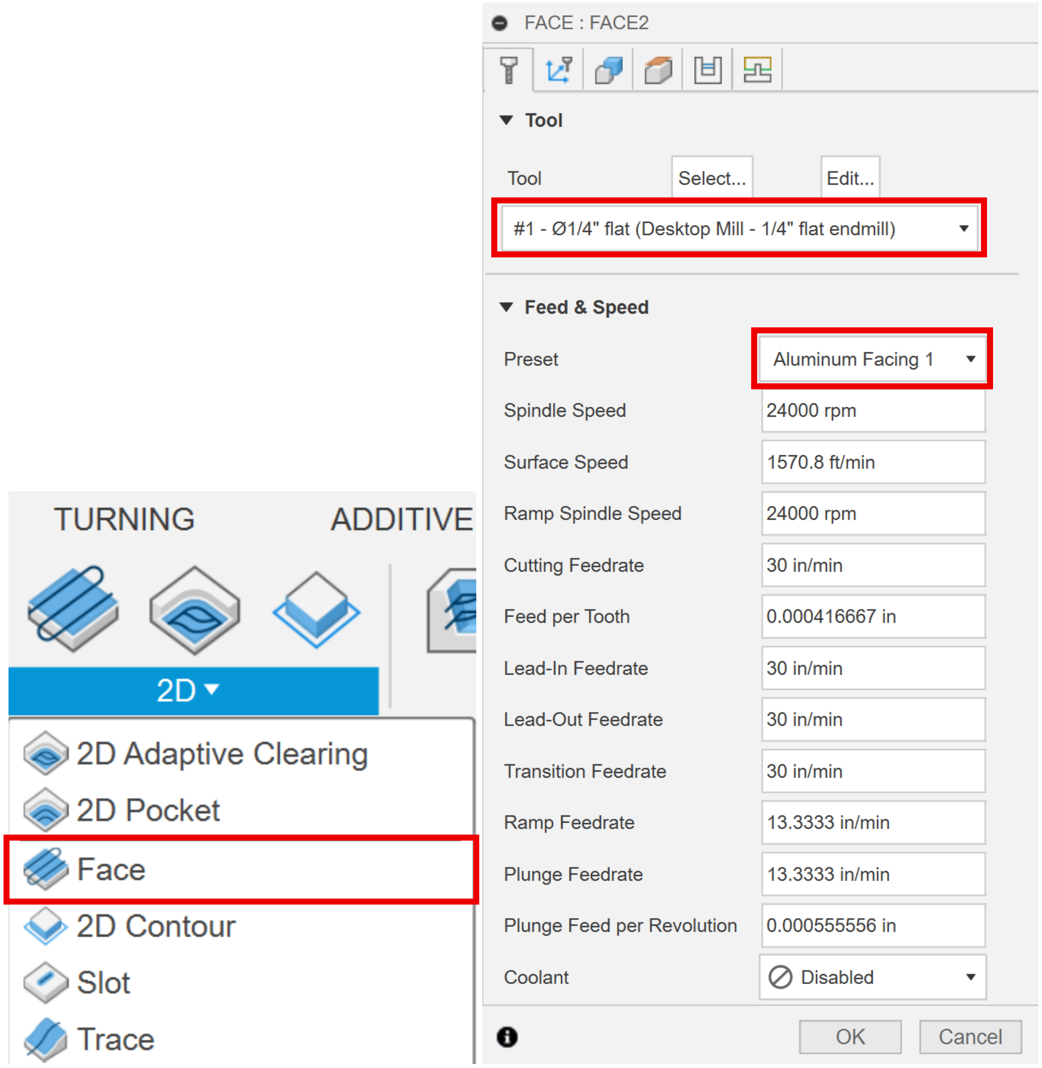
14. In the *passes* tab change the *optimal load* to 0.015 in, the *radial stock to leave* to 0.1 in and the *axial stock to leave* to 0.01 in.



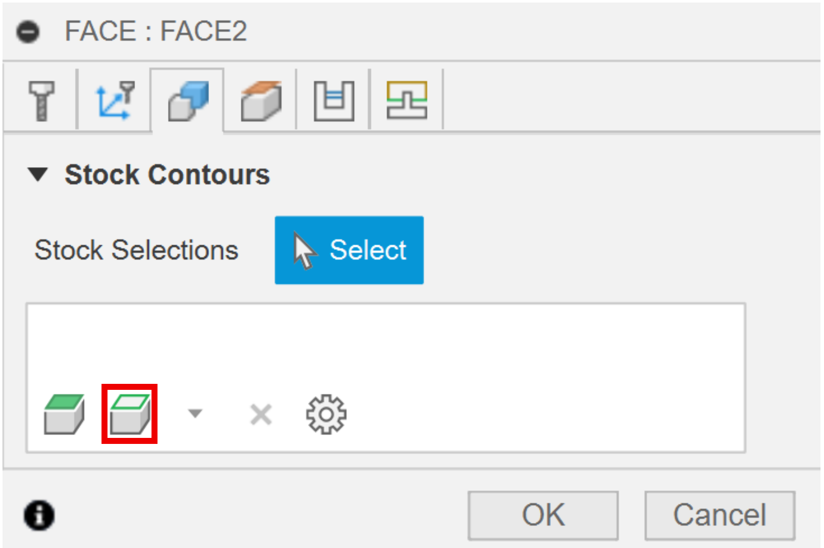
15. In the *linking* tab, change the *maximum stay-down distance* to 2 in, the *stay-down level* to *most*, and *leads & transitions* to 0.01 in.

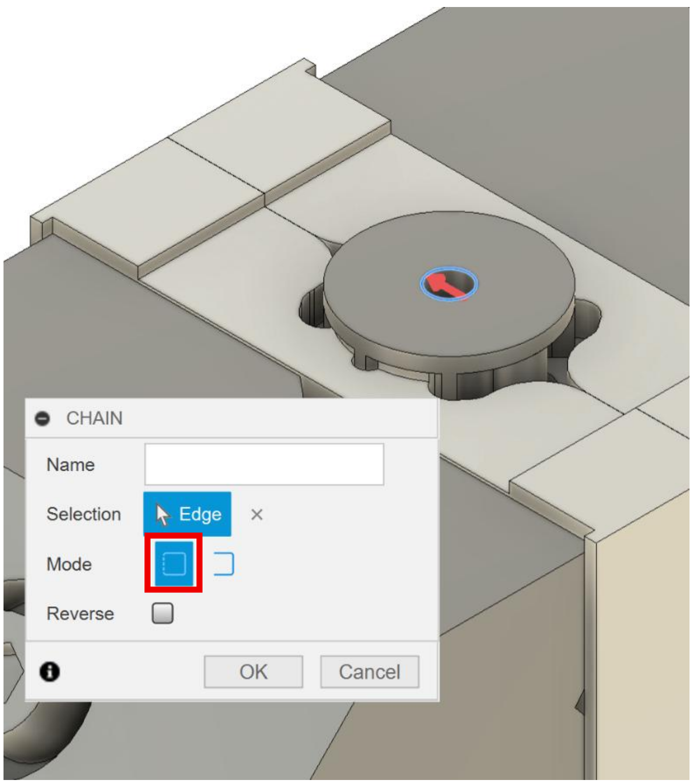


16. Select the *2D Face* tool option, then make sure you select the *¼" tool* and the *Aluminum facing 1* preset.

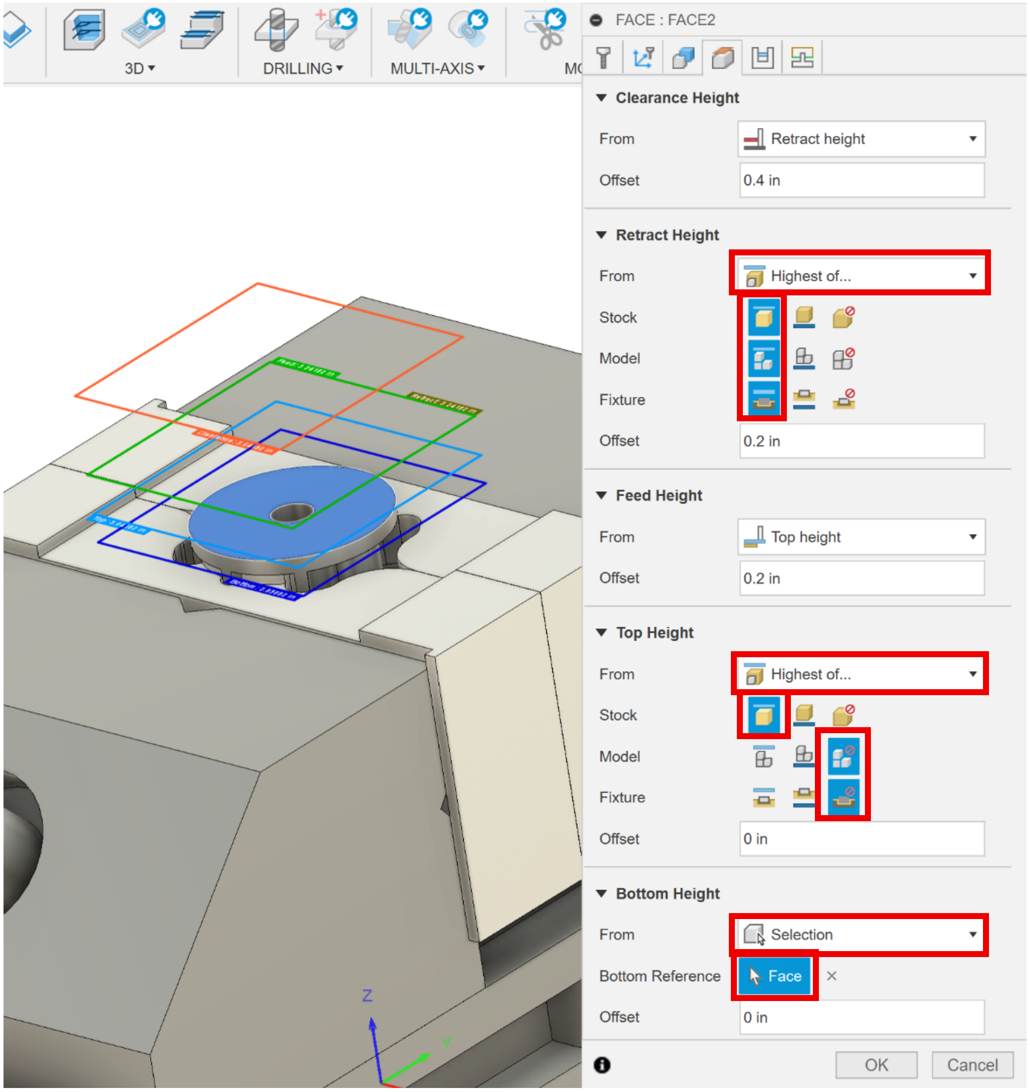


17. In the *geometry* tab, select the *chain* tab then select *closed chain* for *mode* and choose the edge selected in the figure below.

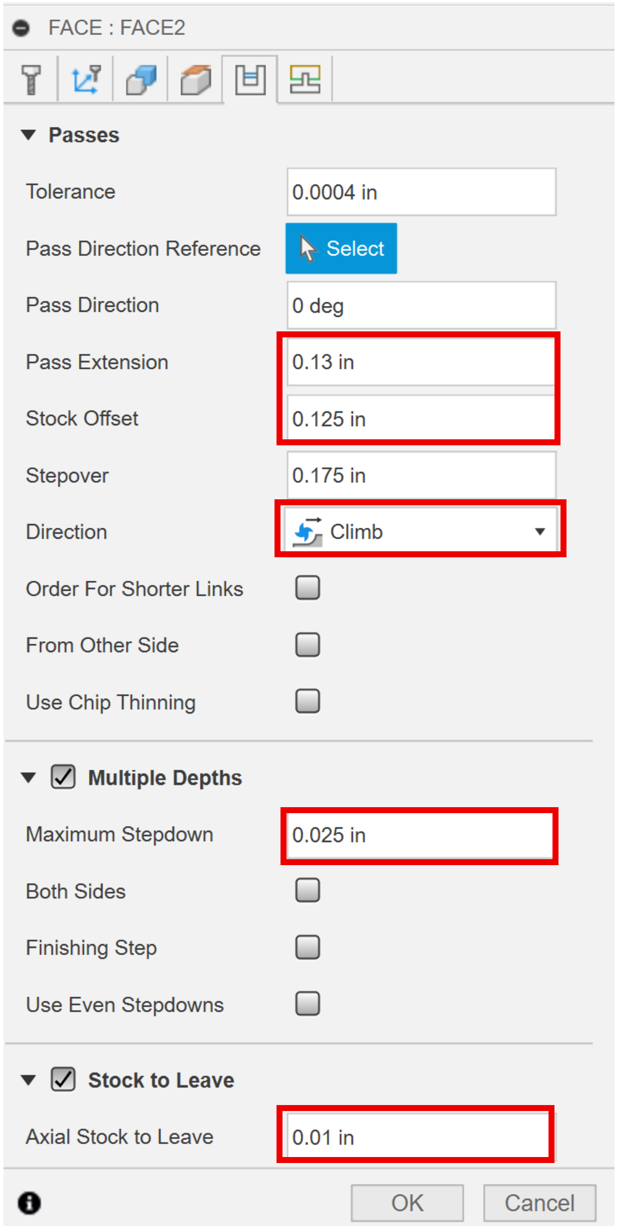




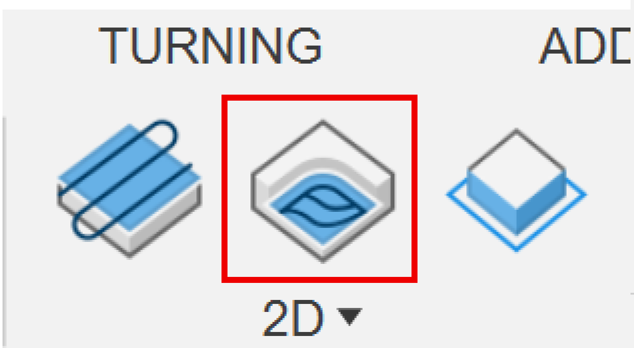
18. In the *heights* tab, for the *retract height* select *highest of…* for *from* then select top for *stock*, *model*, and *fixture*. For the top height select highest of… for from then select top for stock and ignore for model and fixture. For the bottom height select selection for from and then choose the top face of the part.

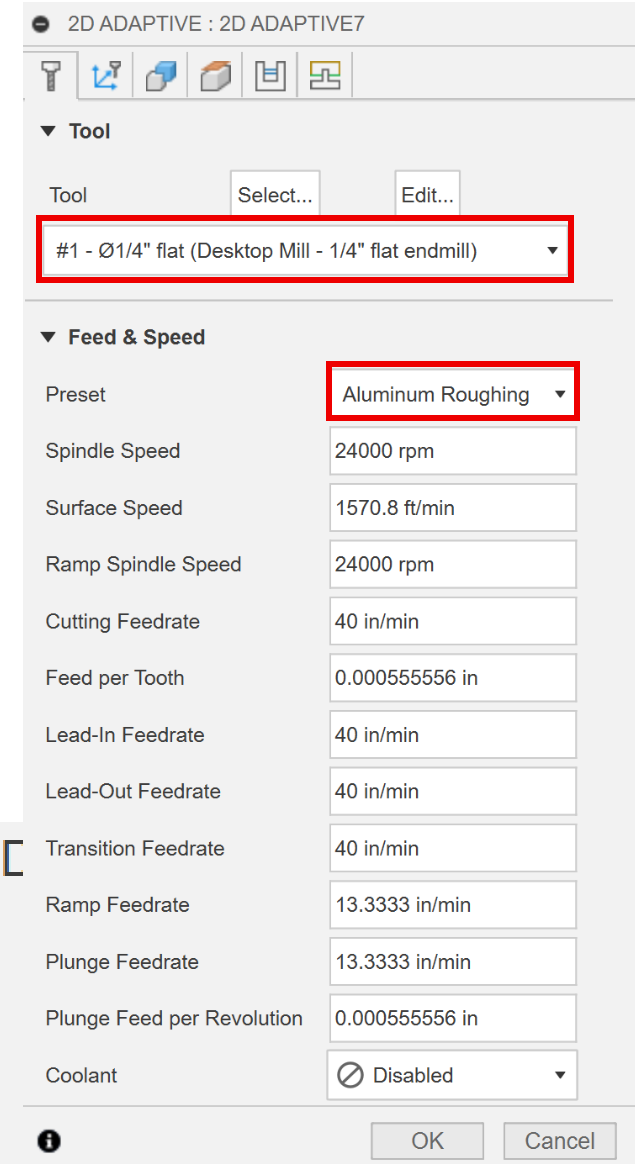


19. In the *passes* tab, change the *pass extension* to 0.13 in, the *stock offset* to 0.125 in and the *direction* to *climb*. Check the *multiple depths* and *stock to leave* boxes. Change the *maximum stepdown* to 0.025 in and the *axial stock to leave* to 0.01 in. Then click *ok*.

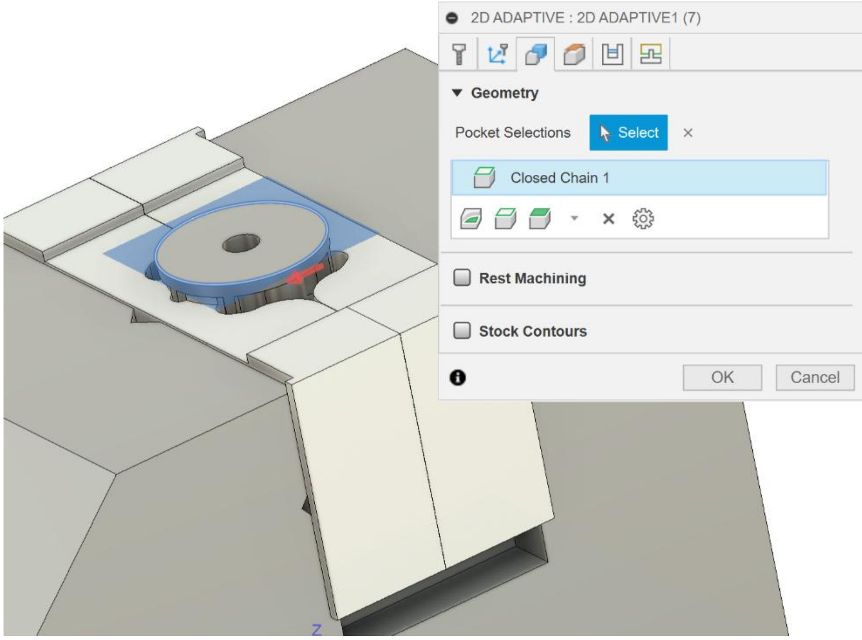


20. Select the *2D Adaptive Clearing* tool option, make sure the tool is set to the *¼" tool* and the preset is set to *Aluminum roughing*.

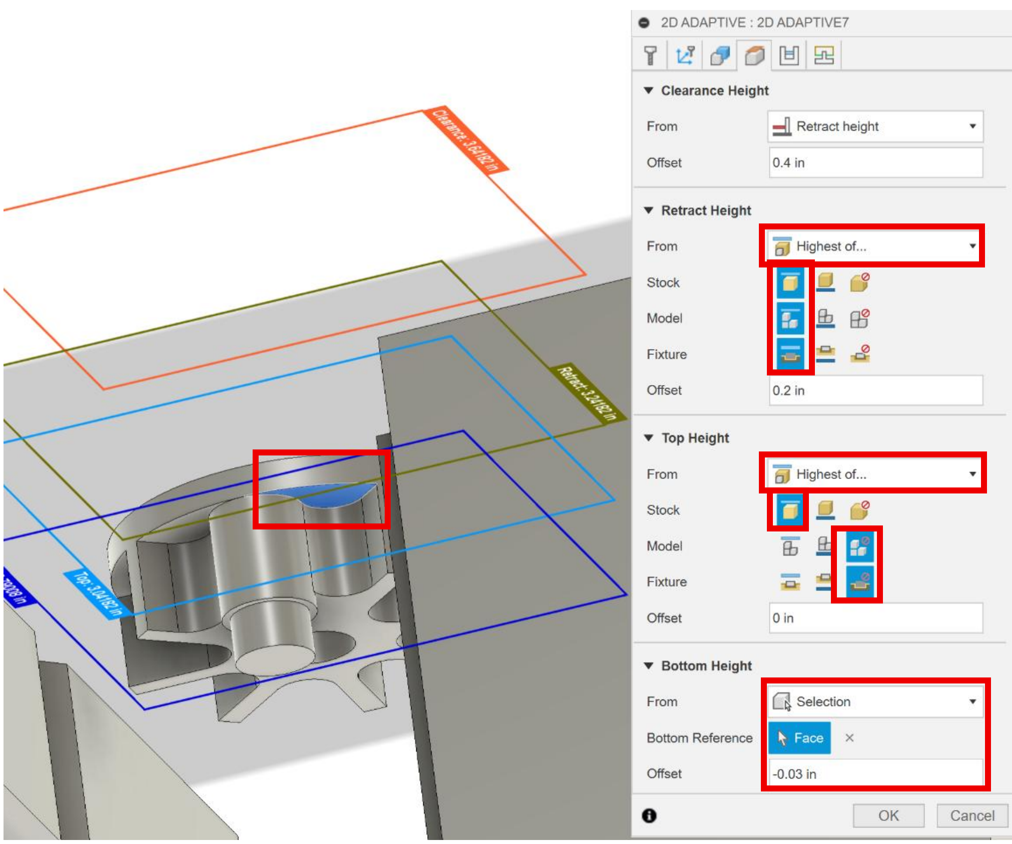

TURNING
ADD
2D


2D ADAPTIVE : 2D ADAPTIVE7
Tool
Tool
Select...
Edit...
#1 - Ø1/4" flat (Desktop Mill - 1/4" flat endmill)
Feed & Speed
Preset
Aluminum Roughing
Spindle Speed
24000 rpm
Surface Speed
1570.8 ft/min
Ramp Spindle Speed
24000 rpm
Cutting Feedrate
40 in/min
Feed per Tooth
0.000555556 in
Lead-In Feedrate
40 in/min
Lead-Out Feedrate
40 in/min
Transition Feedrate
40 in/min
Ramp Feedrate
13.3333 in/min
Plunge Feedrate
13.3333 in/min
Plunge Feed per Revolution
0.000555556 in
Coolant
Disabled
OK
Cancel

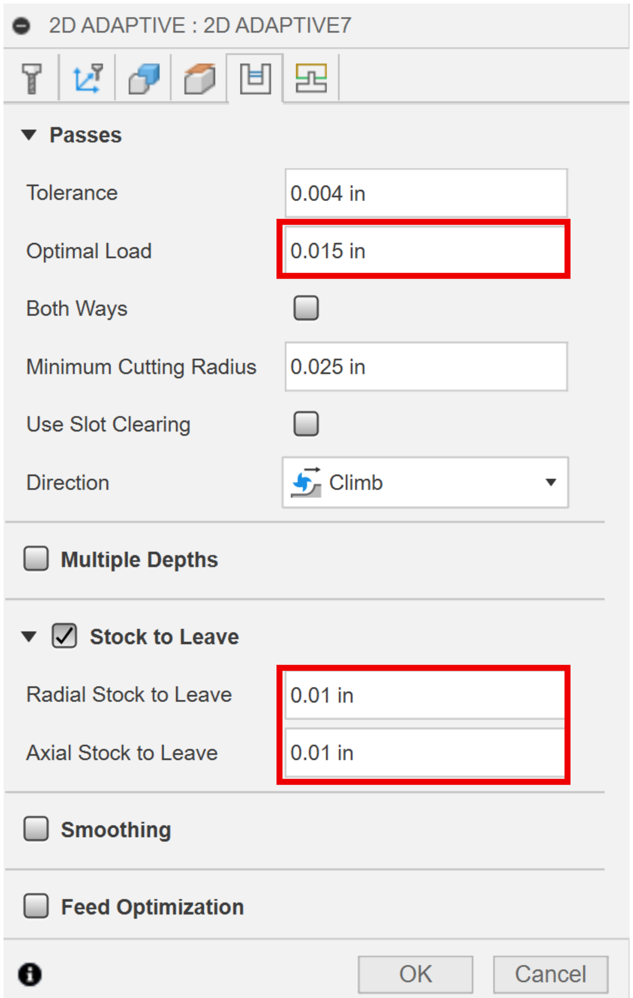
21. In the *geometry* tab, select the edge shown in the figure below.



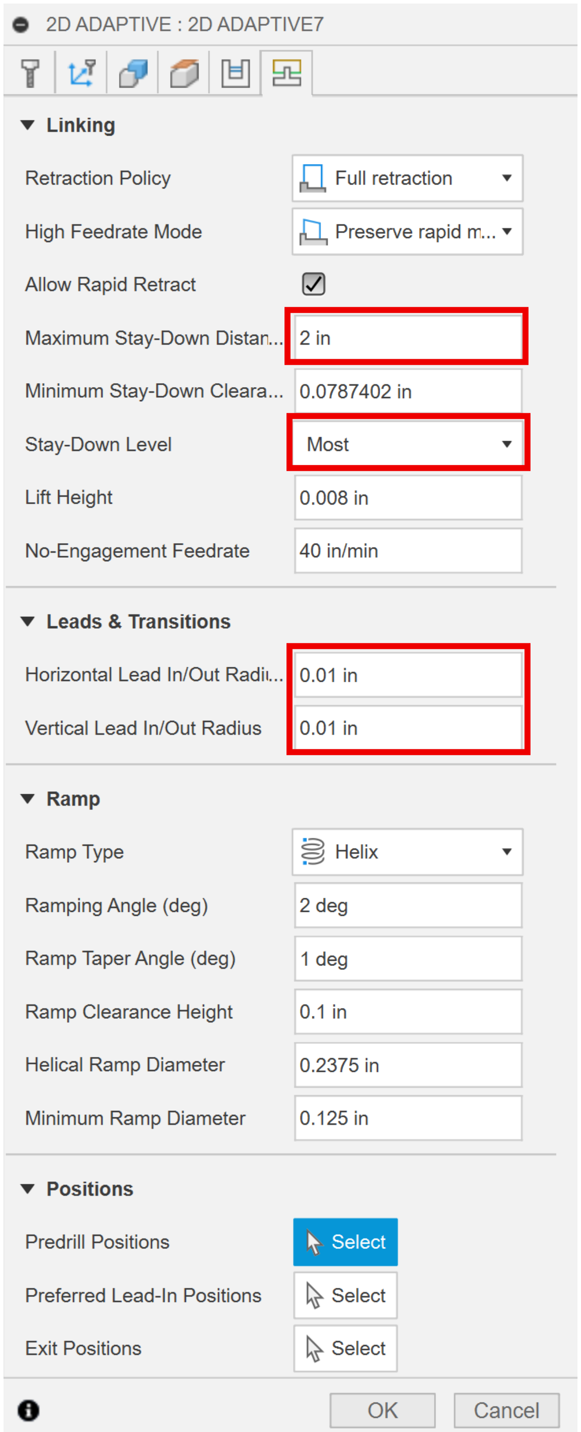
22. In the heights tab, for the retract height select highest of… for from, then top for the stock, model and fixture. For the top height select highest of.. for from, then top for stock and ignore for model and fixture. For the bottom height select selection, then select the face as shown in the figure (you may need to hide the soft jaws the reach this face) and change the offset to -0.03 in.



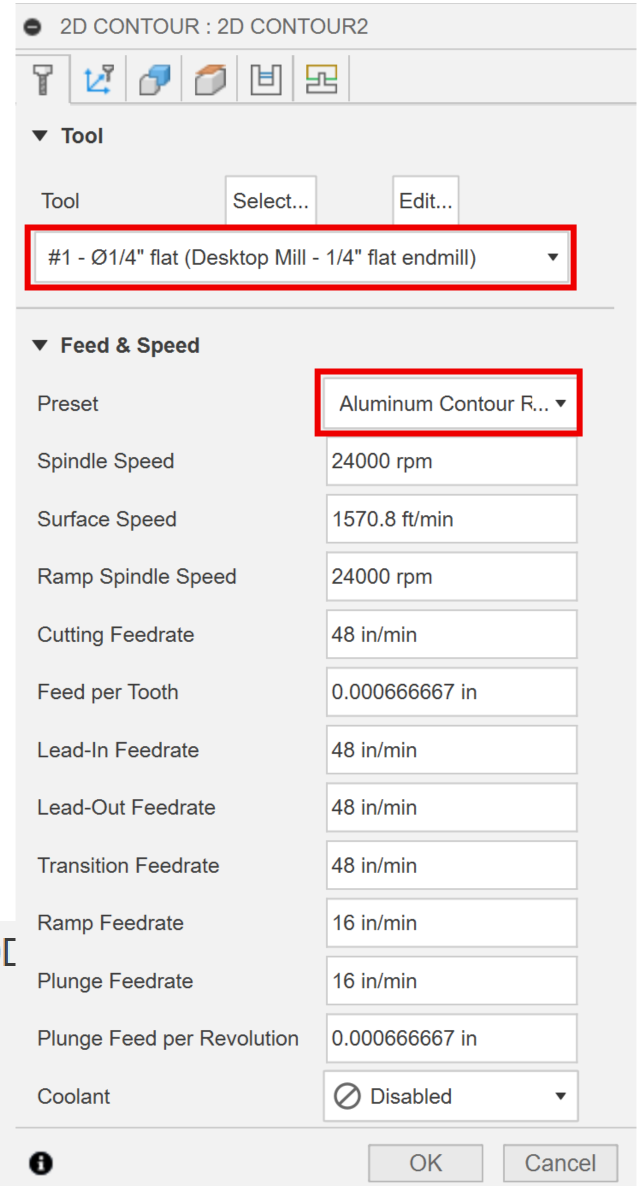
23. In the *passes* tab, change the *optimal load* to 0.015 in, then *axial and radial stock to leave* to 0.01 in.

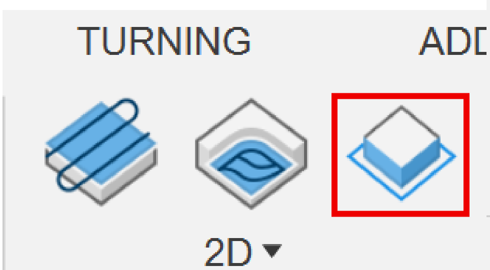


24. In the *linking* tab, change the *maximum stay-down distance* to 2 in, the *stay-down level* to *most*, and *leads & transitions* to 0.01 in. Then click *ok*.

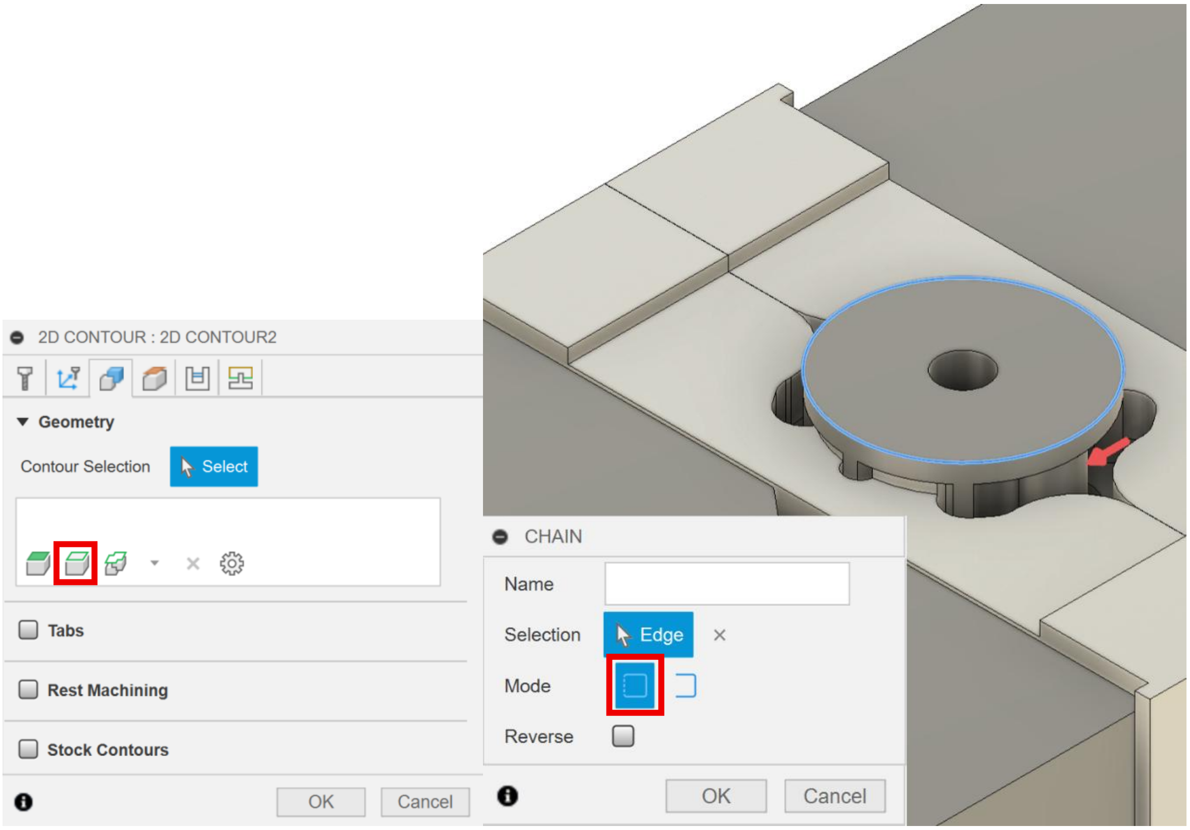


25. Select the *2D contour* tool option, make sure the *¼" tool* and *Aluminum contour roughing* preset are selected.

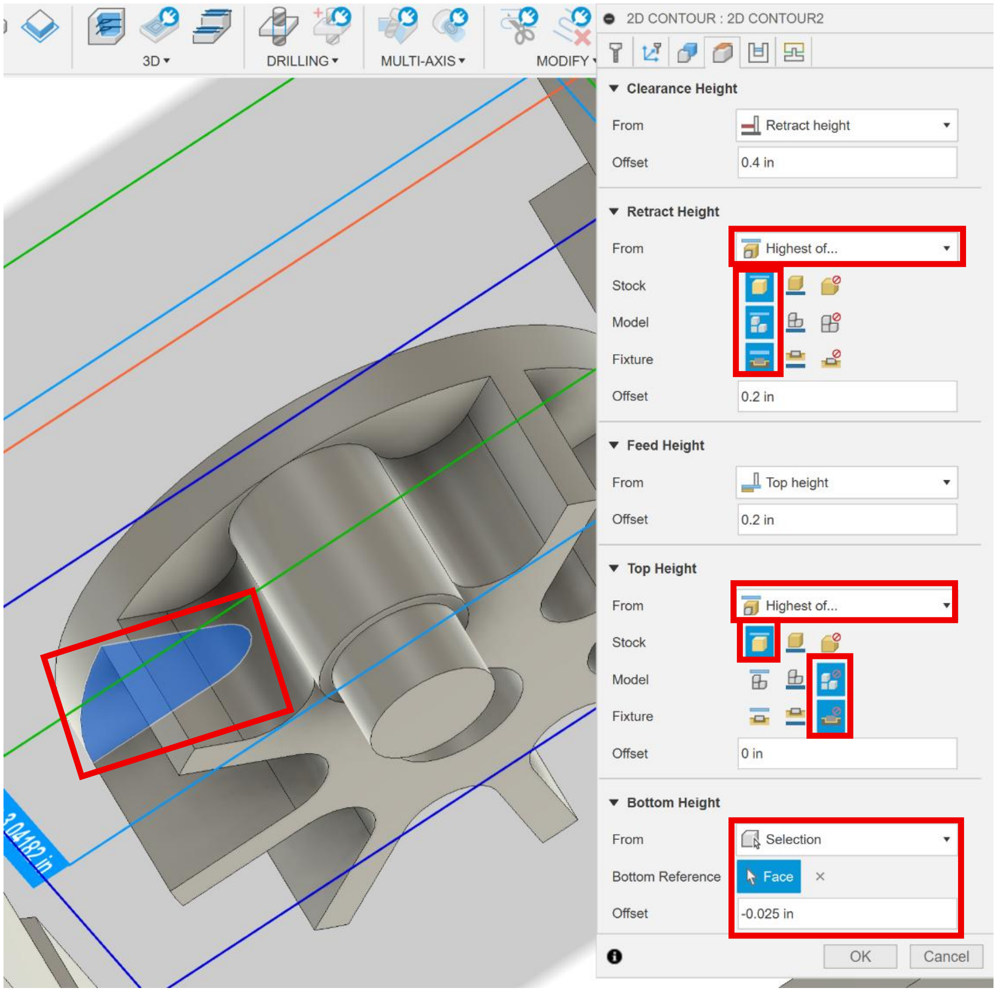

2D CONTOUR : 2D CONTOUR2
Tool
Tool
Select...
Edit...
#1 - Ø1/4" flat (Desktop Mill - 1/4" flat endmill)
Feed & Speed
Preset
Aluminum Contour R...
Spindle Speed
24000 rpm
Surface Speed
1570.8 ft/min
Ramp Spindle Speed
24000 rpm
Cutting Feedrate
48 in/min
Feed per Tooth
0.000666667 in
Lead-In Feedrate
48 in/min
Lead-Out Feedrate
48 in/min
Transition Feedrate
48 in/min
Ramp Feedrate
16 in/min
Plunge Feedrate
16 in/min
Plunge Feed per Revolution
0.000666667 in
Coolant
Disabled
OK
Cancel


TURNING
ADD
2D

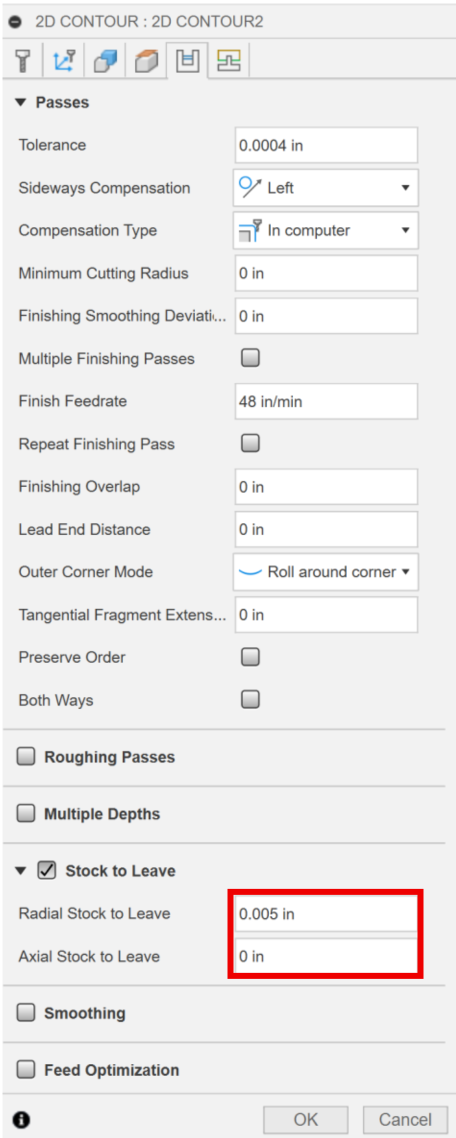
26. In the *geometry* tab, select the *chain* tab, choose the *closed chain* mode and then select the face as shown in the figure.



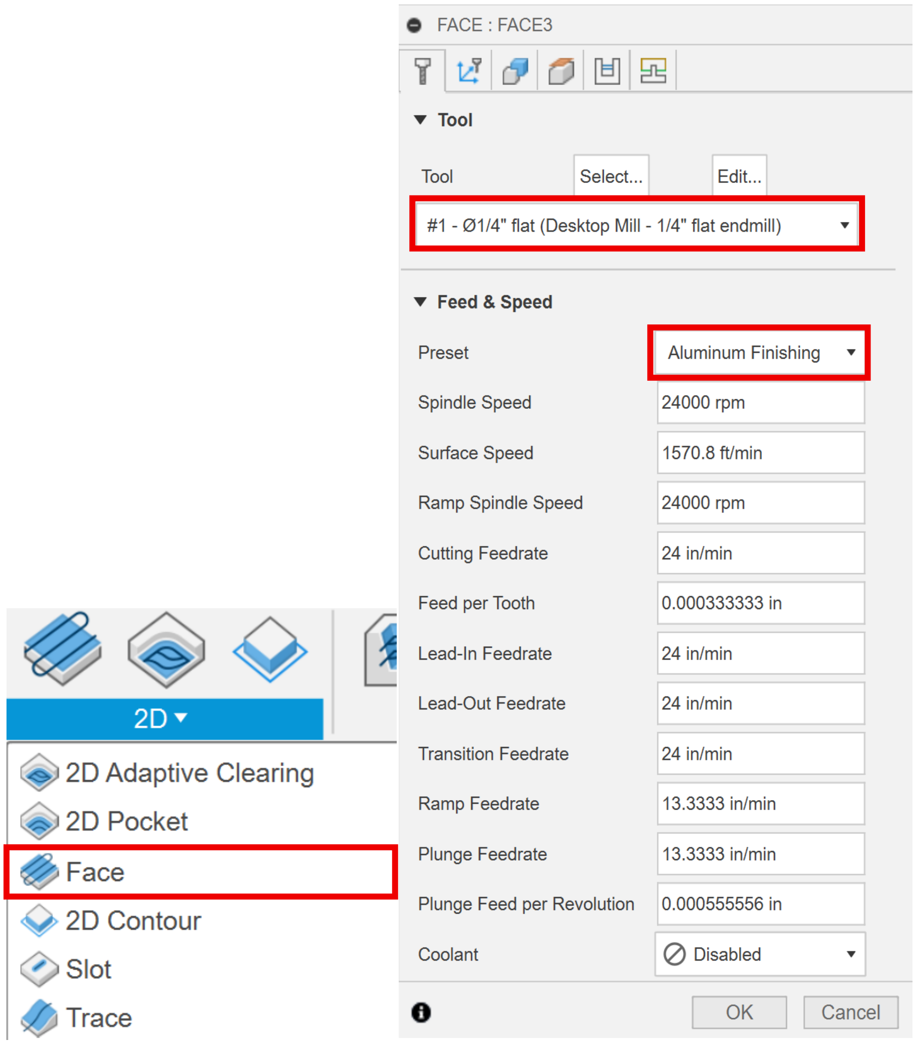
27. In the *heights* tab, for the *retract height* select *highest of…* for *from* then *top* for *stock*, *model*, and *fixture*. For the *top height* select *highest of…* for *from*, then *top* for *stock* and *ignore* for *model* and *fixture*. For the *bottom height* select *selection* for *from* then select the face as shown in the figure (you may need to hide the soft jaws to select this edge) and set the *offset* to -0.025 in.



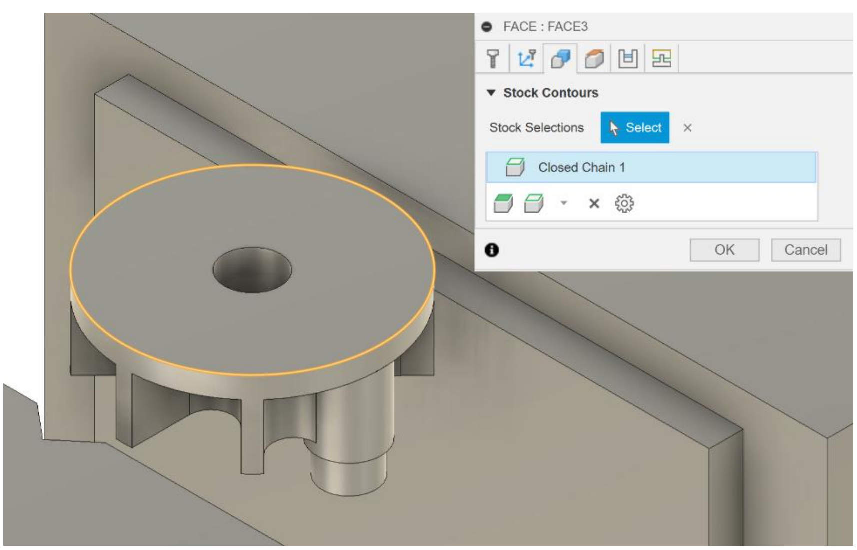
28. In the *passes* tab, check the *stock to leave* box and change the *radial stock to leave* to 0.005 in and the *axial stock to leave* to 0 in. Then click *ok*.



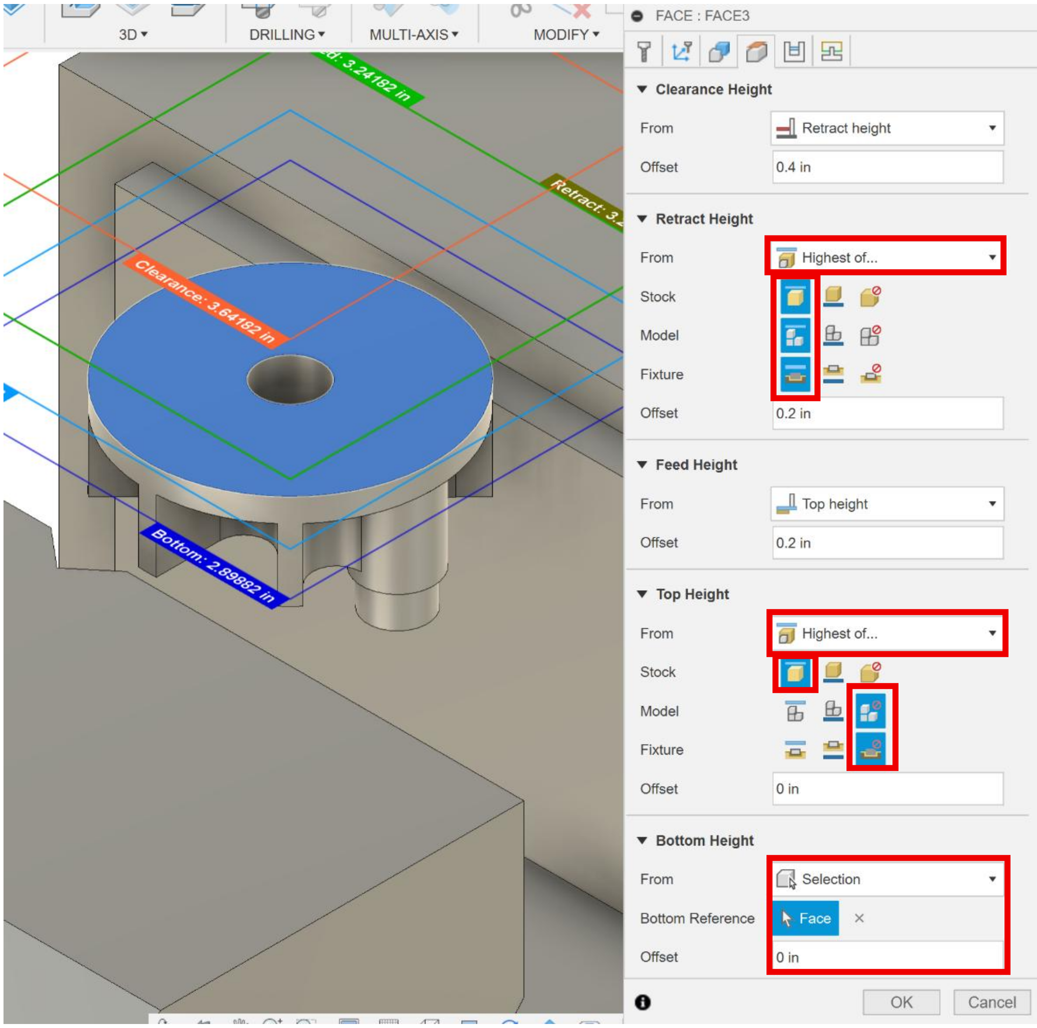
29. Select the *2D Face* tool option, make sure the *¼" tool* and *Aluminum finishing* preset are selected.



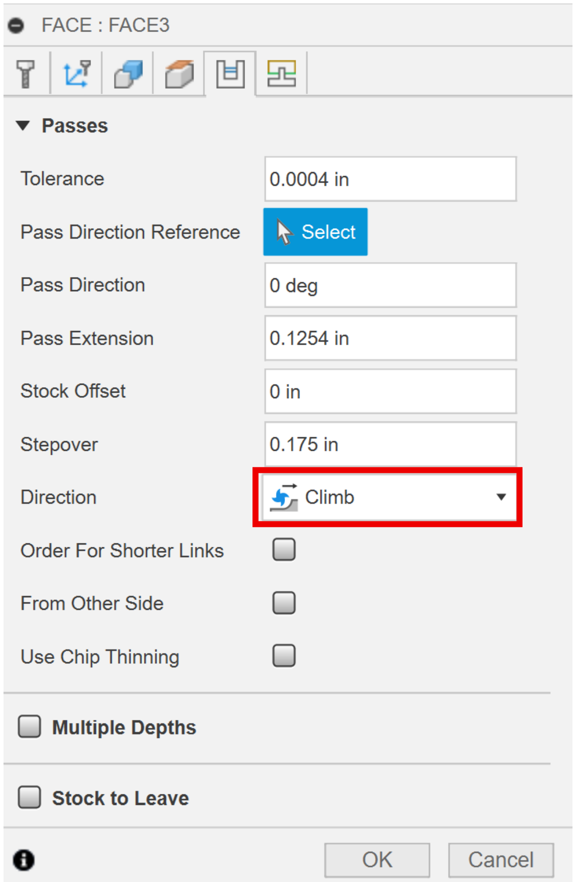
30. In the *geometry* tab, select the outer edge of the top face of the part as shown in the figure below.



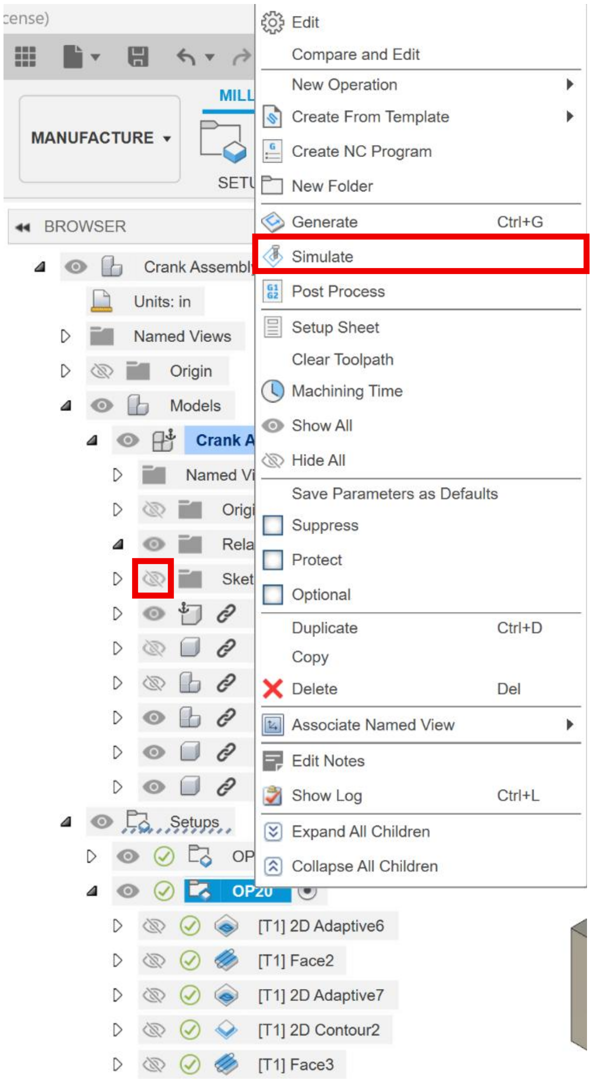
31. In the *heights* tab, for the *retract height* select *highest of…* for *from*, then select *top* for *stock*, *model* and *fixture*. For the *top height* select *highest of…* for *from* then *top* for *stock*, then *ignore* for *model* and *fixture*. For the *bottom height* select *selection* for *from* and select the top face of the part as shown in the figure below.

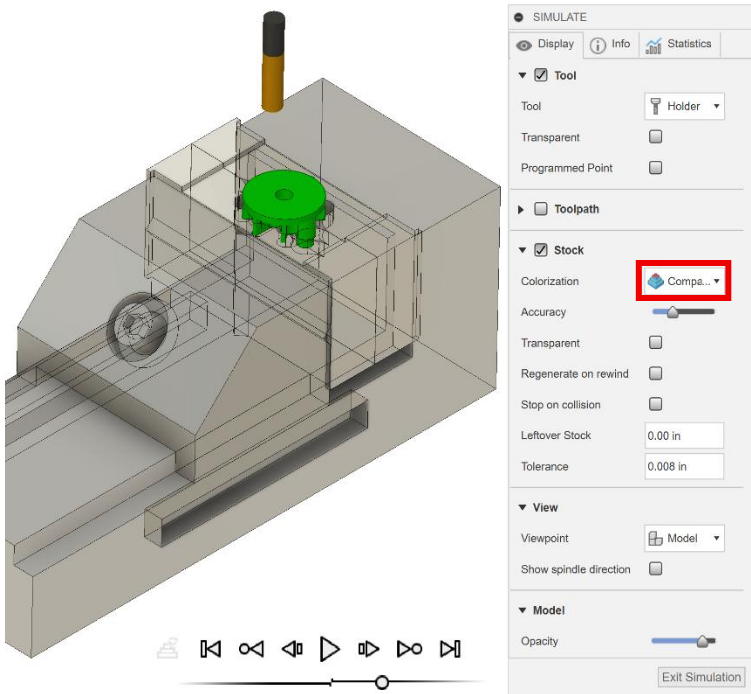


32. In the *passes* tab change the *direction* to *climb*. Then click *ok*.



33. *OP20* should now be finished. Hide *Sketches* in *Base Assembly v1:1.* Right click on *OP20* and select *Simulate*. Under *Colorization* choose *Comparison.* Press the play button at the bottom center of the window. Adjust the *Accuracy* slider until simulation runs smoothly. Once the simulation is complete, the part should be green if programmed correctly. Note: If *Accuracy* is set to the lowest value, blue may appear around the part.


MANUFACTURE
BROWSER
Crank Assembl
Units: in
Named Views
Origin
Models
Setups
OP20
[T1] 2D Adaptive6
[T1] Face2
[T1] 2D Adaptive7
[T1] 2D Contour2
[T1] Face3
Edit
Compare and Edit
New Operation
Create From Template
Create NC Program
New Folder
Generate Ctrl+G
Simulate
Post Process
Setup Sheet
Clear Toolpath
Machining Time
Show All
Hide All
Save Parameters as Defaults
Suppress
Protect
Optional
Duplicate Ctrl+D
Copy
Delete Del
Associate Named View
Edit Notes
Show Log Ctrl+L
Expand All Children
Collapse All Children


SIMULATE
Display
Info
Statistics
Tool
Tool
Holder
Transparent
Programmed Point
Toolpath
Stock
Colorization
Compa...
Accuracy
Transparent
Regenerate on rewind
Stop on collision
Leftover Stock
0.00 in
Tolerance
0.008 in
View
Viewpoint
Model
Show spindle direction
Model
Opacity
Exit Simulation

IV: Setup and Probing

1. Place the crank stock centered on 1 1/8” parallels. The stock should be resting with the widest face down on the parallels and the flatter, rolled edge along the X-axis as seen below. Close the jaws but do not tighten.

2. Use a 6mm Allen key to tighten the bolt in the center of the front vise jaw (if it is a different vise style, follow manufacturer instructions to clamp the part). This should close on the part and clamp tightly. As you tighten the vise, apply force down on the part. This will make sure that the part is evenly seated on the parallels. Check this by gently shaking the parallel; if it shakes noticeably against the part, try again (NOTE: an uneven surface may not seat perfectly against a flat parallel).

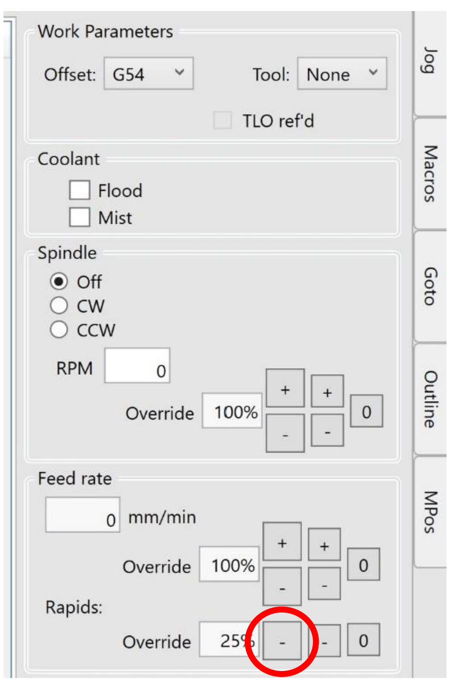
3. Gather the probe and attach the probing tip. Slide the probe into the spindle with a ¼” collet and tighten by hand. This should hold the probe safely. Be sure that the spindle is turned off by the emergency stop or door switch.


4. With the electrical box on and connected to both the computer and machine tool, plug the probe into the electrical box and GENTLY move the probe tip. The light should change from green to red and the computer software should show a “P” alarm while the probe tip senses contact. Both conditions must be met to ensure probe operation.

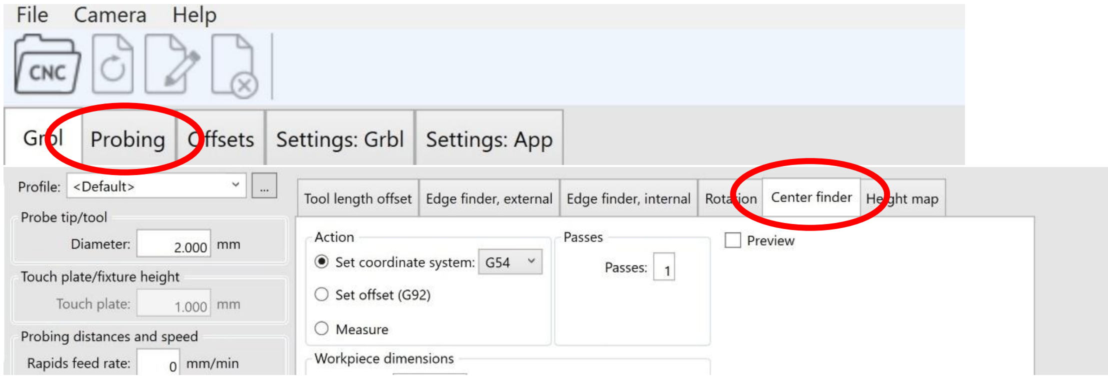
5. Carefully jog the probe over the part. Do not allow the probe tip to touch any surface while jogging, as excessive force or deflection can permanently damage the probe tip. Carefully jog the probe until it is as centered as visually possible over the part and 5-7 mm from the top surface of the part.

6. Before probing, reduce the rapids rate to 25%. Rapid movement rate can be changed with the minus sign buttons in the "Rapids" section shown below.

   NOTE: this is an important step. This is done so that any errors in probing can be safely caught before any damage is done.

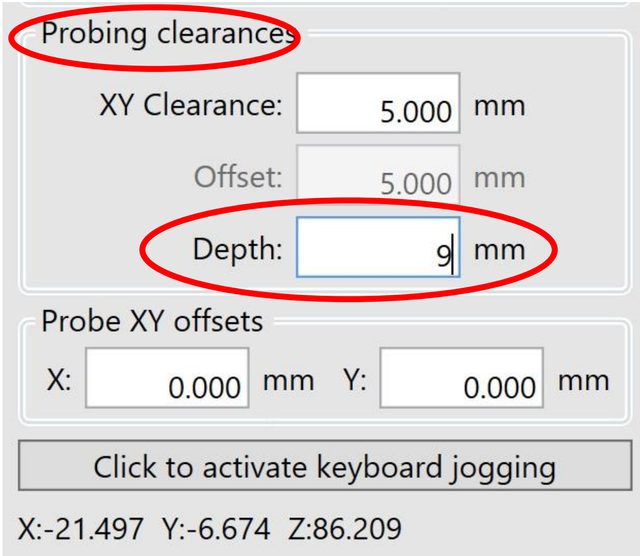


7. Move to the “Probing” tab in the ioSender software. From there, move to the “Center finder” option.

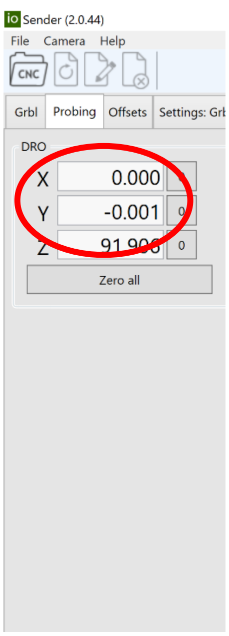


8. Ensure that the “Action” is set to “Set coordinate system” in G54. Uncheck the “Lock” box. Measure the stock and type in the workpiece dimensions in millimeters. This should be roughly 92mm in the X-direction and 40mm in the Y-direction. Select the external option. Select the external bore option for probing the outside surfaces of the stock.

9. In the “Probing clearances” section, set the probing depth. This value will determine how far down the probe travels to touch the workpiece. It is equal to the distance between the current probe tip position and the place on the part that will be probed-- it may be necessary to roughly measure this distance before probing to ensure correct positioning and mitigate errors.

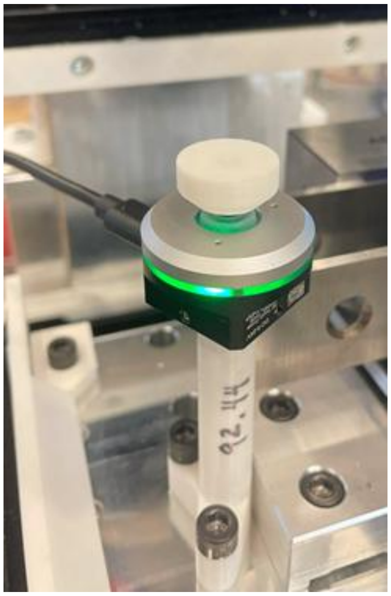


10. Double check inputs. Errors are much more difficult to catch quickly in operation. Press "Start" and watch the probing cycle very closely. Emergency stop the machine if anything unexpected occurs and safely try again. The machine should move across the part to touch each side once. This will set the X and Y offsets for a coordinate system centered on the rectangular block. The X- and Y- offsets should be zeroed.



11. We will now set the Z height for the coordinate system. This is based on the surface of the X-carriage and will include the tool offset. Jog the probe safely away from the part and remove it from the spindle.

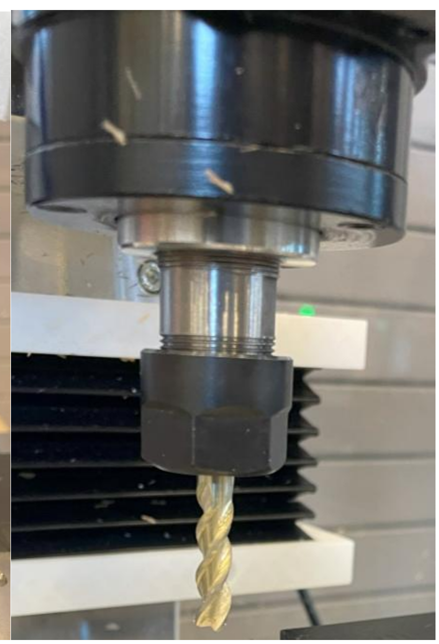
12. Gently replace the probe tip with the probe touch plate. Insert the shaft of the probe into the holder on the left side of the vise. If the signal cable was disconnected, plug it into the probe and test again to be sure the signal is working.



13. Press the emergency stop. **Never handle tools in the spindle without the emergency stop engaged.**

    Fully unscrew the collet from the spindle. Gently press out the ¼” collet and replace with a 1/8” collet. Press until the collet “clicks” into place and is correctly seated.

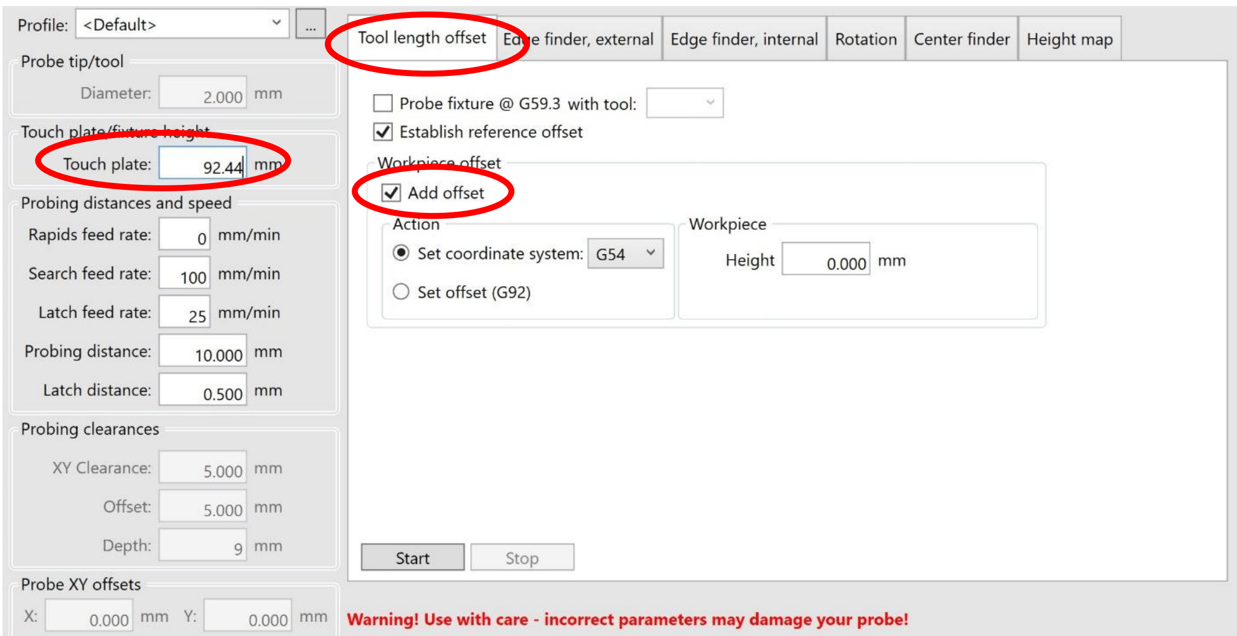
    Screw the collet onto the spindle loosely, then insert the 1/8” endmill into the spindle and secure tightly using the provided wrenches. The tool should protrude just past the start of the flutes.

14. NOTE: Trying to jog any axis while the emergency stop is engaged will move the coordinate system without moving the machine. Do not jog until the emergency stop is disengaged.

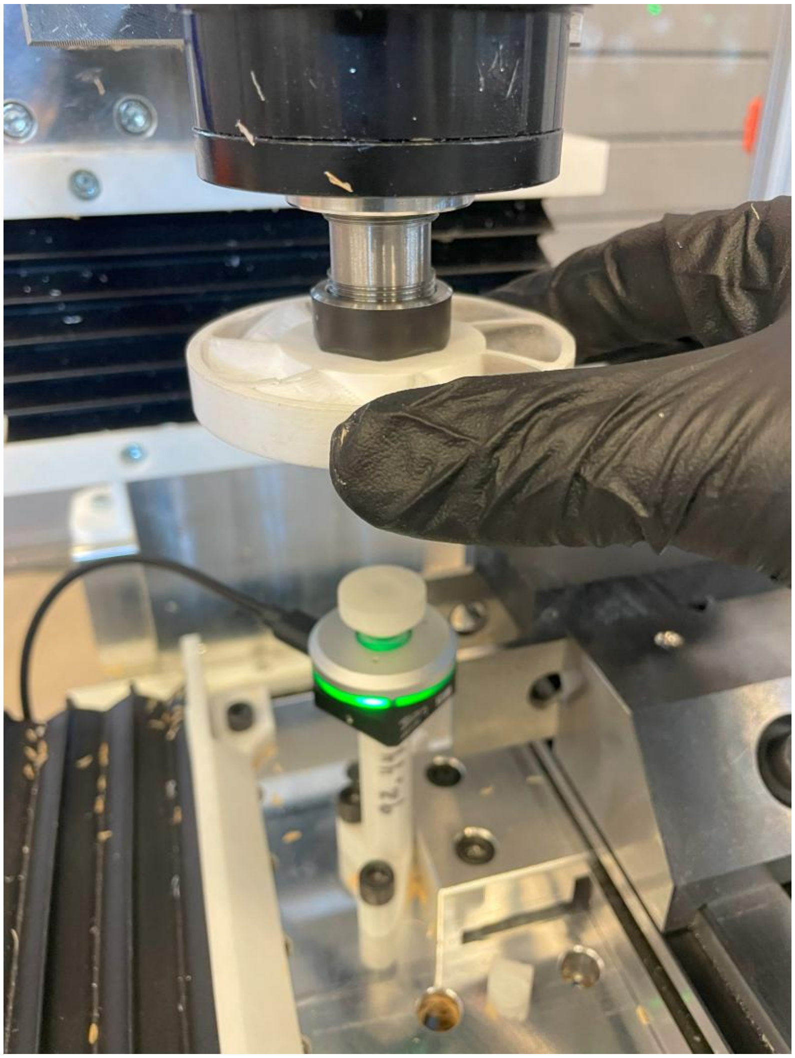
    Carefully jog the spindle until the tool rests approximately 5mm above and centered on the touch plate.

15. Move back to the "Probing" tab and select "Tool length offset". Check the "Add offset" box and set the touch plate height to 92.44 mm (or the otherwise determined accurate distance from the touch plate to the X-carriage surface – see *Troubleshooting*).



16. Double check all parameters and press start. The tool should slowly approach the touch plate and gently press it twice. This will set the Z-coordinate at the bottom of the vise / top of the X-carriage.

17. Jog the tool out of the way and remove the probe completely from the enclosure. If the machine tool is not equipped with compressed air, slide the plastic chip fan onto the spindle over the tool as seen below.



18. Jog the tool so that it starts the operation **above** the part.

    The machine tool should now be ready to run the first operation.

    REVIEW AND FOLLOW GENERAL INSTRUCTIONS AND CHECKLIST.

    Be ready to feed hold the machine IMMEDIATELY after pressing cycle start. This will allow the spindle to reach the desired speed before milling begins (this usually takes no longer than 5 seconds – the spindle should be making a constant sound and not "ramping up").
    ALWAYS ALLOW THE SPINDLE TO GET UP TO SPEED BEFORE ENTERING A CUT.

    After pressing feed hold and letting the spindle reach its target RPM, you can resume the operation with cycle start.

19. Watch the entire operation carefully.

OP20

20. Once OP10 is complete, jog the tool out of the way. Emergency stop the machine and open the door. Clear any chips as necessary. Carefully slide the chip fan off the spindle. Remove the 1/8” endmill with the wrenches.

21. Use the Allen key to loosen the vise jaw and remove the cut part.

22. Place one aluminum soft jaw on each vise face, with the patterned side facing up and towards the other jaw (see below).

23. Place the crank part centered on the geometry of the aluminum soft jaws. The part should rest naturally centered and flat between the soft jaws. Close the jaws but do not tighten.

24. To prevent the jaws from moving when clamping a part (and because this vise has no threaded holes), place an appropriate plastic spacer between the soft jaw faces at the bottom of the jaws. The spacer should be about the same width as the gap in between the jaw faces, but loose enough to move.

25. Use a 6mm Allen key to tighten the bolt in the center of the front vise jaw (if it is a different vise style, follow manufacturer instructions to clamp the part). This should close on the part and clamp tightly. As you tighten the vise, apply force down on the part. This will make sure that the part is evenly seated on the parallels. Check this by gently shaking the parallel; if it shakes noticeably against the part, try again (NOTE: an uneven surface may not seat perfectly against a flat parallel).

    Check by hand to see if the part and spacer are tightly held. If the part is loose and the spacer is tight, choose a spacer of slightly smaller width. If the part is tight but the jaws have moved from flat against the jaw face, choose a slightly wider spacer. DO NOT attempt to run parts without good spacing and flat jaws: this could damage parts, tools, and the machine.

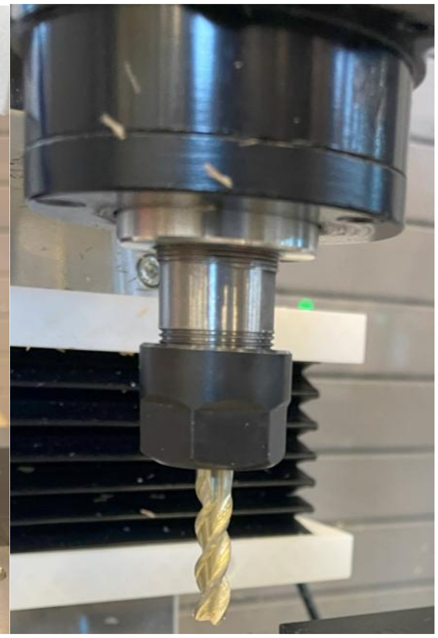
26. Press the emergency stop. **Never handle tools in the spindle without the emergency stop engaged.**

    Fully unscrew the collet from the spindle. Gently press out the 1/8” collet and replace with a 1/4” collet. Press until the collet “clicks” into place and is correctly seated.

27. Follow steps 3-10 to probe the part. The X- and Y- measurements should remain the same as in probing the first operation. For this setup, ensure that the probing depth is set so that the flat faces of the stock contact the probe tip and not the probe shaft while probing.

28. Press the emergency stop. **Never handle tools in the spindle without the emergency stop engaged.**

    Insert the ¼” endmill into the spindle and secure tightly using the provided wrenches. The tool should protrude just past the start of the flutes.



29. Follow steps 11-16 to set the Z-offset.

30. Repeat step 18.

31. Watch the entire operation carefully.

32. Once OP20 is complete, jog the tool out of the way. Emergency stop the machine and open the door. Clear any chips as necessary. Carefully slide the chip fan off the spindle. Remove the 1/4” endmill with the wrenches.

33. Use the Allen key to loosen the vise jaw and remove the cut part.